

University of Dhaka
Department of Robotics & Mechatronics Engineering

Development of an Origami-Inspired Inchworm Robot with Minimal Actuation for Locomotion in Constrained Spaces

RME 4200: Project



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Proposed By

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Problem & Motivation

Rescue missions, industrial inspection, and exploration often involve

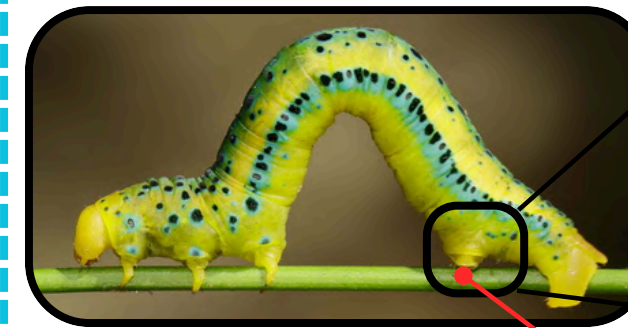
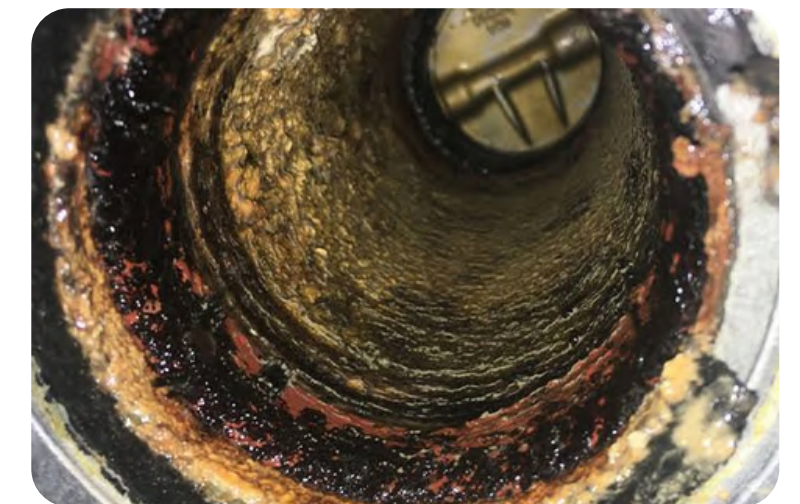
- **narrow, cluttered or irregular spaces**
- where conventional wheeled or rigid robots have **limited accessibility and maneuverability**.

Is there any solution to this problem?

- **Natural solution: inchworm.**
- This bio organism moves through **constrained spaces with ease, using a simple looping gait**

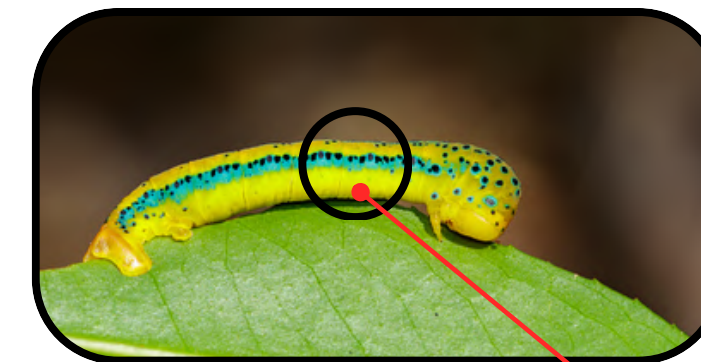
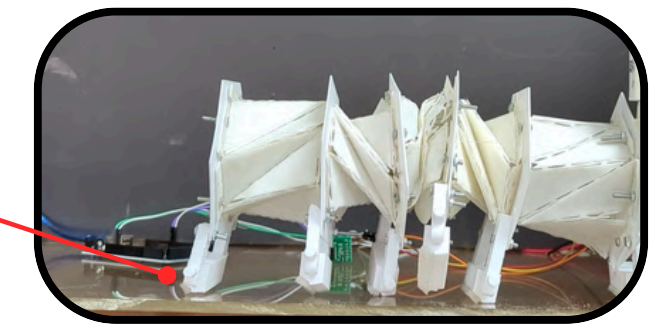
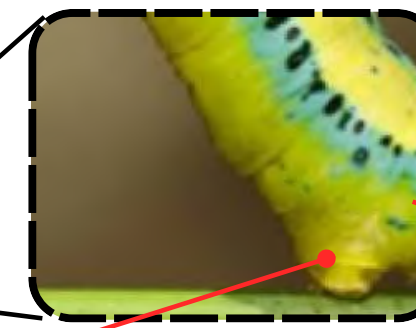
How can we implement this locomotion of into a mechanism?

- Relying on **“mechanical intelligence”** instead of electronics”.
- Origami allows a structure with a **large functional volume** to be compressed into a **tiny space through simple folding**.
- **Doesn't require actuator** for every degree of freedom



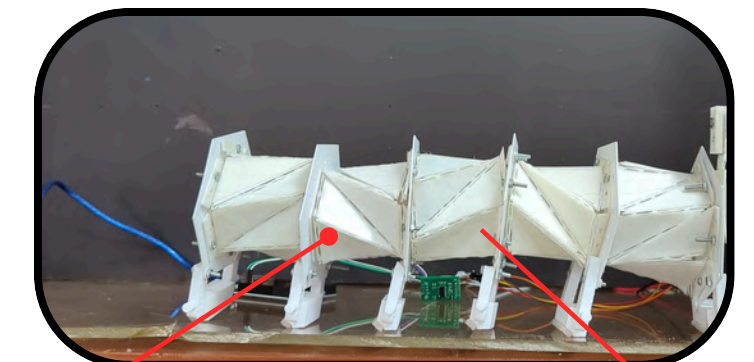
Setae (Rear Prolegs)

Anchors to the surface and provides high friction



Compliant Joints

Flexible joints that allows yaw, pitch



Kresling Origami

Creates peristalsis twists by rotating the entire body

Relevant Works

Actuation Strategies

- **Electrostatic**

Lightweight and compact, **but requires high-voltage power.**

- **Pneumatic**

High force and deformation, but requires **external pumps and tubing.**

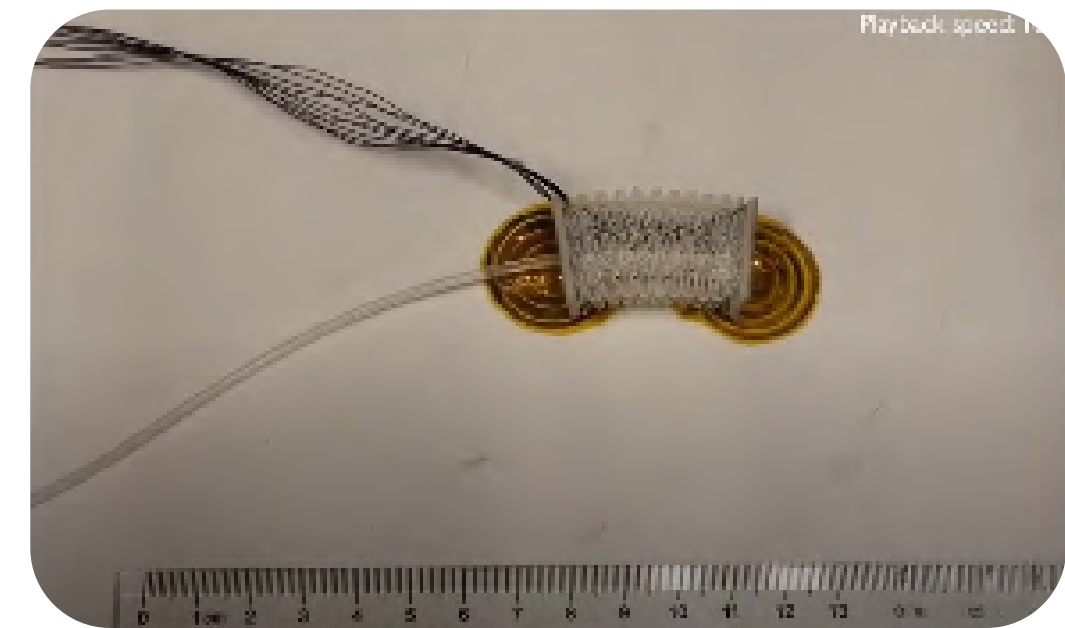
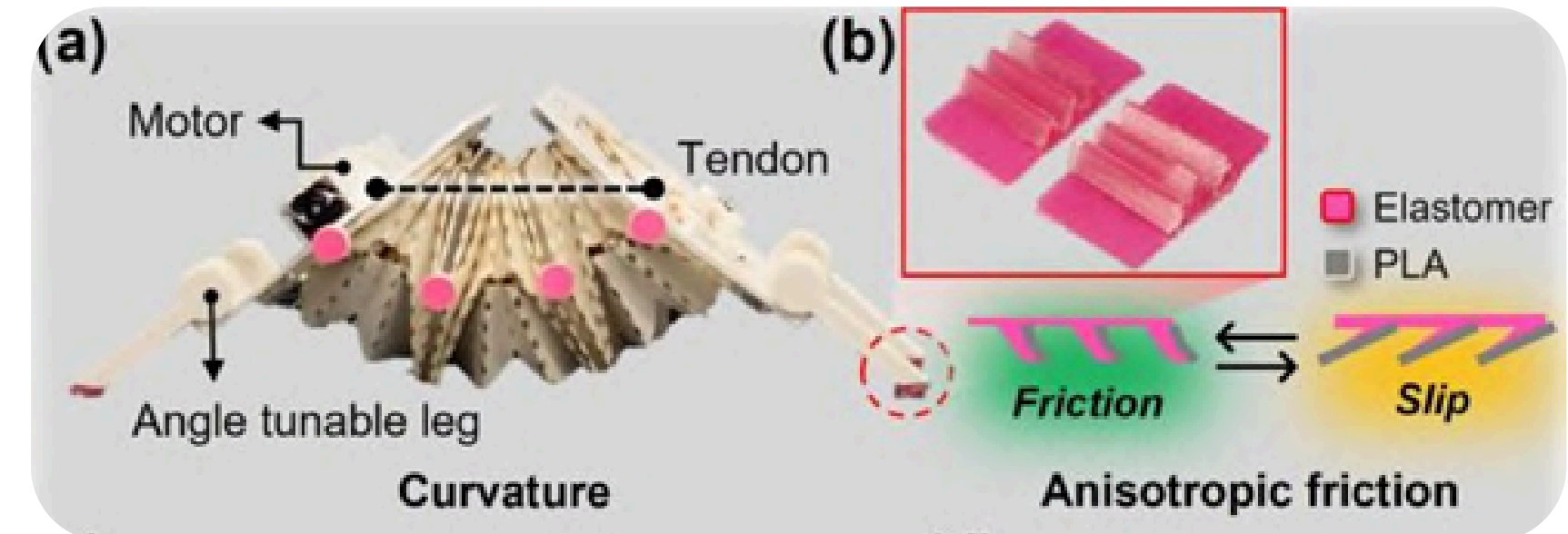
- **Magnetic**

Wireless actuation, but requires a **continuous external magnetic field** and constrained workspace.

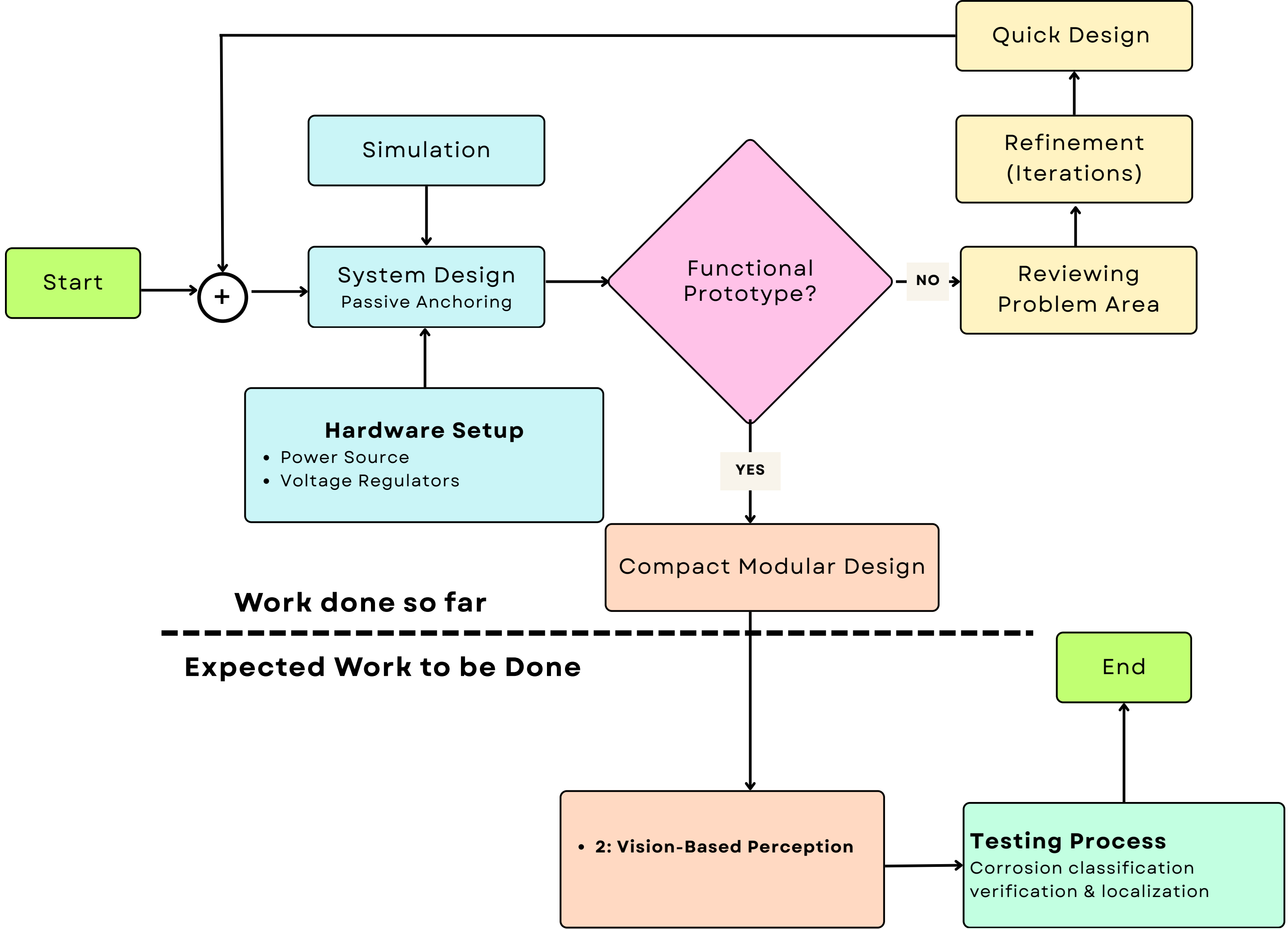
- **Tendon-driven**

Compact, self-contained and motor-driven, with **efficient force transmission** from a **single onboard actuator.**

Kim et al., Hu et al. developed **a single-motor tendon-driven mirrored Kresling** modules suppress parasitic rotation. And the **anisotropic friction** enables crawling and **demonstrated locomotion on flat and inclined PVC surfaces.**

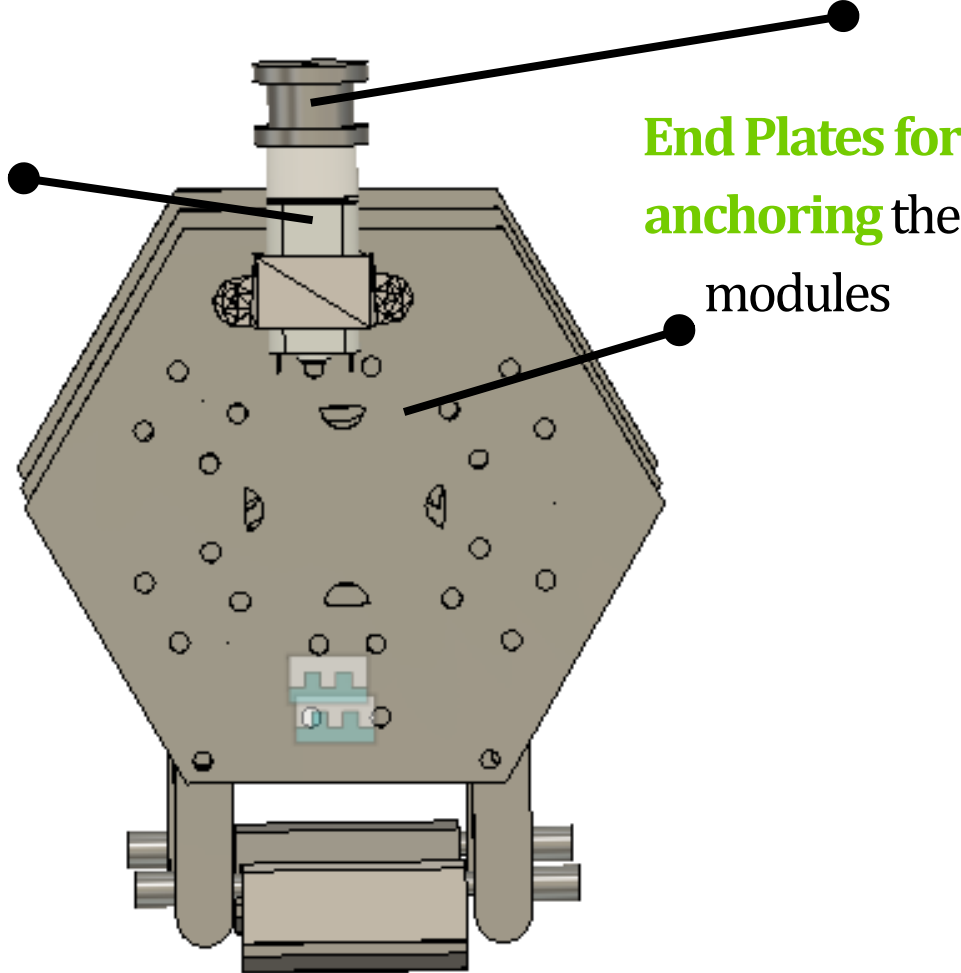


Methodology



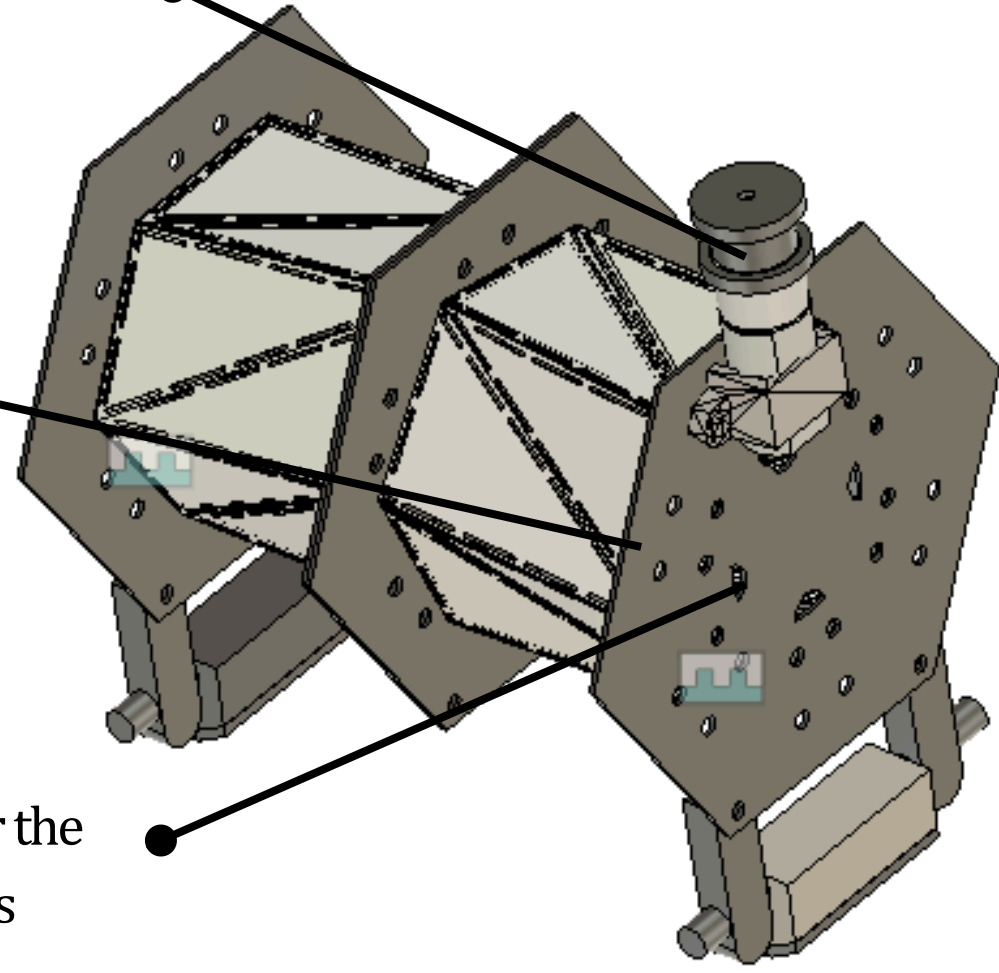
Mechanical Architecture

Single actuator for tendon driven locomotion



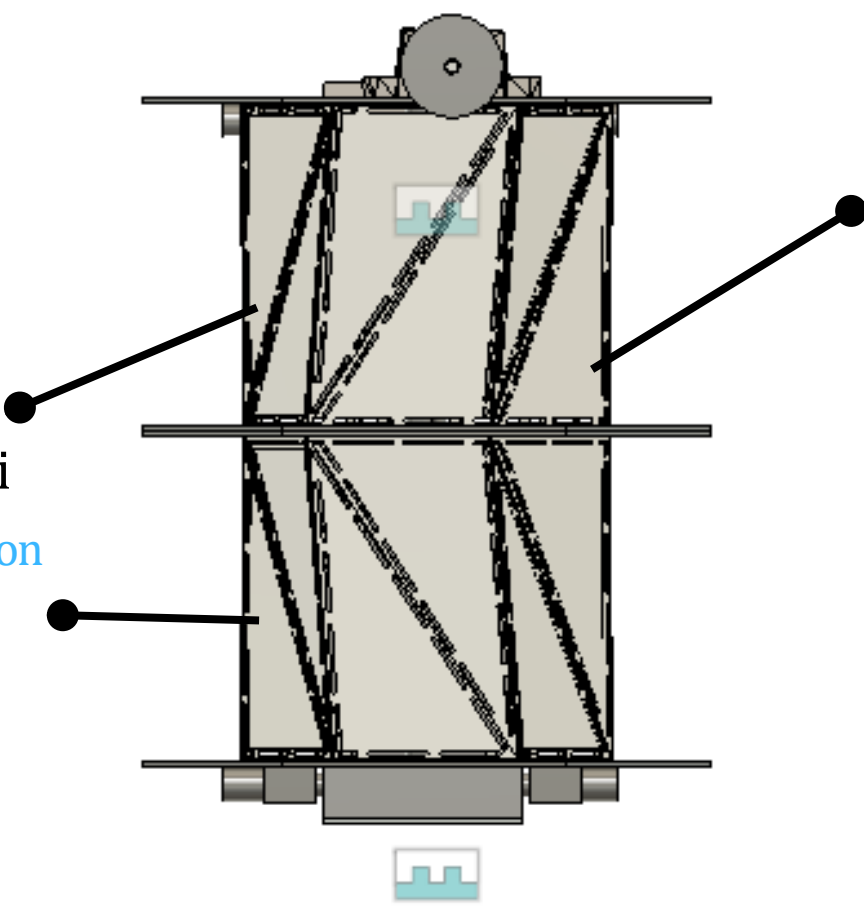
Spool for actuating the modules with tendons

End Plates for anchoring the modules



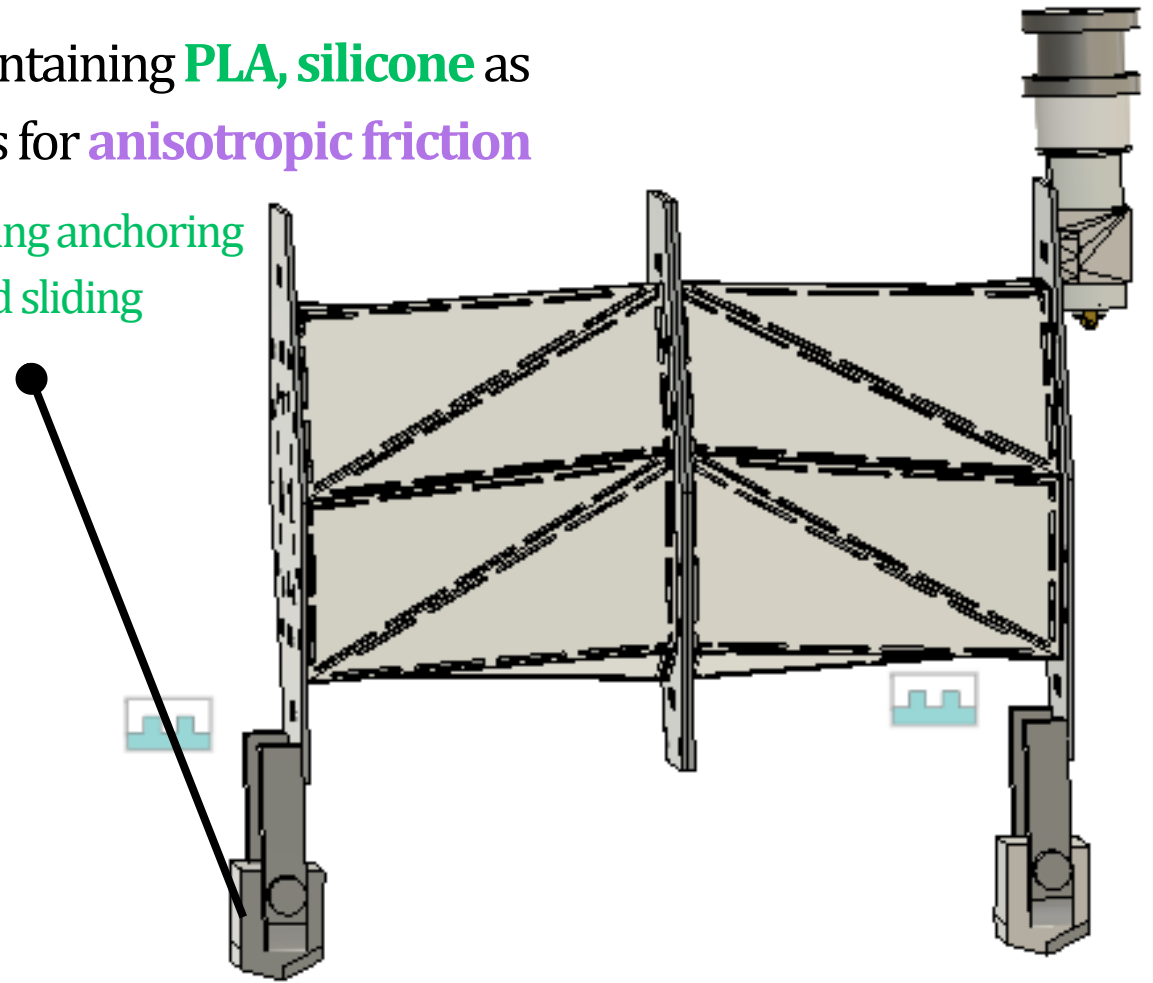
Routes for the tendons

Mirrored Kresling Origami
Axial contraction with twist cancellation



Foot containing PLA, silicone as surfaces for anisotropic friction

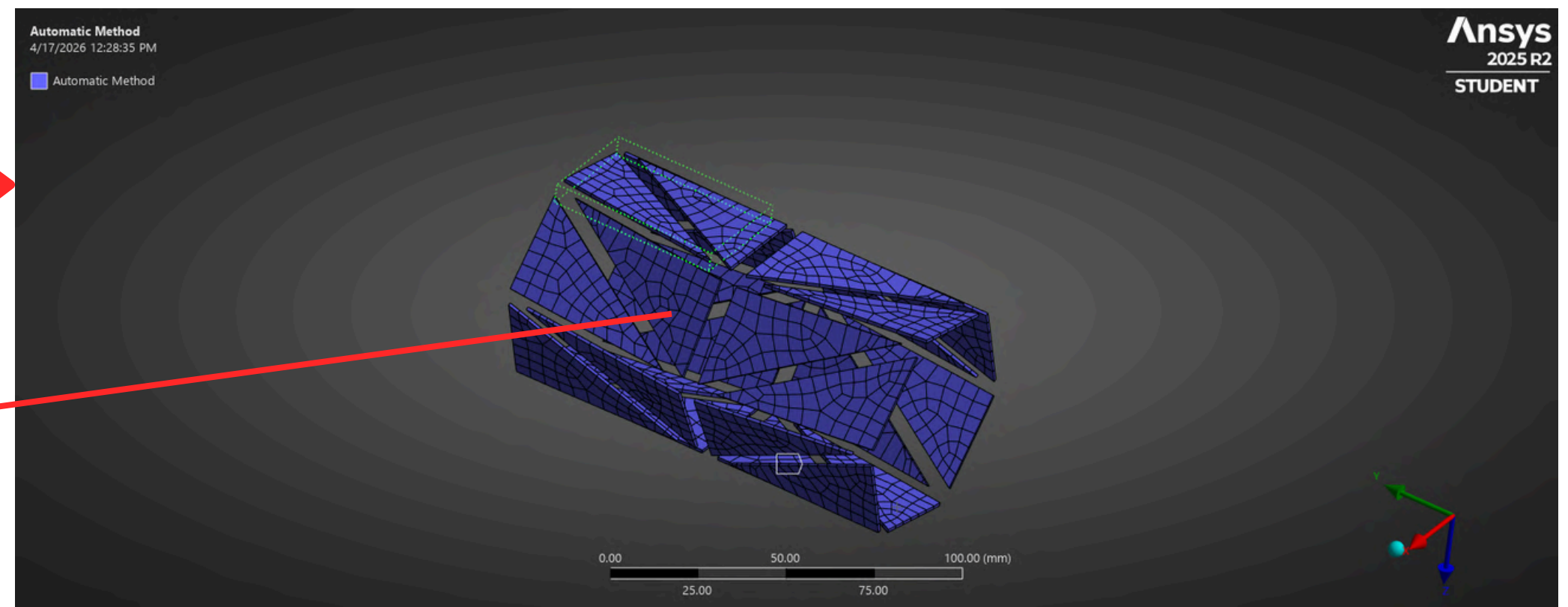
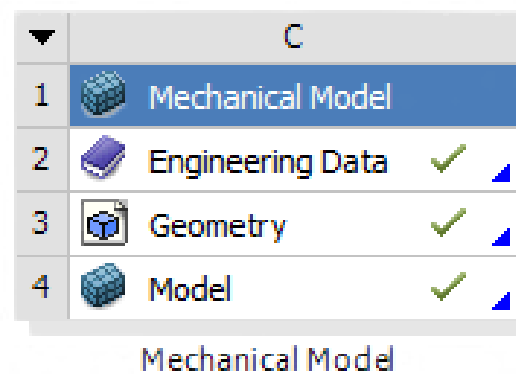
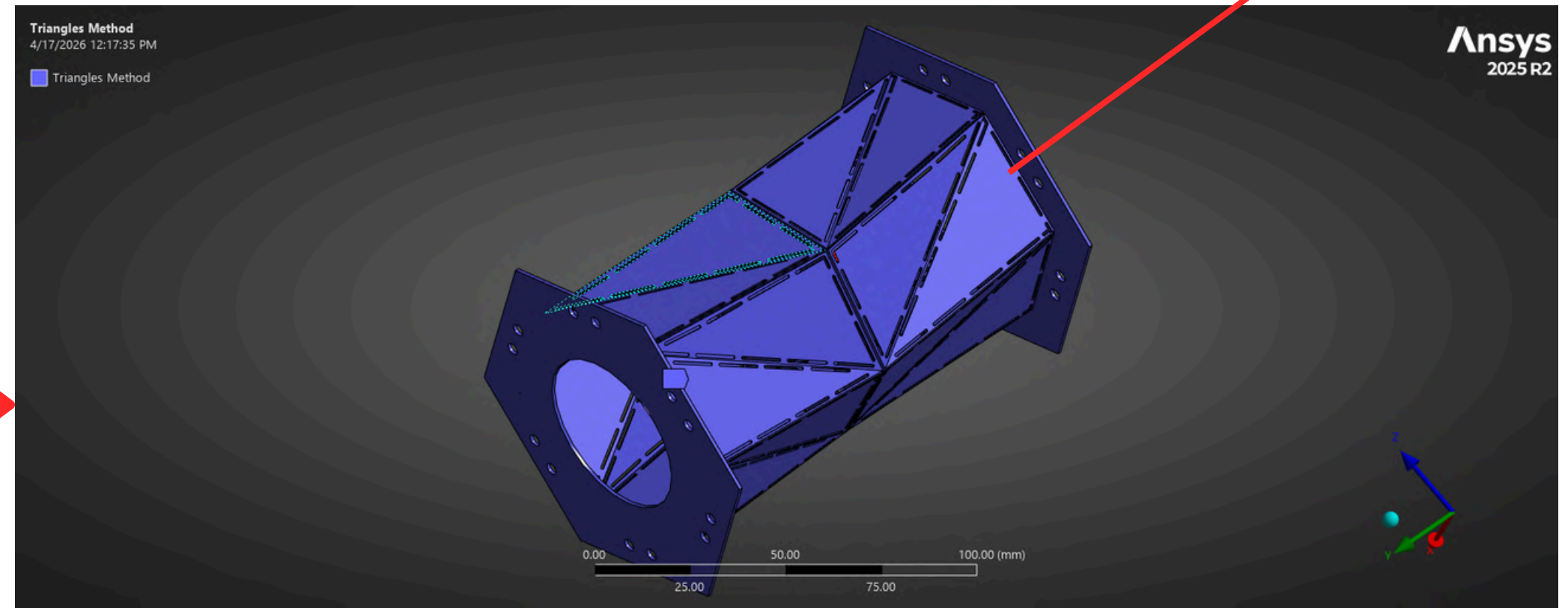
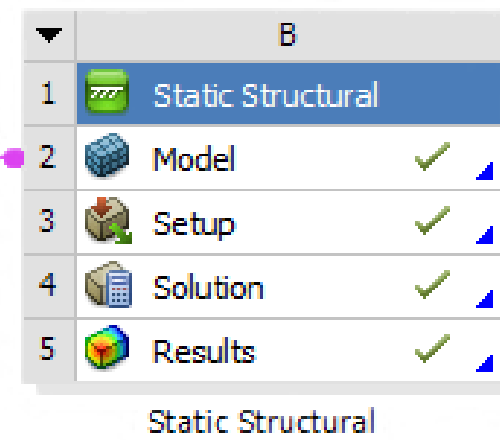
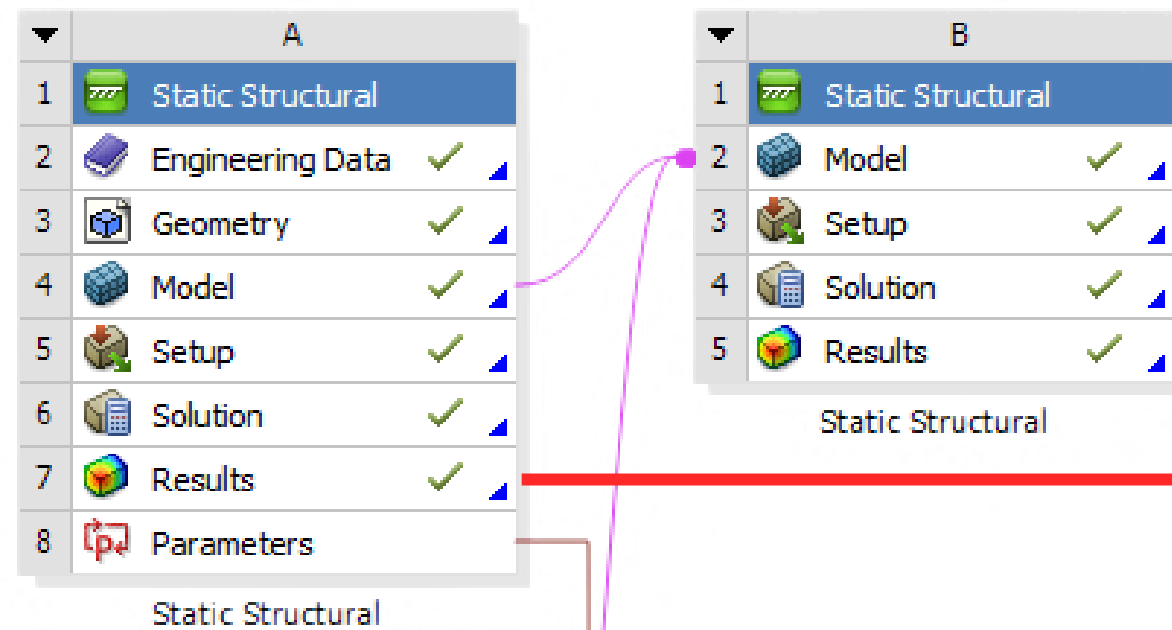
Alternating anchoring and sliding



How was this architecture developed and optimized?

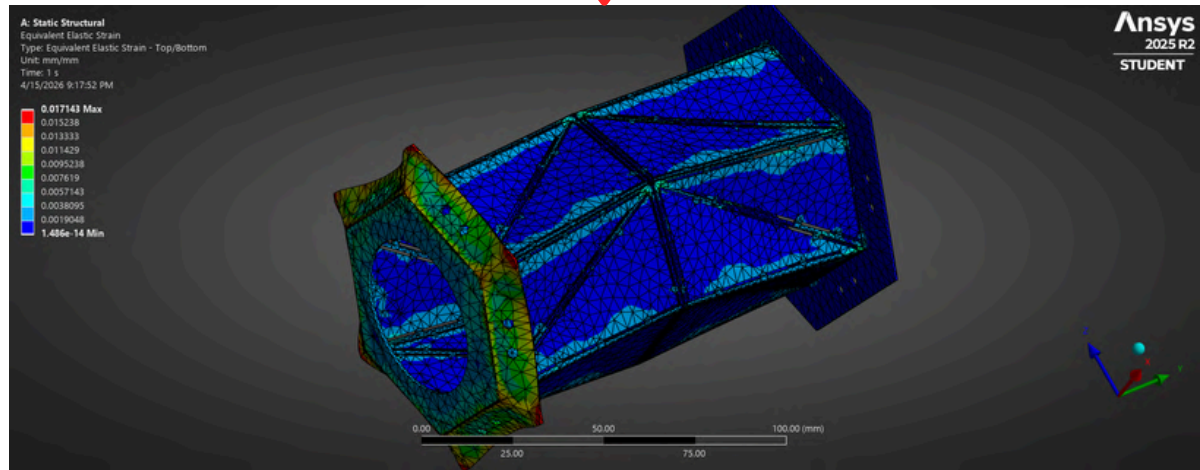
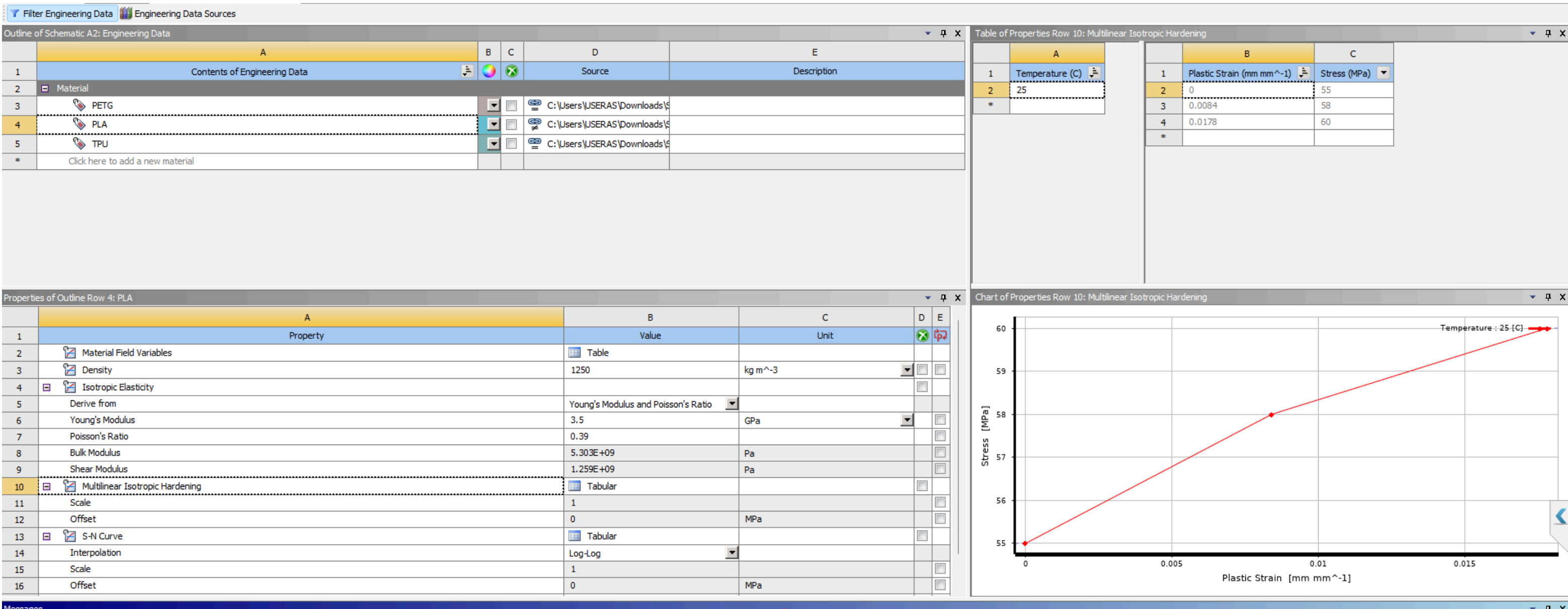
Material optimization: Kresling Origami Chassis

Outer body

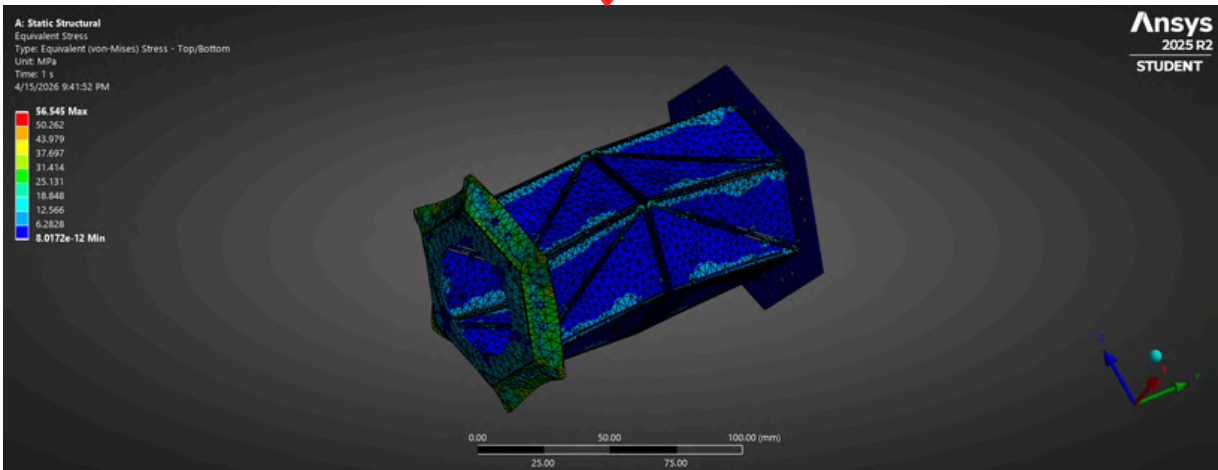


Facets

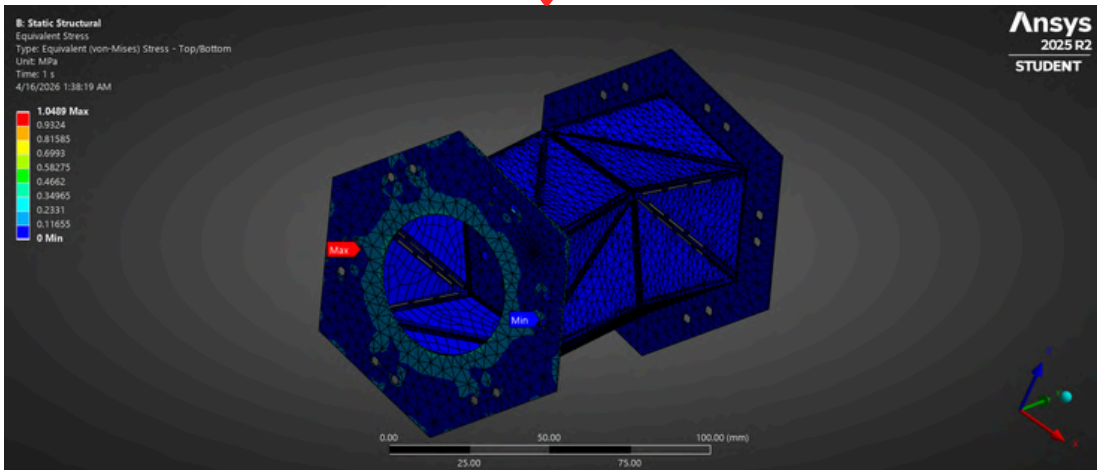
Which material will be the most suitable for the chassis?



PLA-PETG

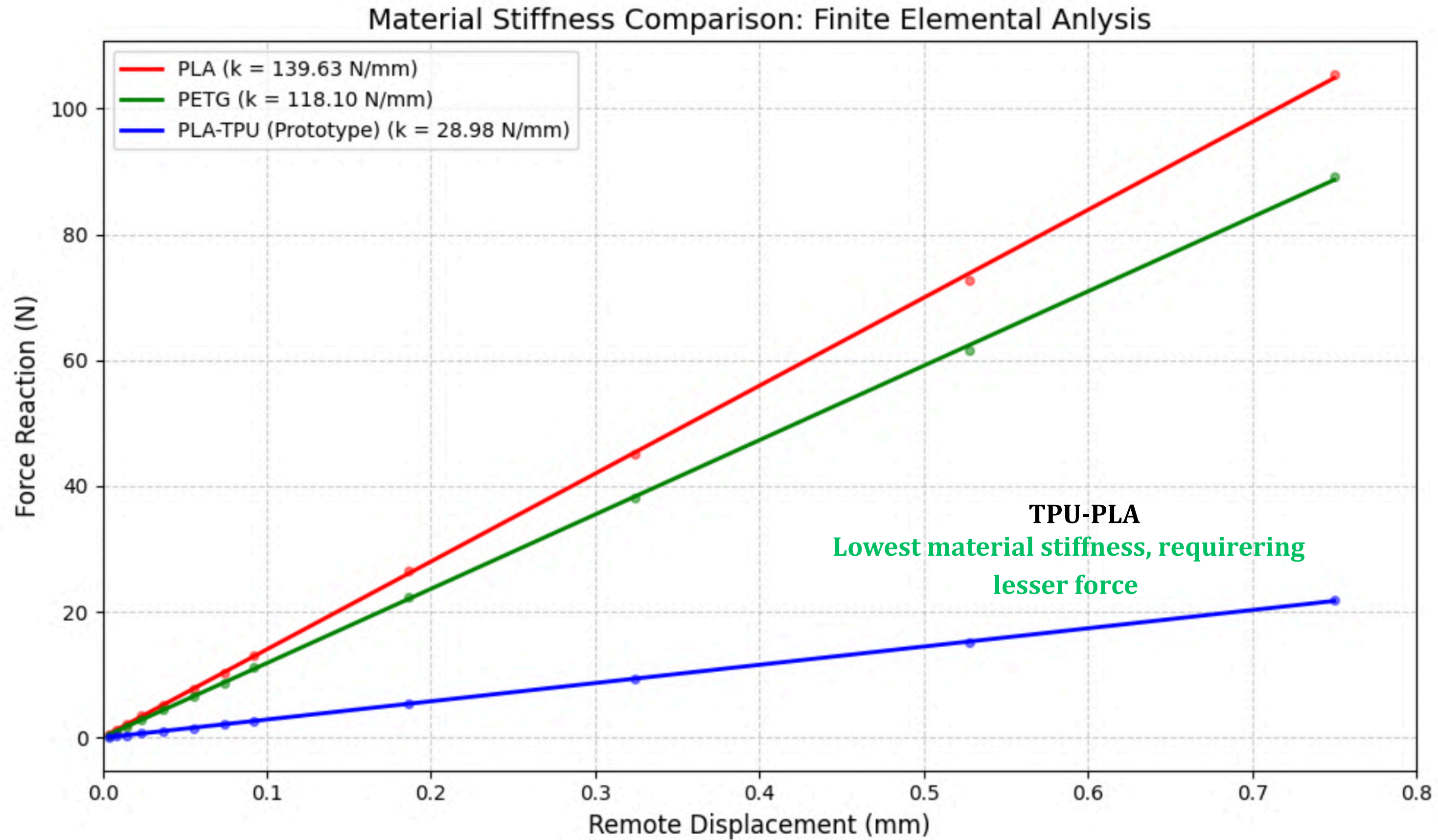


TPU-PETG



TPU-PLA

Trade-off between flexibility and structural integrity

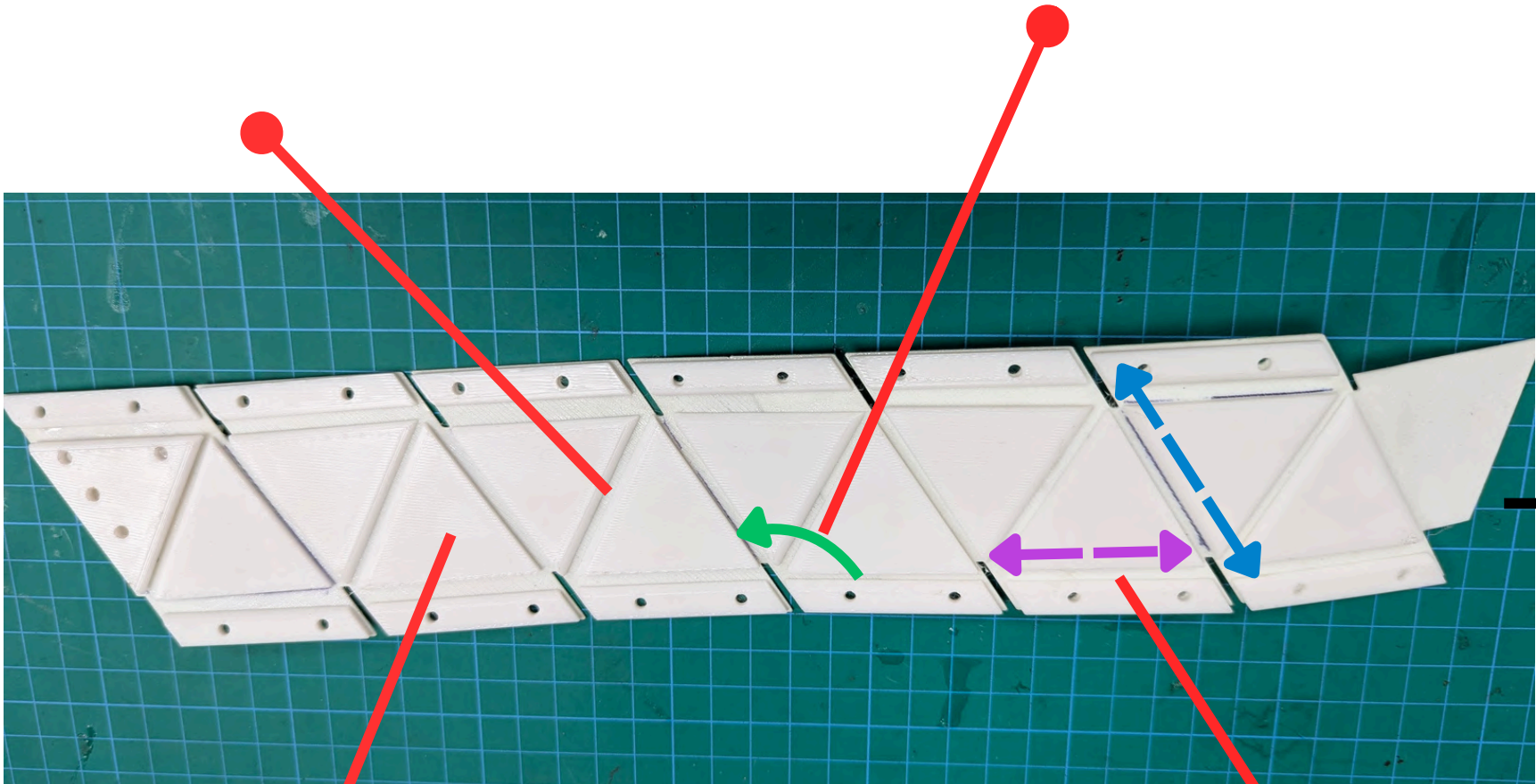


Dimension optimization: Kresling Origami Chassis

Iteration 1: *PLA-TPU Sheets*

2mm Tolerance for living hinges

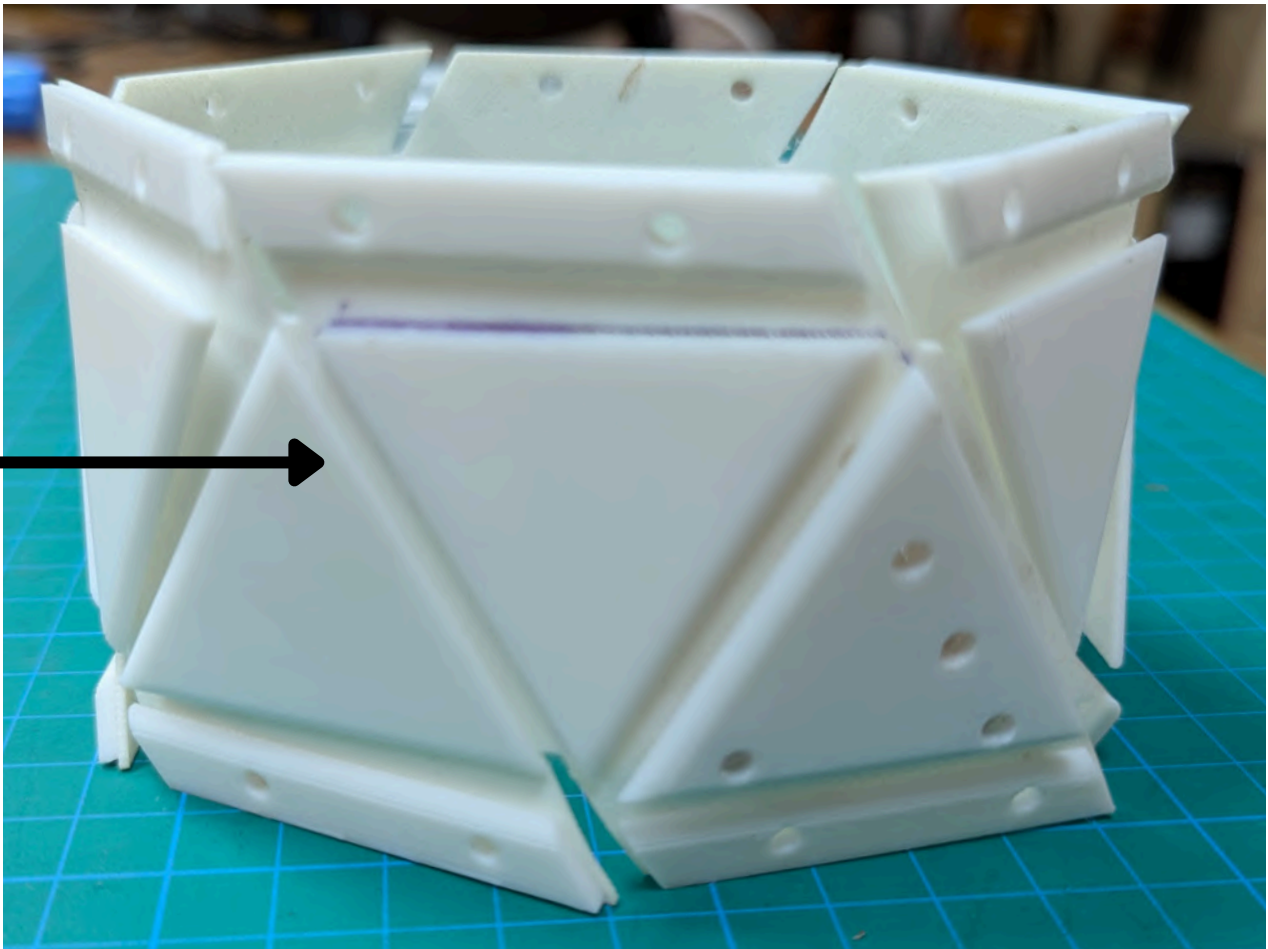
Fold Angle(60°)



PLA Facets (2mm)

TPU Sheet (4mm)

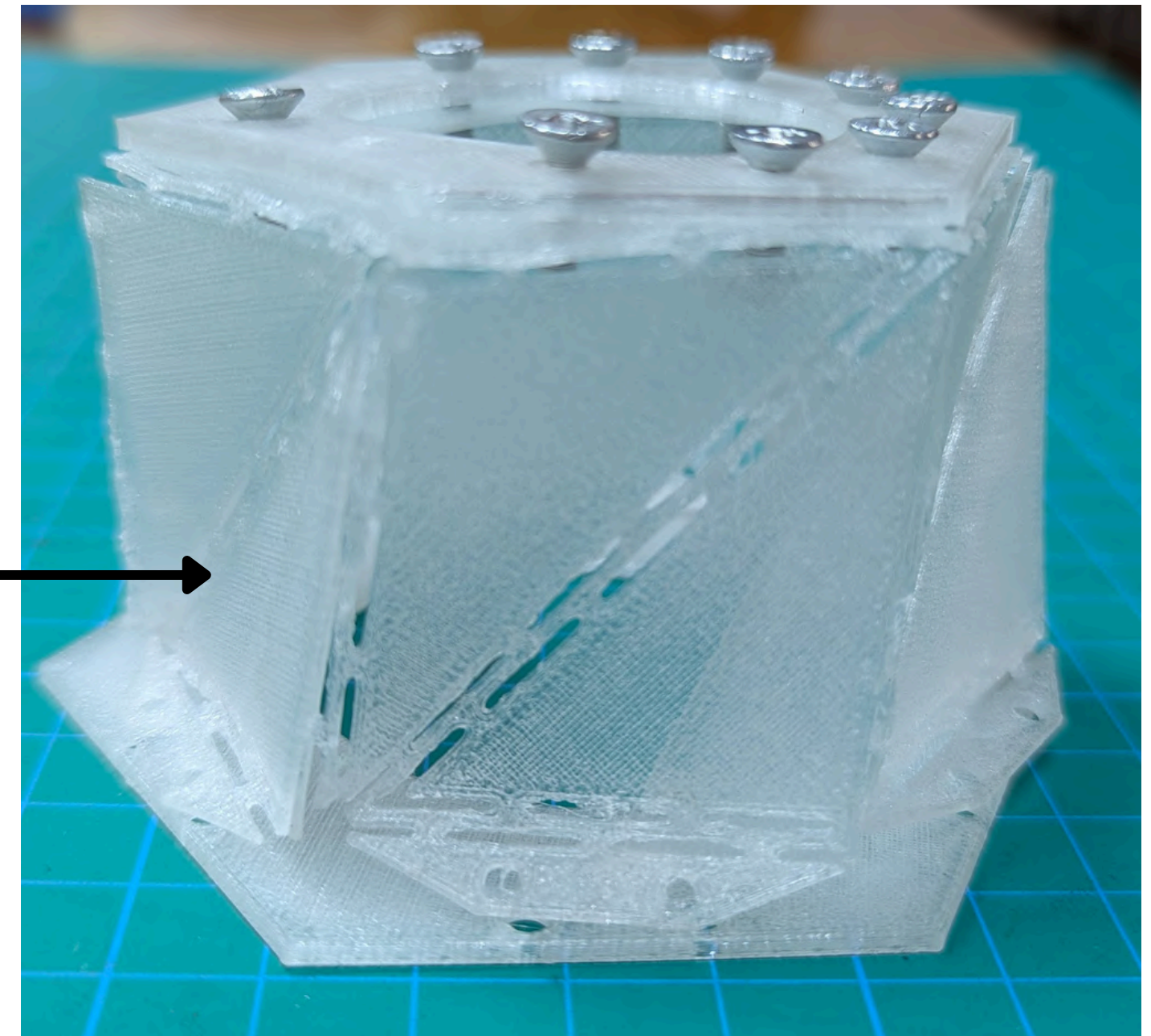
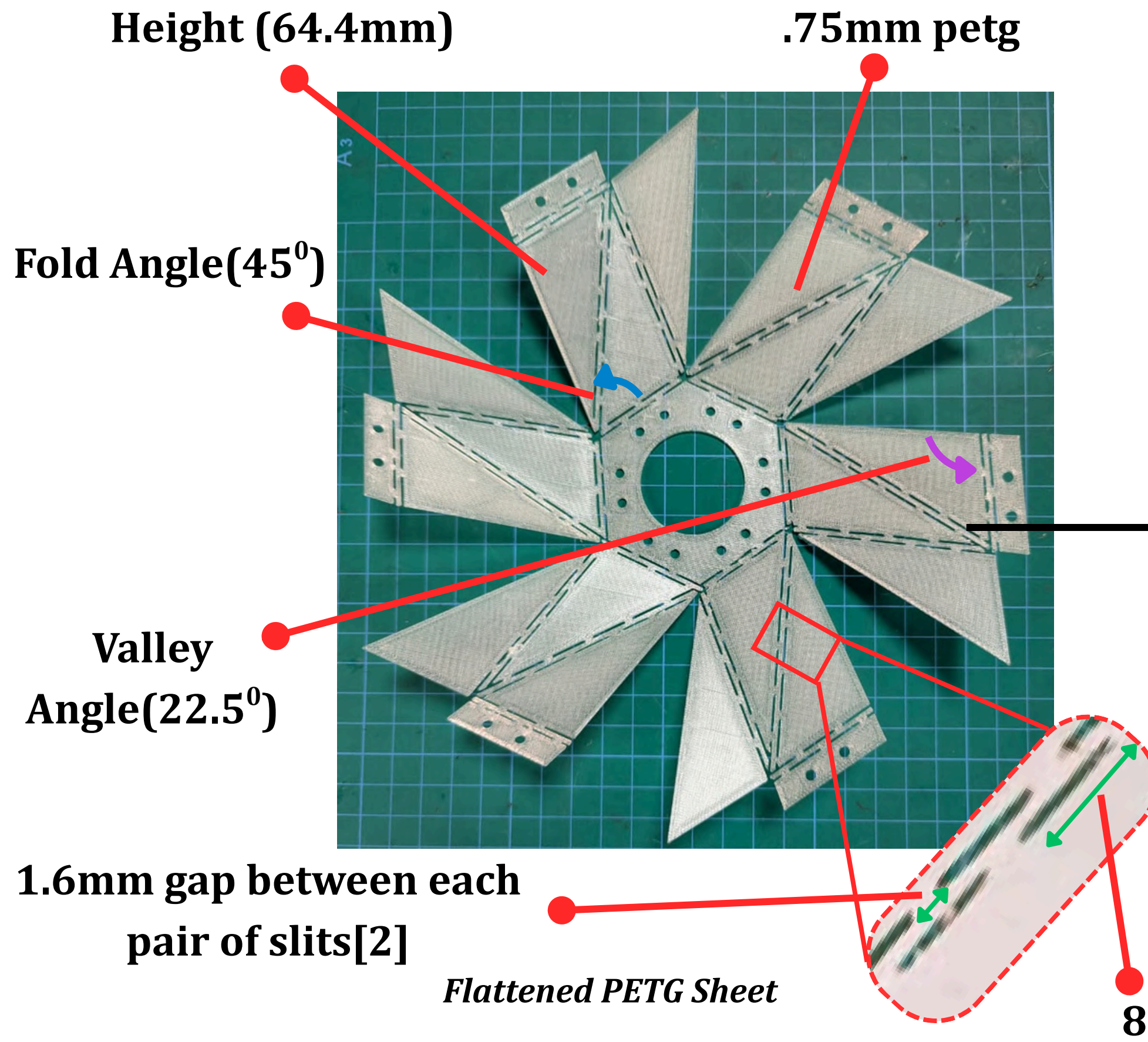
Flattened PLA-TPU Sheet



Folded PLA-TPU Unit Module

Dimension optimization: Kresling Origami Chassis

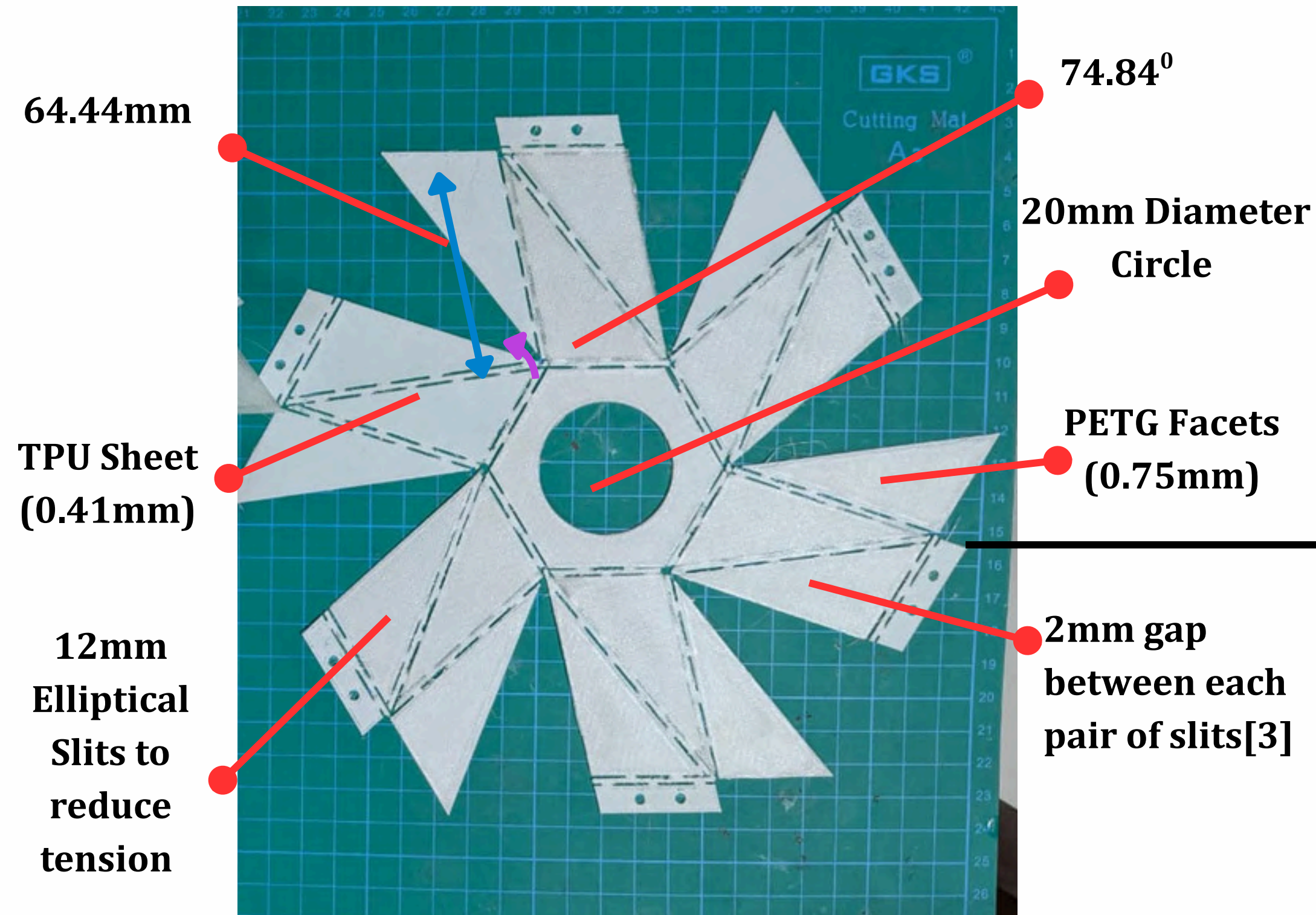
Iteration 2: *PETG Sheets*



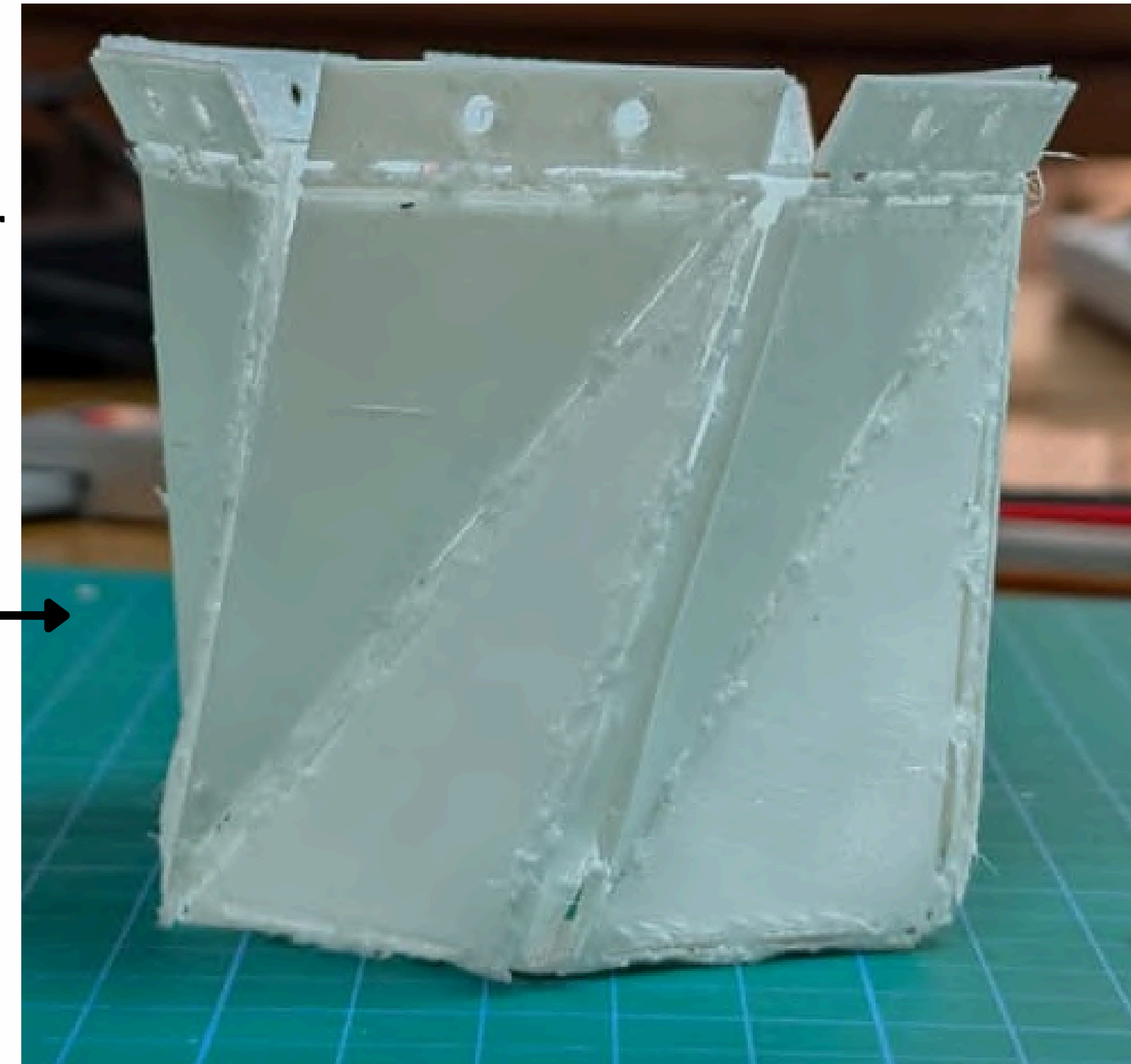
Folded PETG Unit Module

Dimension optimization: Kresling Origami Chassis

Iteration 3: *TPU-PETG Sheets*



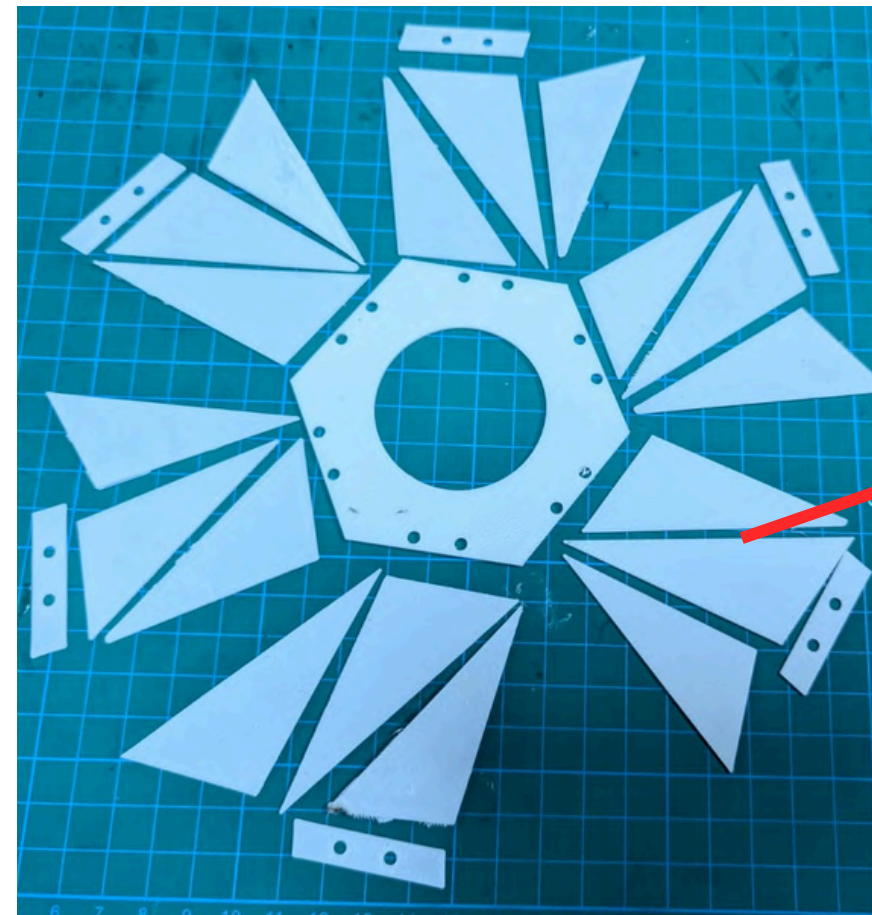
Flattened TPU-PETG Sheet



Folded TPU-PETG Unit Module

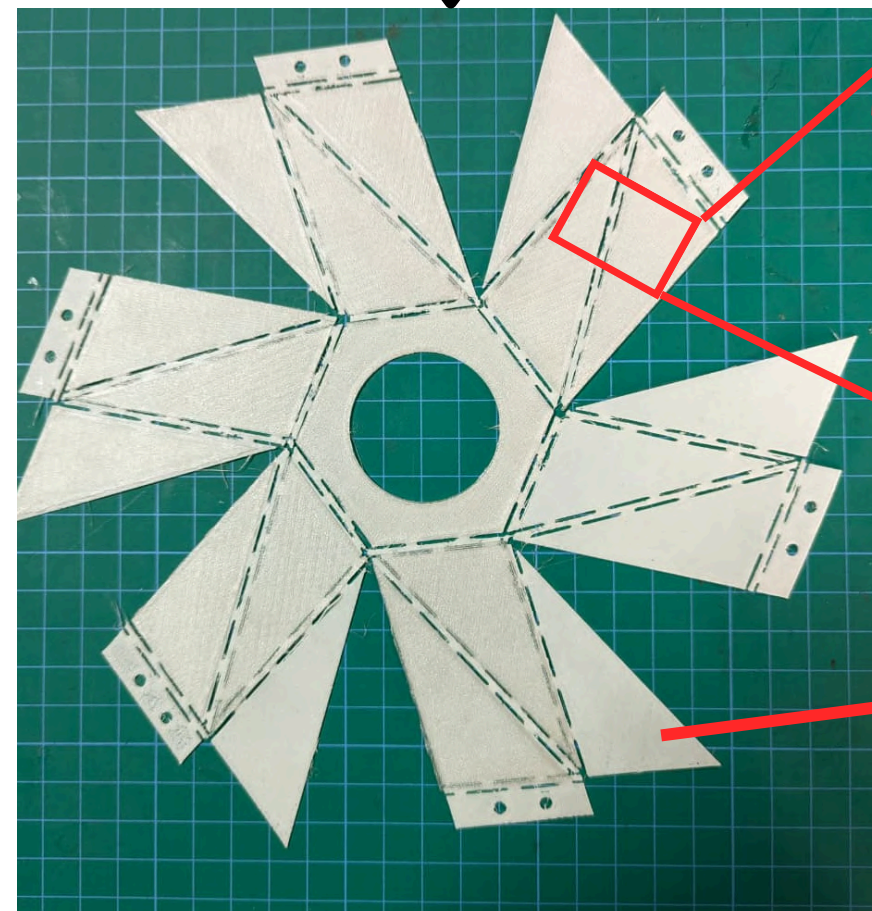
Dimension optimization: Kresling Origami Chassis

Iteration 4: *TPU-PLA Sheets*

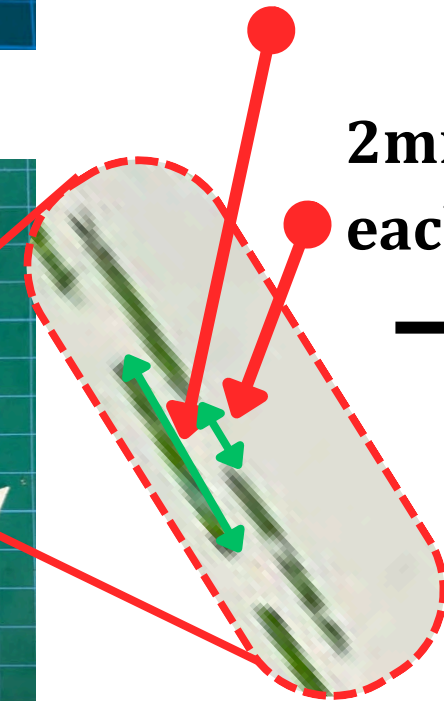


PLA Facets (width
1mm)

12 mm Elliptical Slits to reduce
tension

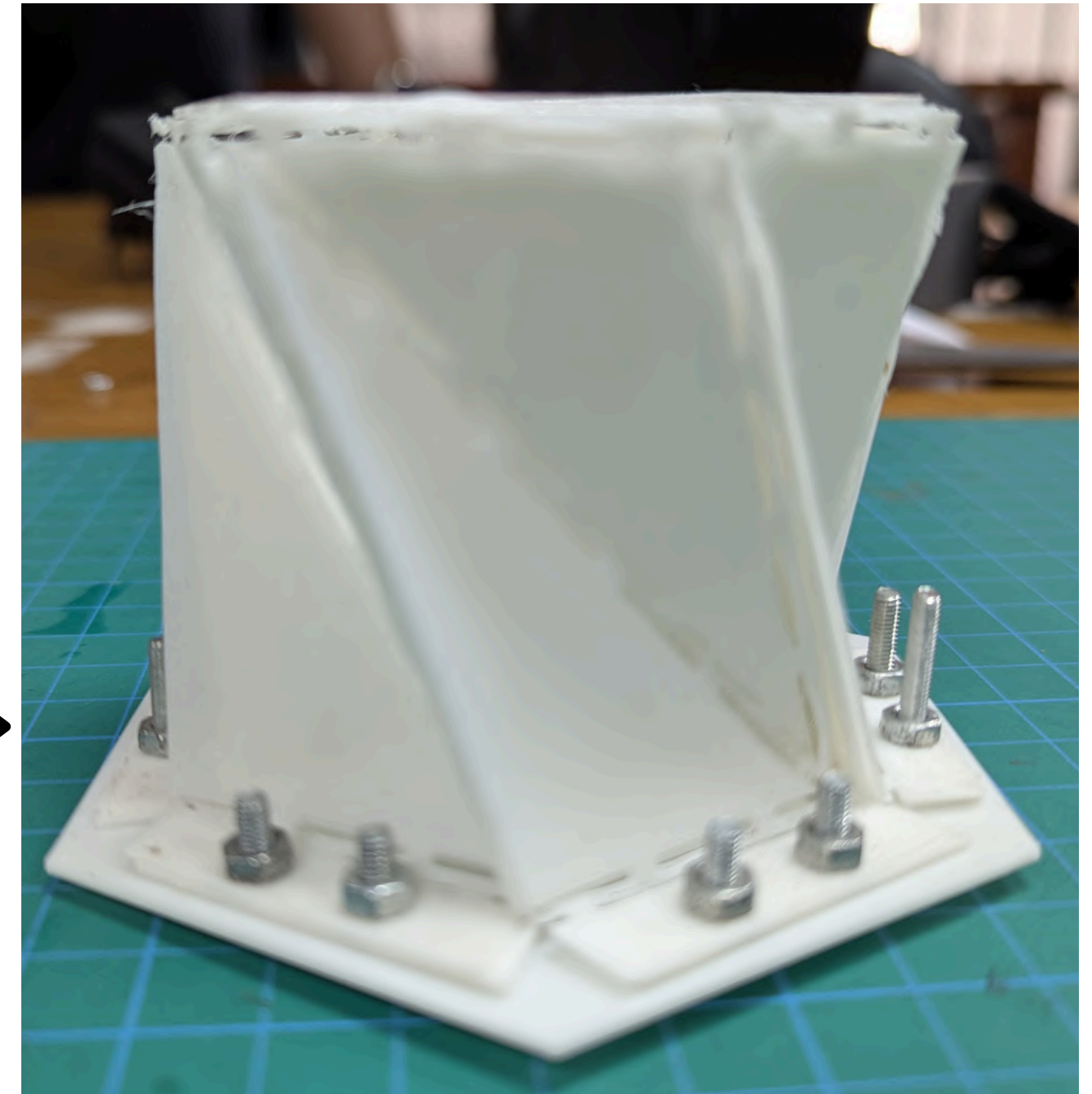


Flattened TPU-PLA Sheet



2mm gap between
each pair of slits

TPU Sheet (width
.41mm)



Folded TPU-PLA Unit Module

How is the module actuated?

Kinematic Modeling via Denavit-Hartenberg (D-H) Convention

$$\mathbf{H}_{\text{Kresling}} = \begin{bmatrix} \cos \varphi & -\sin \varphi & 0 & 0 \\ \sin \varphi & \cos \varphi & 0 & 0 \\ 0 & 0 & 1 & H \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{T}_0^n = \prod_{i=1}^n \mathbf{H}_i^{i-1}, \quad \mathbf{H}_i^{i-1} = \begin{bmatrix} \mathbf{R}_i & \mathbf{p}_i \\ \mathbf{0}^\top & 1 \end{bmatrix}$$

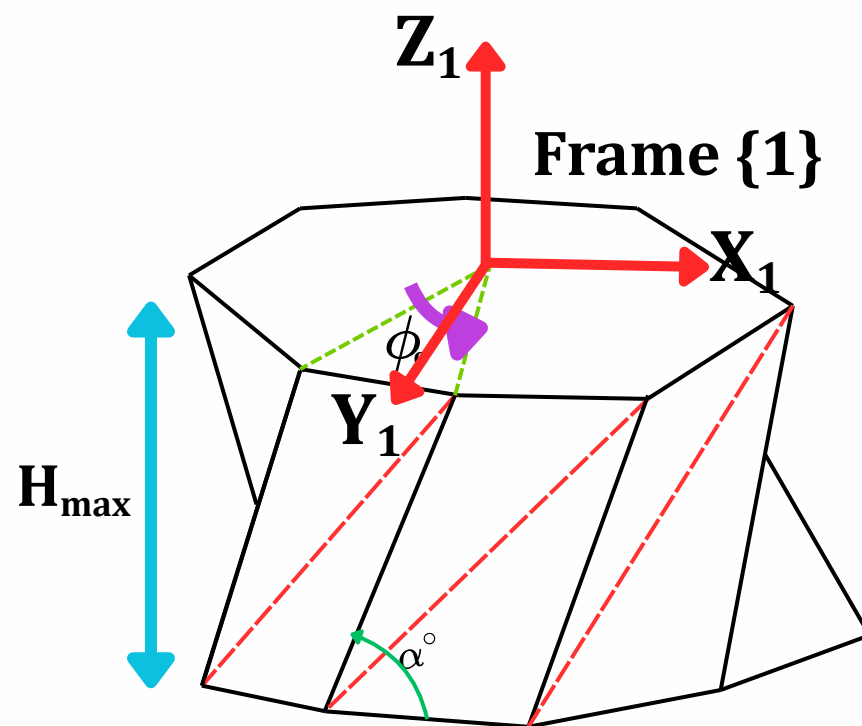
$$\mathbf{H}_{(1)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & H_{\max} \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad H_{\max} = 63.1 \text{ mm}$$

Two states

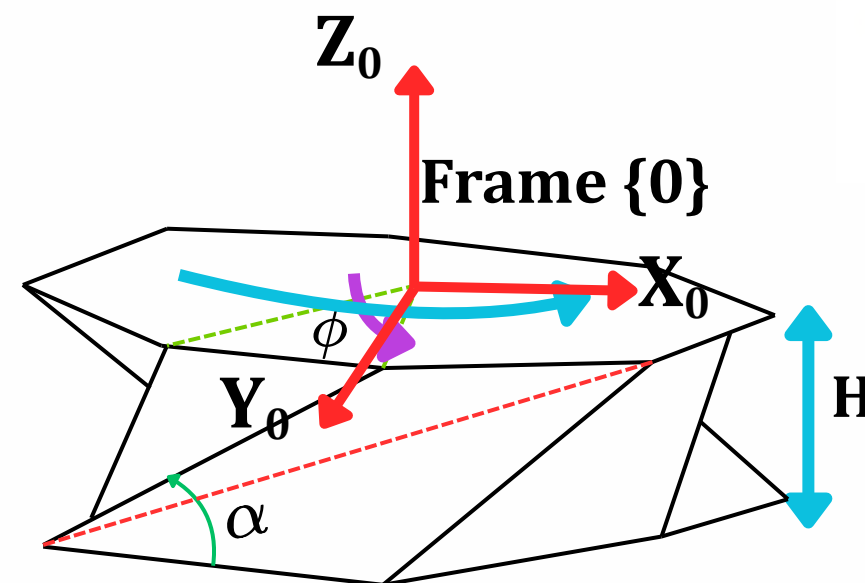
State 1 — Extended (Stiff Link).

State 0 — Contracted (Screw Joint).

$$\mathbf{H}_{(0)} = \begin{bmatrix} \cos \varphi & -\sin \varphi & 0 & 0 \\ \sin \varphi & \cos \varphi & 0 & 0 \\ 0 & 0 & 1 & H(t) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



State 1: Rigid Pure Translation



State 0: Screw Transformation

How is the module assembled?

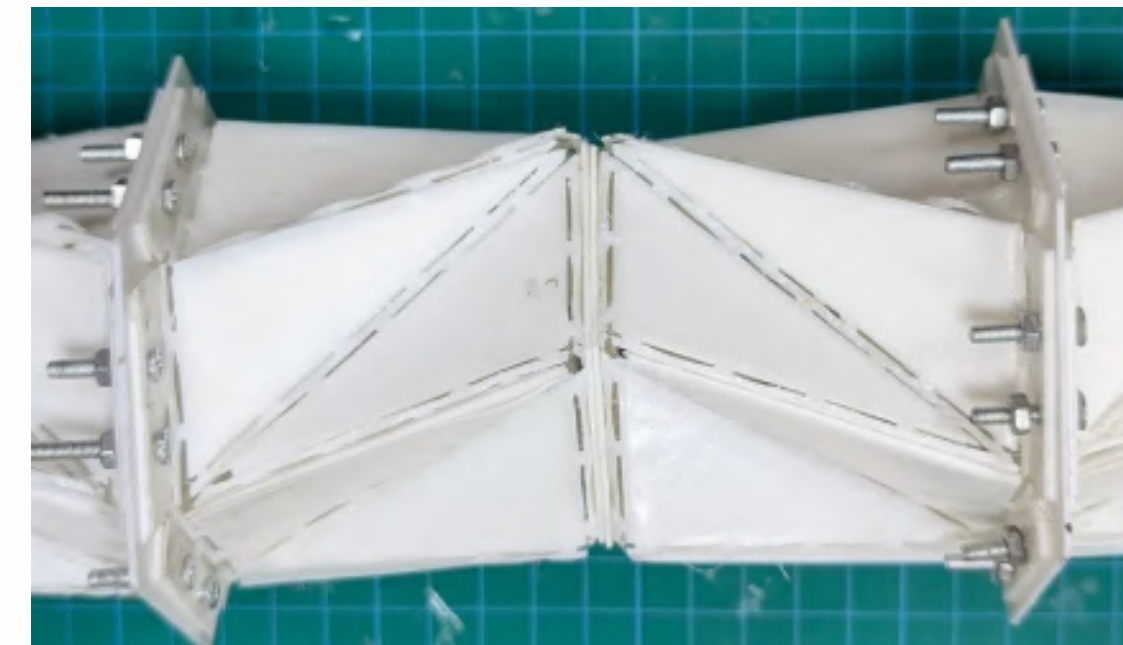
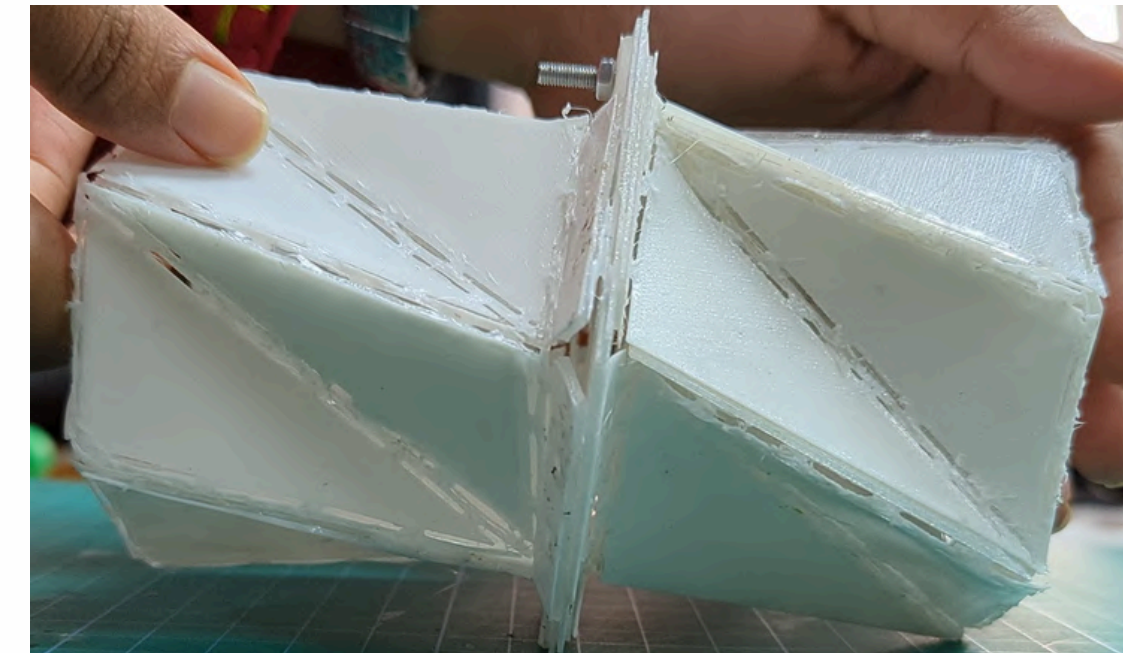
Twist Cancellation: The Kresling Dipole

$$\mathbf{T}_{\text{dipole}} = \mathbf{H}_{(0)}^{\text{CW}}(+\varphi) \cdot \mathbf{H}_{(0)}^{\text{CCW}}(-\varphi) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2H(t) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

D-H Parameter Table [Odd links: Revolute; Even links: Prismatic. $a_i = \alpha_i = 0$ for all links[5]]

$$\mathbf{T}_{\text{total}} = \mathbf{T}_{\text{dip1}} \cdot \mathbf{T}_{\text{dip2}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 4H(t) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Link No.	Unit	d_i (mm)	θ_i
1	U1 (CW)	0	$+\varphi(t)$, $[0^\circ \rightarrow 50^\circ]$
2	U1 (CW)	$H(t)$	0
3	U2 (CCW)	0	$-\varphi(t)$, $[0^\circ \rightarrow -50^\circ]$
4	U2 (CCW)	$H(t)$	0



ϕ_{CW} Φ_{CW} α

clockwise rotation

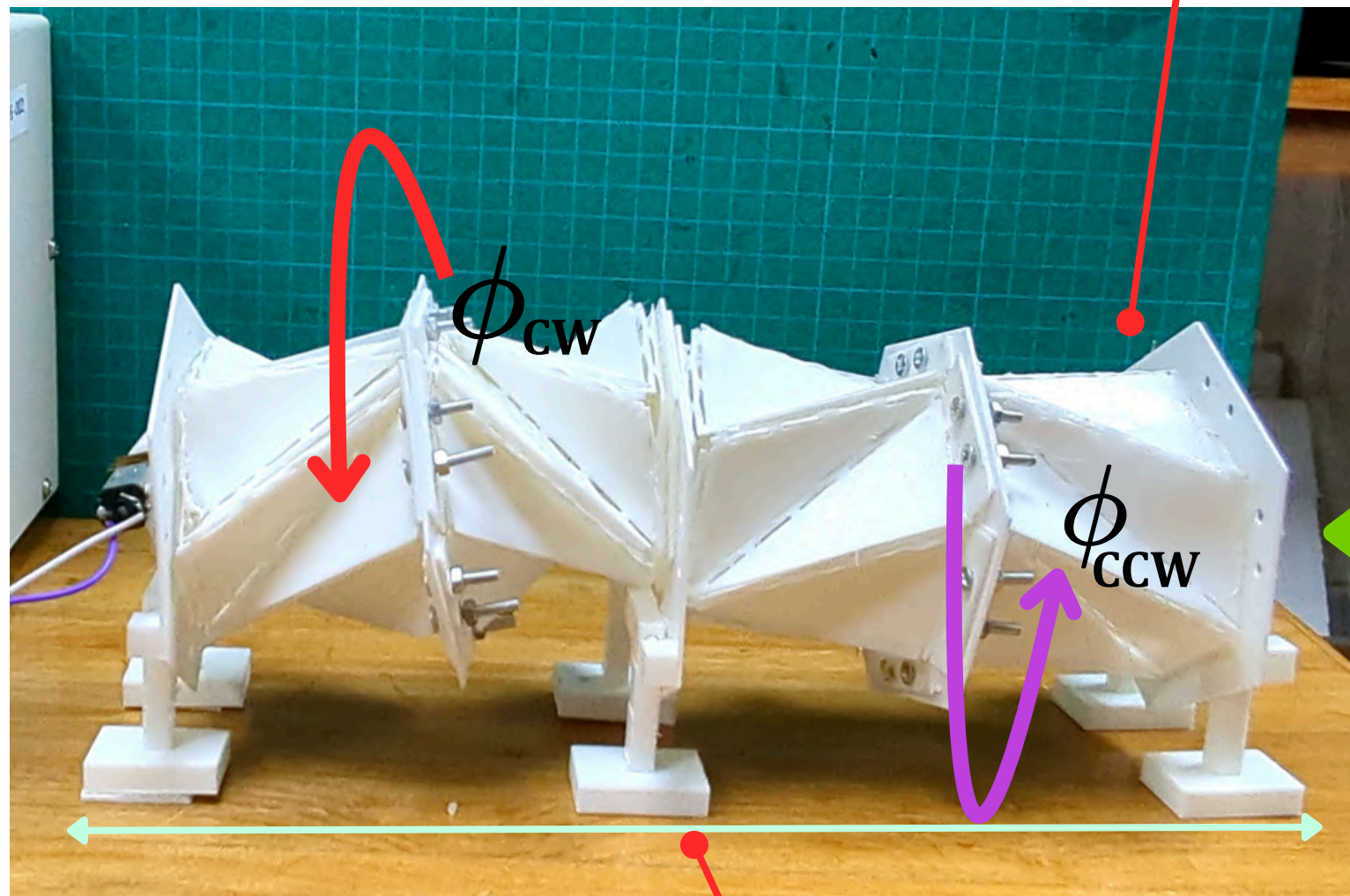
ϕ_{cw}

ϕ_{cw}

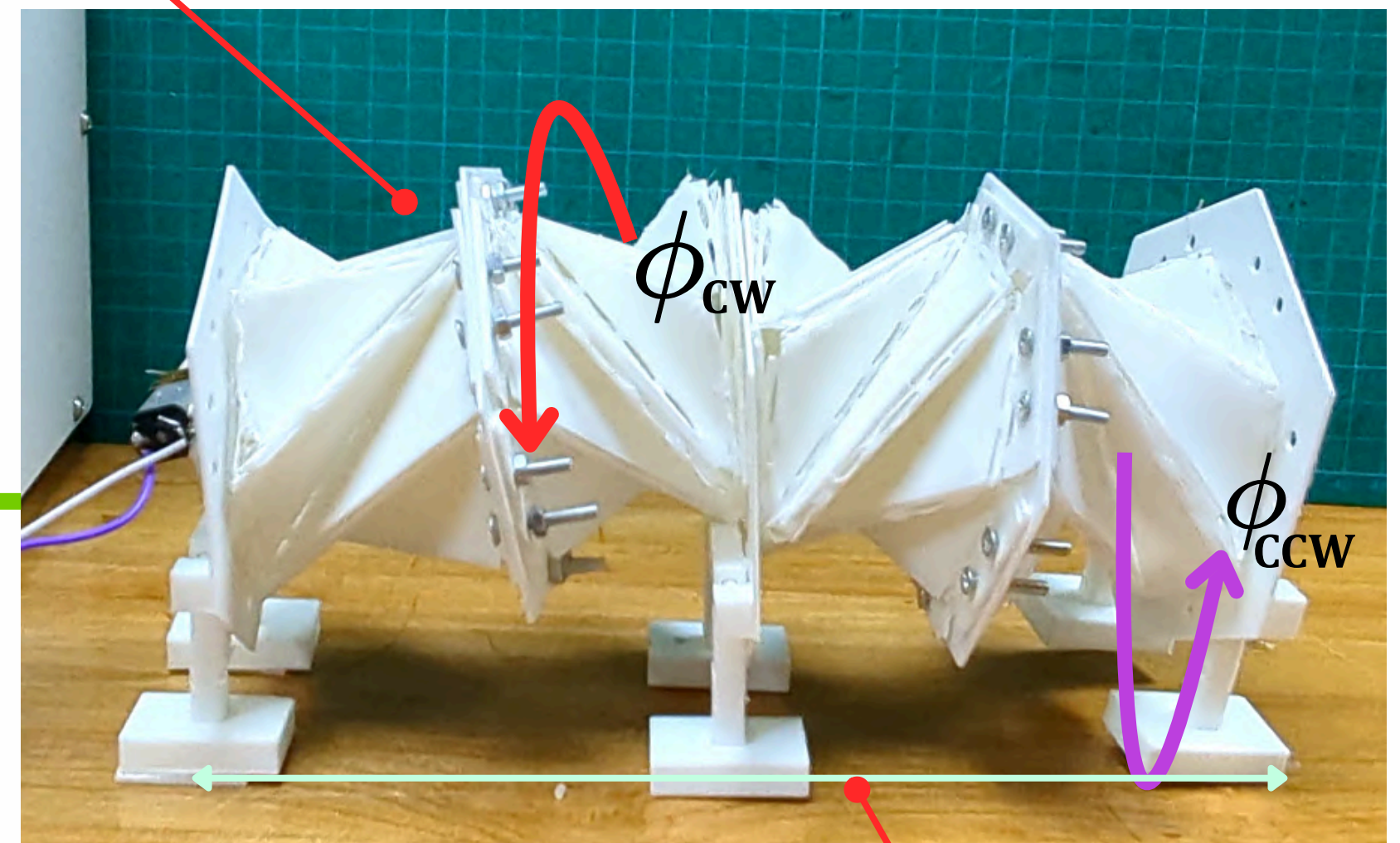
α


 ϕ

Symmetric Twist (Twist Cancellation)



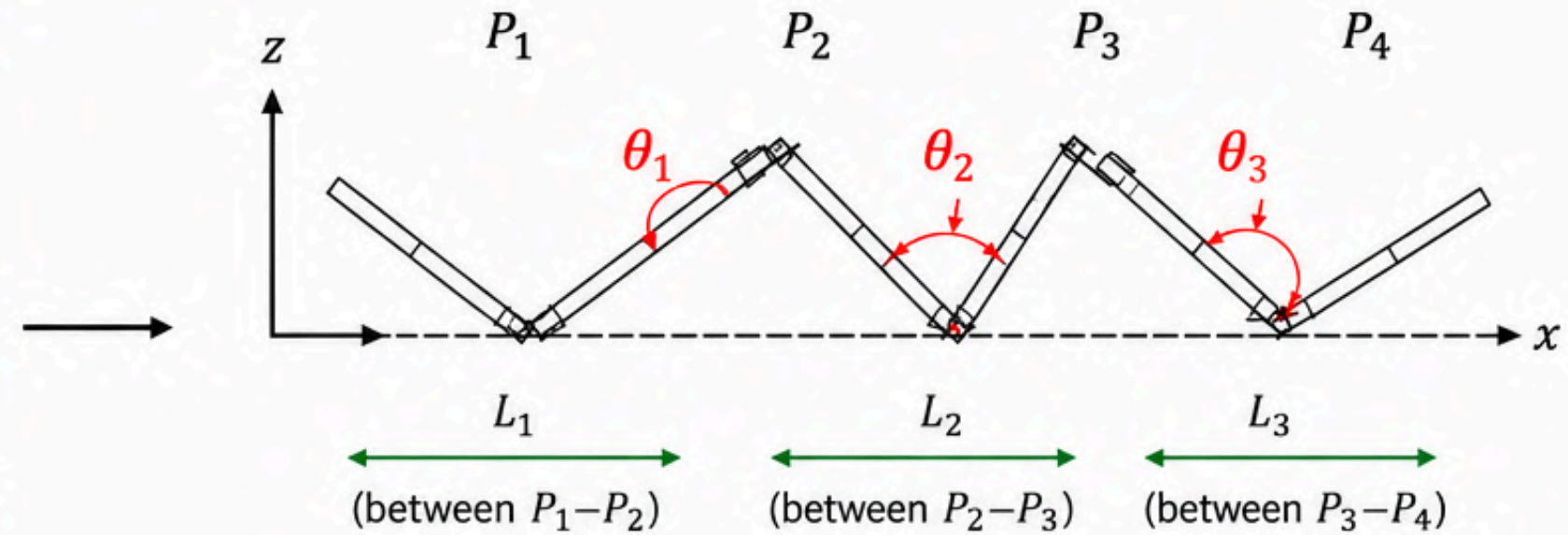
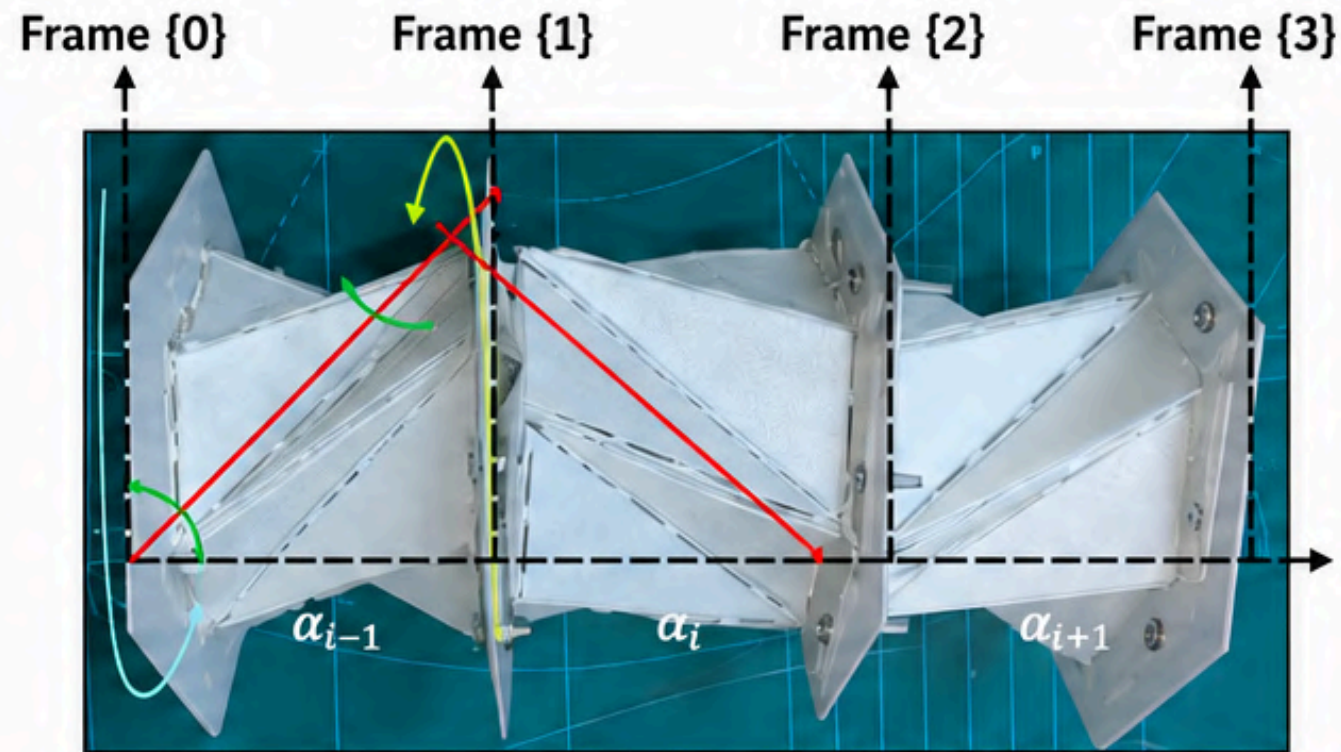
$H_0 = 300\text{mm}$



$H = 221\text{mm}$

Pure Vertical Translation, Zero Net Twist $\Delta\phi = \phi_{cw} - \phi_{ccw} = 0^\circ$

How is the assembled chassis actuated?



Valley angles: $\theta_1, \theta_2, \theta_3$

Plate rotations (about local z at hinges): ϕ_1, ϕ_2, ϕ_3

D-H PARAMETERS (initial)

Joint i	α_{i-1} (rad)	a_{i-1} (m)	d_i (m)	θ_i (rad)
1	0	L_1	0	θ_1
2	0	L_2	0	θ_2
3	0	L_3	0	θ_3

Assuming the length to be the only variable

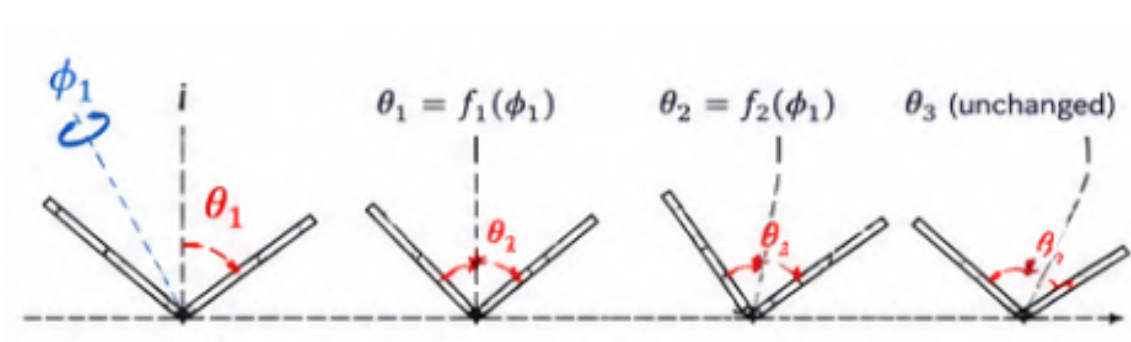
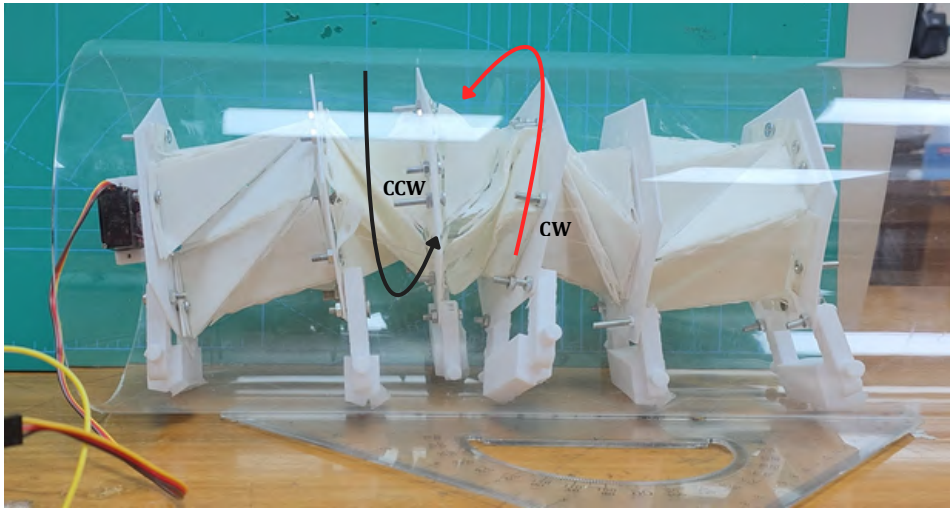
Initial Homogeneous Transformation

$${}^0T_1 = \begin{bmatrix} c_1 & -s_1 & 0 & L_1 c_1 \\ s_1 & c_1 & 0 & L_1 s_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^1T_2 = \begin{bmatrix} c_2 & -s_2 & 0 & L_2 c_2 \\ s_2 & c_2 & 0 & L_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2T_3 = \begin{bmatrix} c_3 & -s_3 & 0 & L_3 c_3 \\ s_3 & c_3 & 0 & L_3 s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

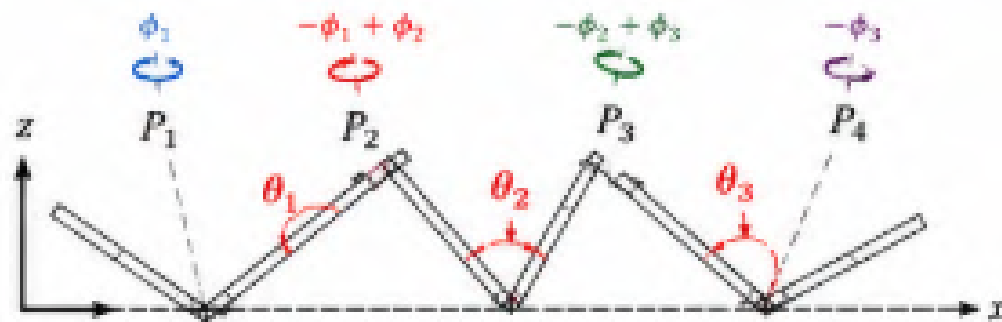
$$c_i = \cos \theta_i, \quad s_i = \sin \theta_i$$

Changes due to valley angle due to plate rotation



Transformation due to rotation of ϕ

*



Rotation due to plate rotation ϕ

D-H PARAMETERS (after ϕ_1)				
Joint i	α_{i-1} (rad)	a_{i-1} (m)	d_i (m)	θ_i (rad)
1	0	L_1	0	$\theta_1(\phi_1)$
2	0	L_2	0	$\theta_2(\phi_1)$
3	0	L_3	0	θ_3

θ_1 and θ_2 change, and the distance is +x

$${}^0T_1(\phi_1) = \begin{bmatrix} c_1 & -s_1 & 0 & L_1 c_1 \\ s_1 & c_1 & 0 & L_1 s_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^1T_2(\phi_1) = \begin{bmatrix} c_2 & -s_2 & 0 & L_2 c_2 \\ s_2 & c_2 & 0 & L_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2T_3(\phi_1) = \begin{bmatrix} c_3 & -s_3 & 0 & L_3 c_3 \\ s_3 & c_3 & 0 & L_3 s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_z(\phi_1) = \begin{bmatrix} c_{\phi_1} & -s_{\phi_1} & 0 & 0 \\ s_{\phi_1} & c_{\phi_1} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad R_z(-\phi_1 + \phi_2) = \begin{bmatrix} c_{-\phi_1 + \phi_2} & -s_{-\phi_1 + \phi_2} & 0 & 0 \\ s_{-\phi_1 + \phi_2} & c_{-\phi_1 + \phi_2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_z(-\phi_2 + \phi_3) = \begin{bmatrix} c_{-\phi_2 + \phi_3} & -s_{-\phi_2 + \phi_3} & 0 & 0 \\ s_{-\phi_2 + \phi_3} & c_{-\phi_2 + \phi_3} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad R_z(-\phi_3) = \begin{bmatrix} c_{-\phi_3} & -s_{-\phi_3} & 0 & 0 \\ s_{-\phi_3} & c_{-\phi_3} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

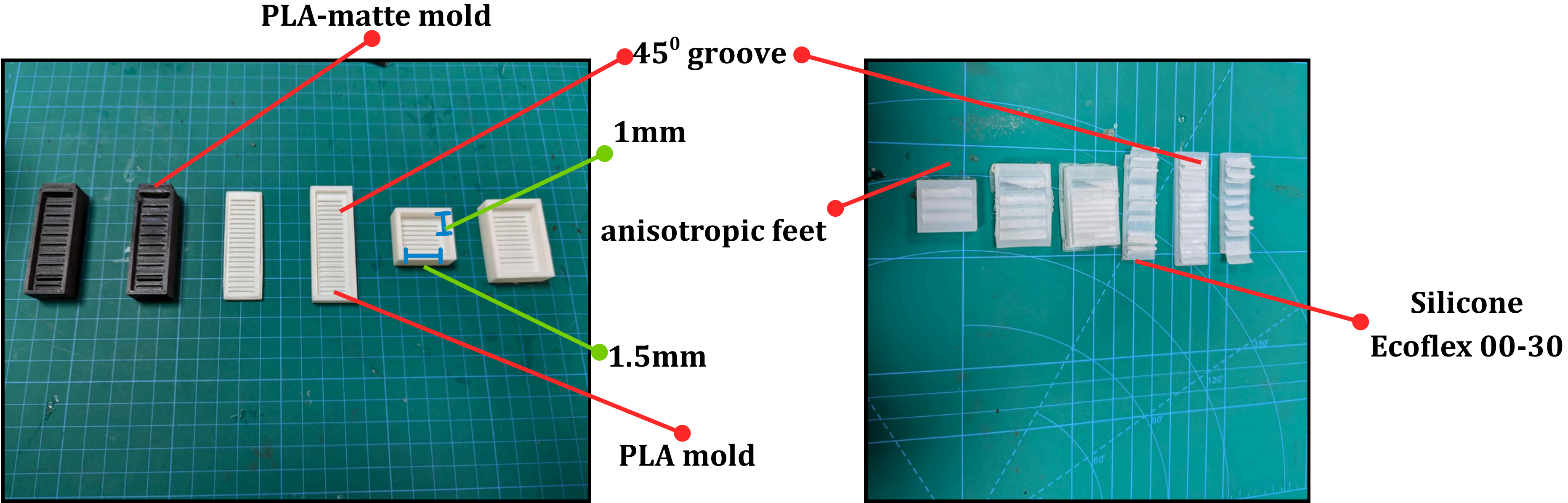
$c_x = \cos x, \quad s_x = \sin x$

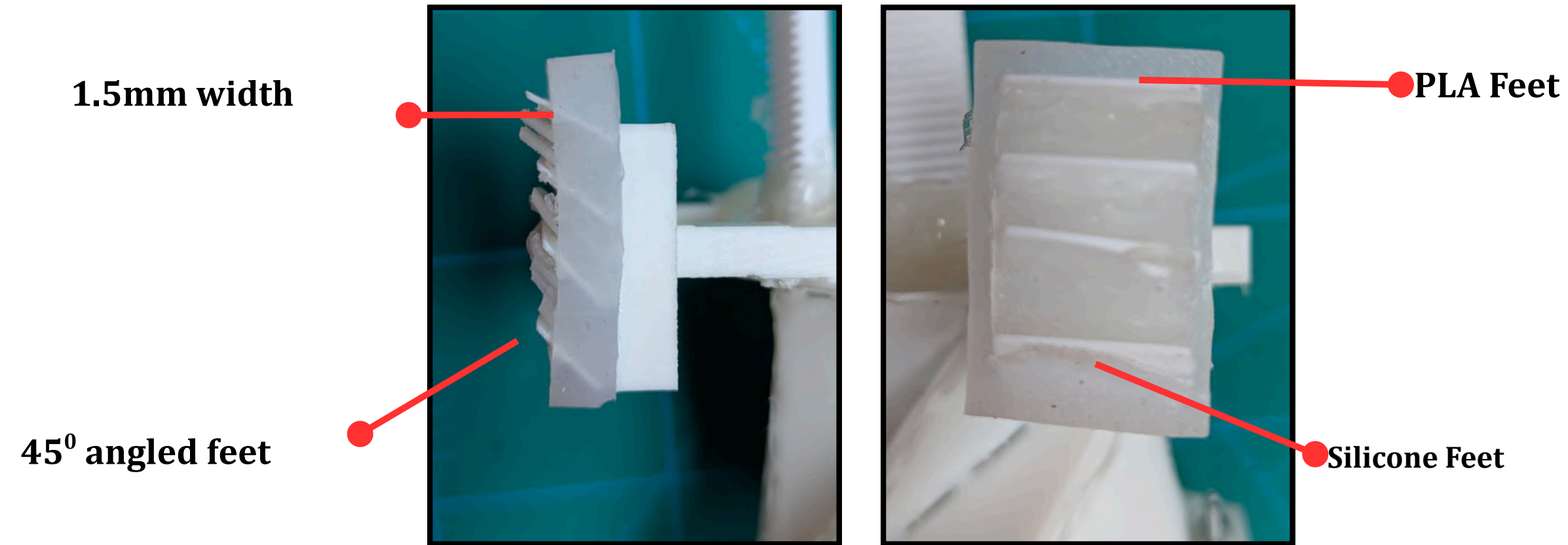
Total Transformation

$${}^0T_3^{total} = \underbrace{(R_z(\phi_1) \quad {}^0T_1)}_{\text{Plate } P_1} \underbrace{(R_z(-\phi_1 + \phi_2) \quad {}^1T_2)}_{\text{Plate } P_2} \underbrace{(R_z(-\phi_2 + \phi_3) \quad {}^2T_3)}_{\text{Plate } P_3} \underbrace{R_z(-\phi_3)}_{\text{Plate } P_4}$$

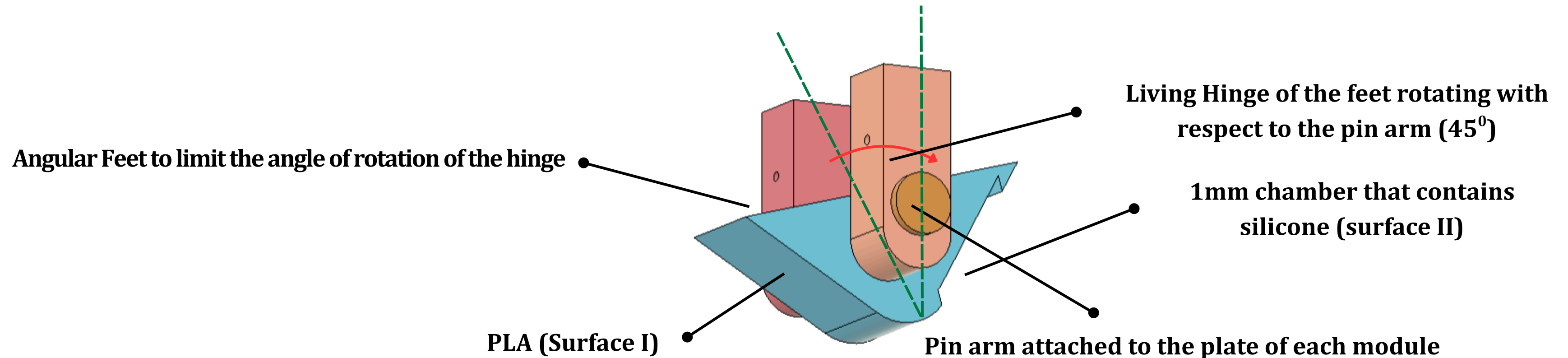
Optimized feet for aiding locomotion: Iterations

Iteration 1: PLA-Silicone alternataively placed with equal spacing



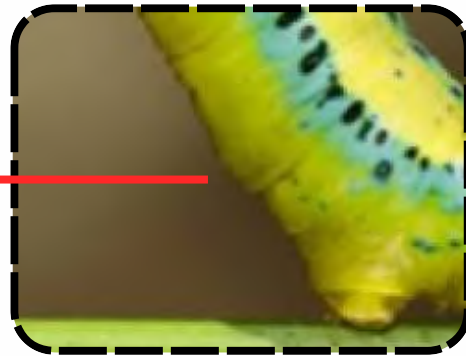


Iteration 2: PLA-Silicone in alternative contact area with a living hinge



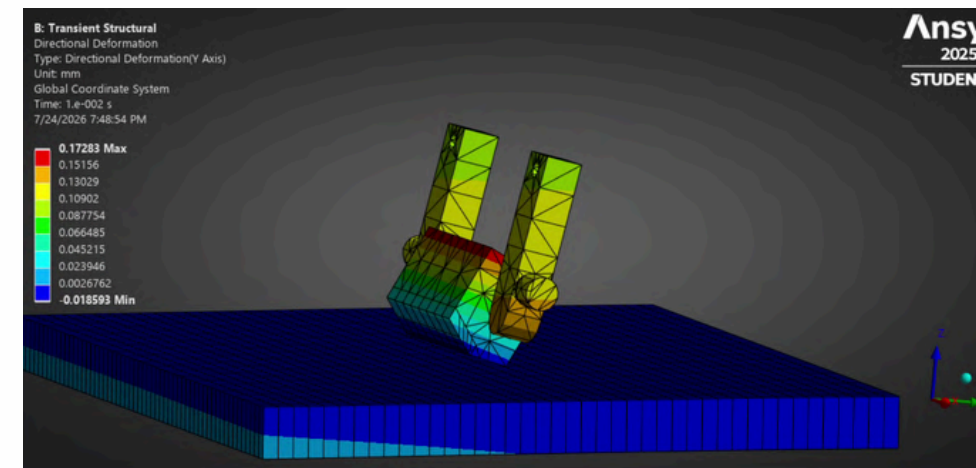
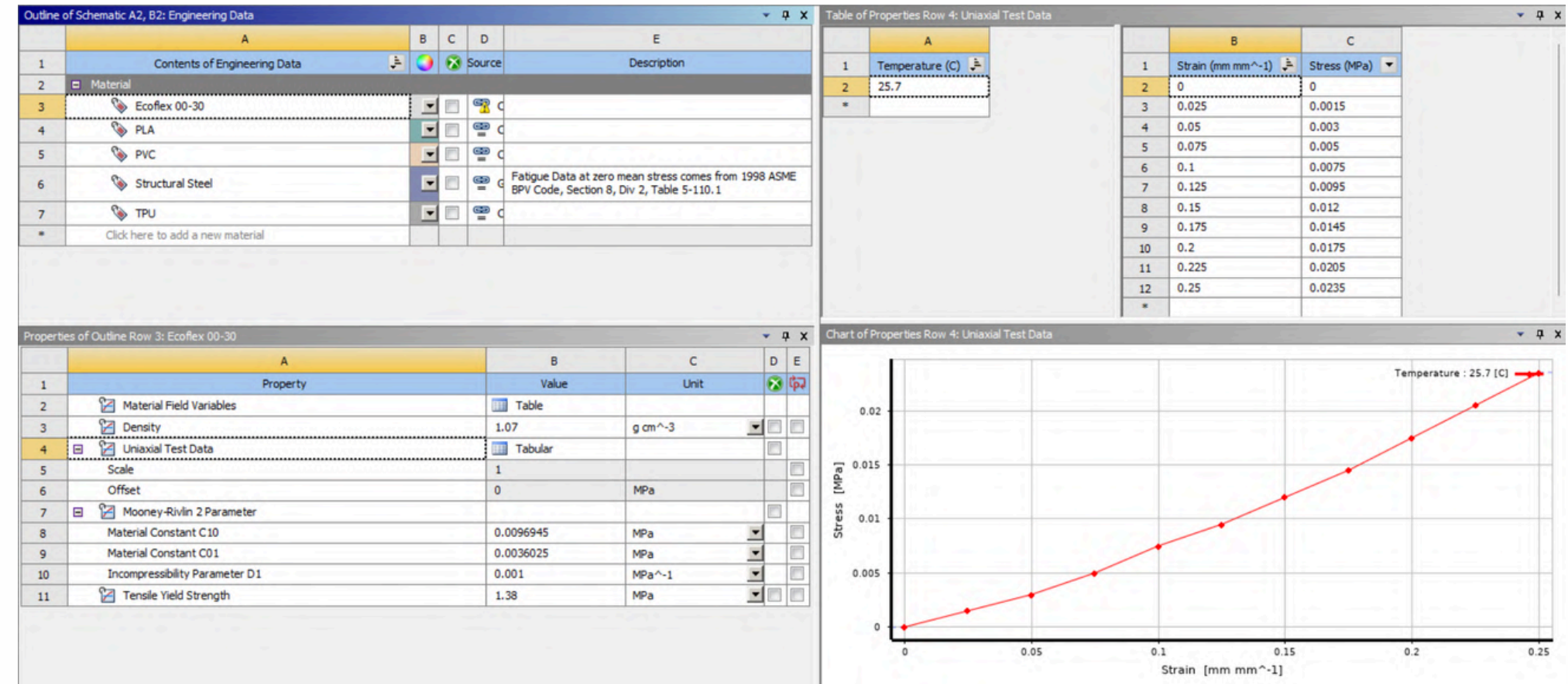
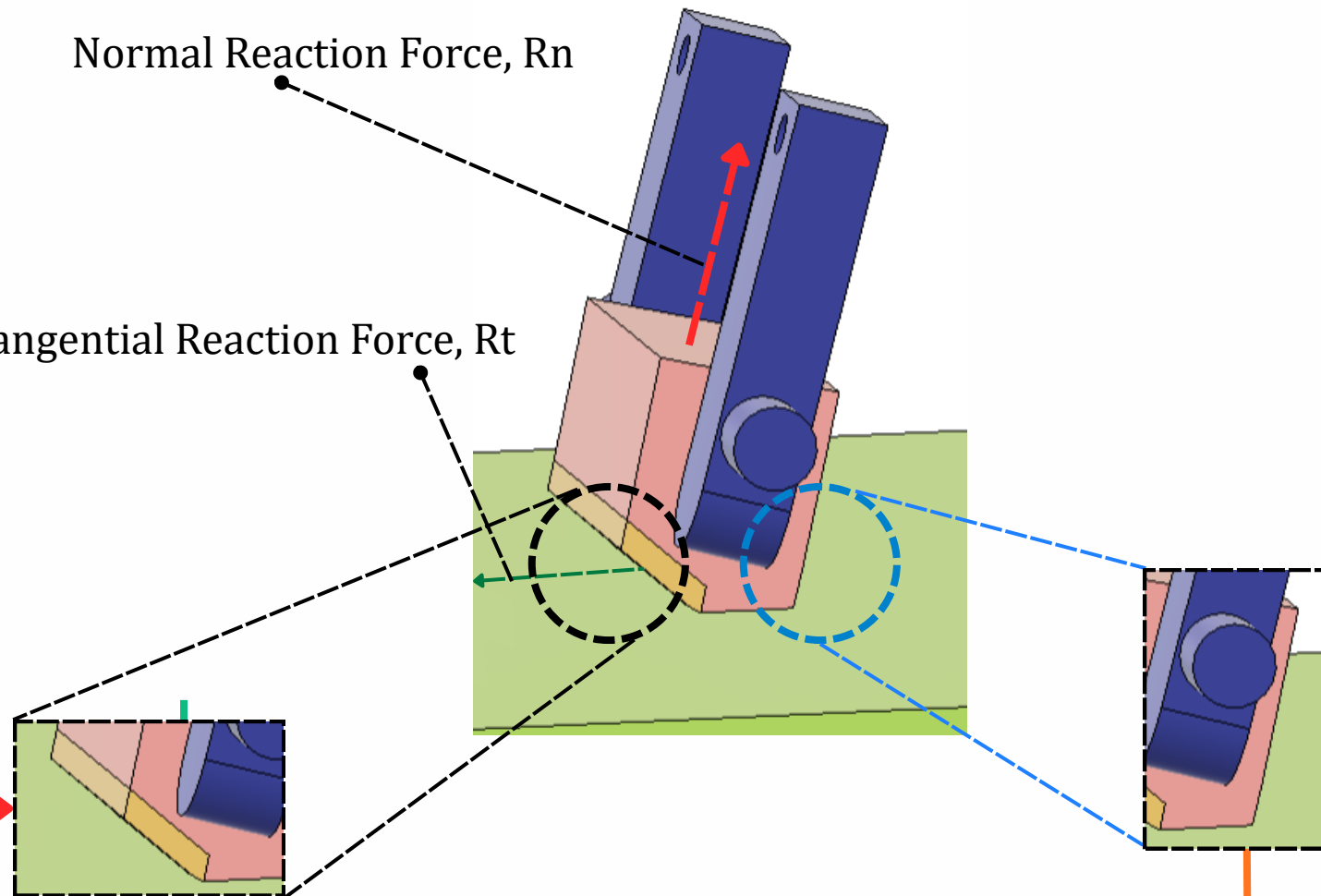
How is the characteristics of an inchworm setae implemented for this robot?

Setae for anchoring

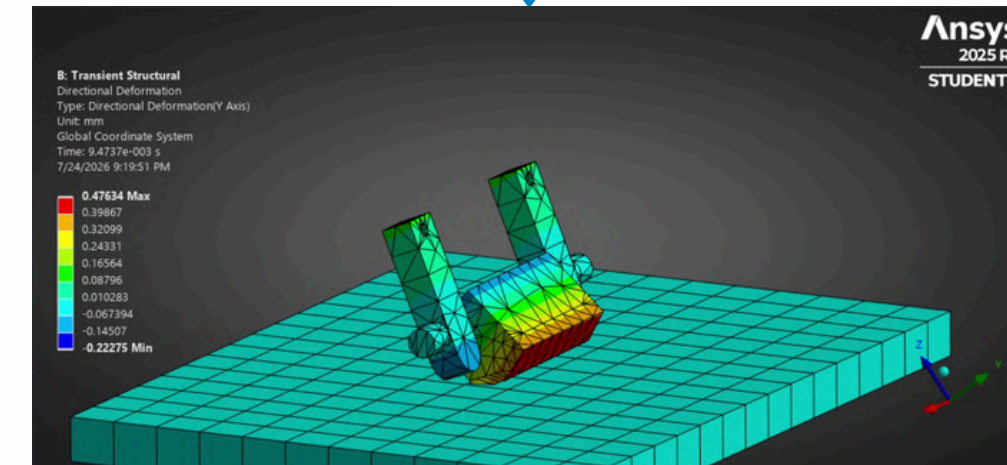


Normal Reaction Force, R_n

Tangential Reaction Force, R_t

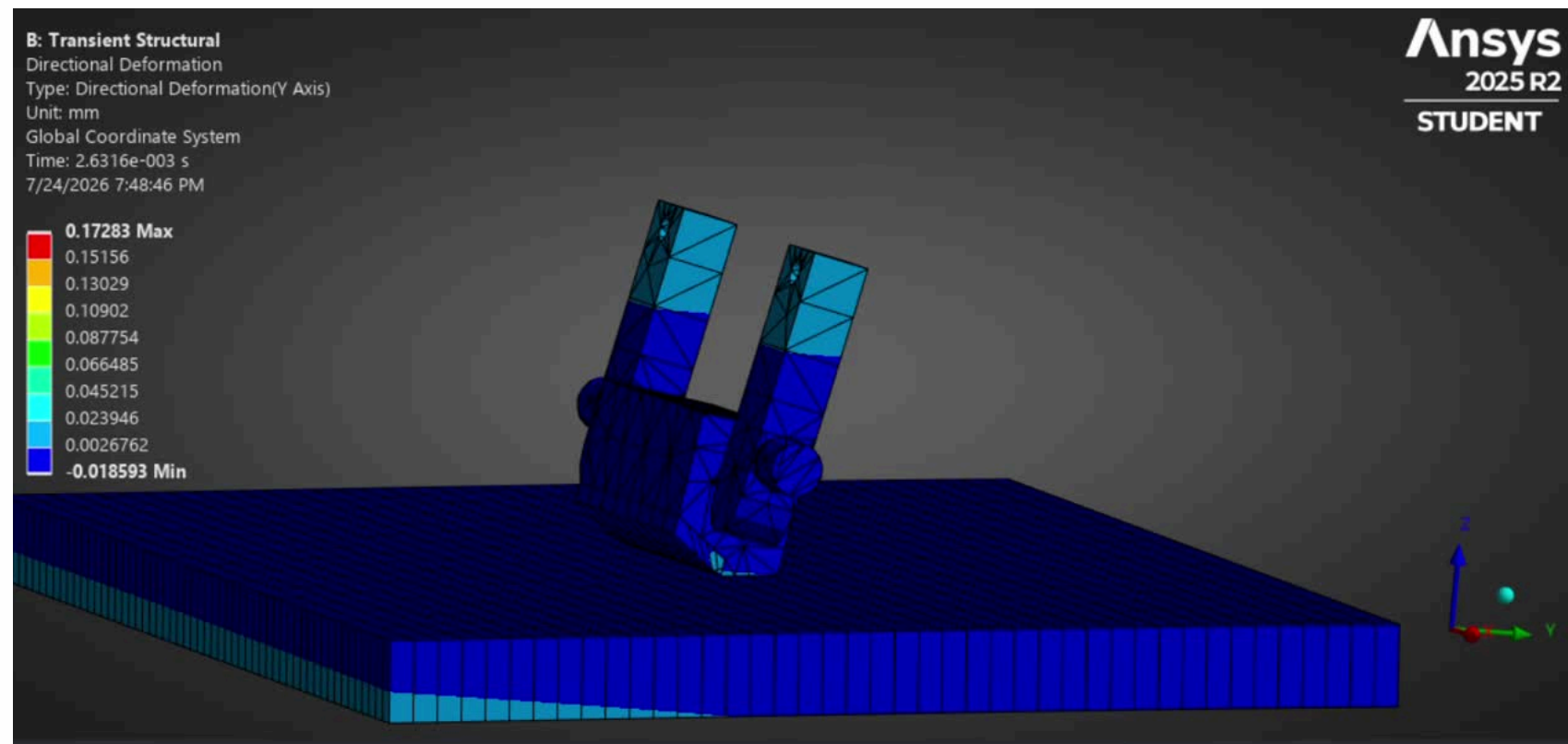


Silicone-Acrylic surface friction analysis

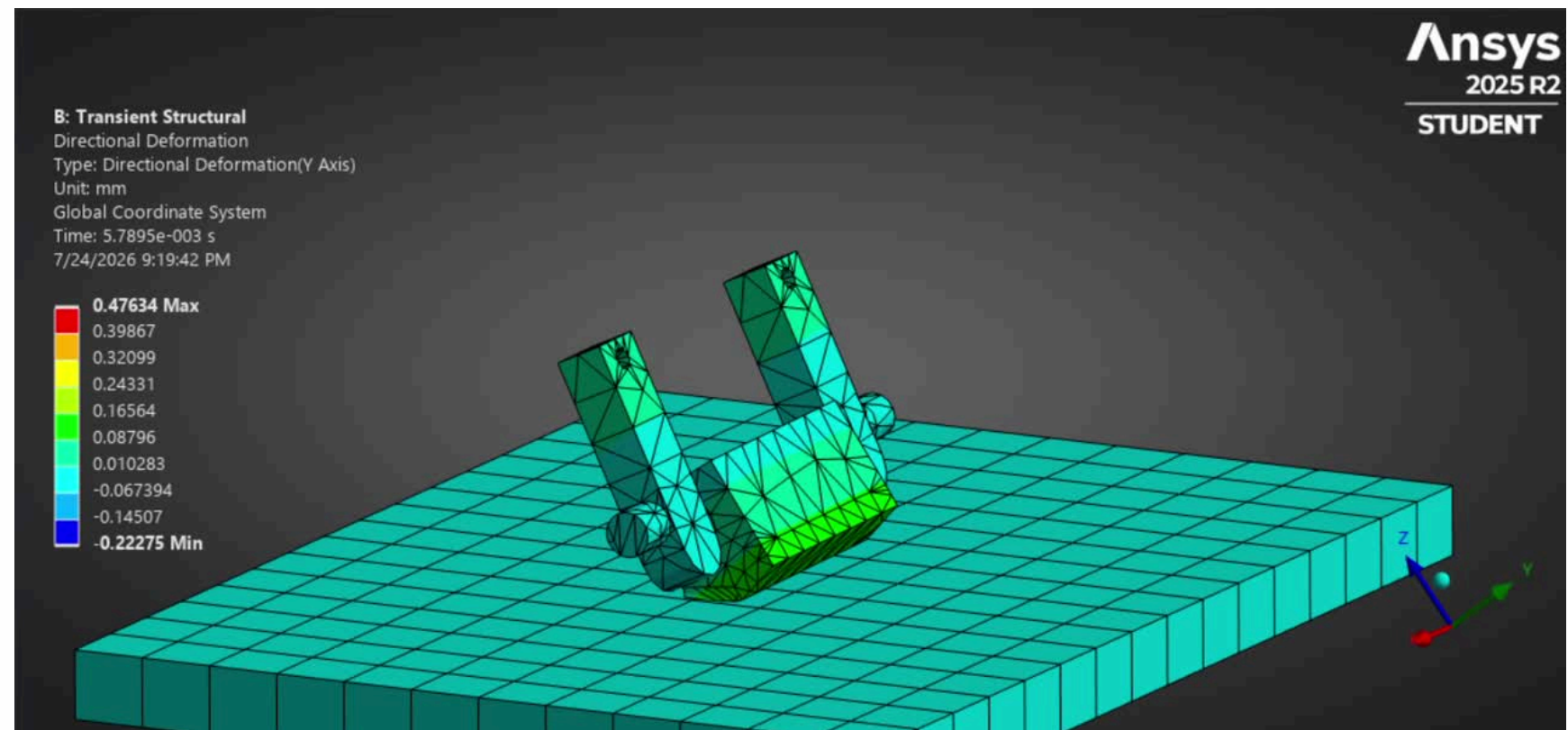


PLA-acrylic surface friction analysis

Passive Anchoring Mechanism:Anisotropic Friction Validation



PLA-acrylic surface friction analysis



Silicone-Acrylic surface friction analysis

Parameter	Silicone-Acrylic	PLA-Acrylic
Static Friction coeffcients	0.6	0.1
Kinemtic friction coefficients	0.1061	0.02356
Mobilization ratio	17.70%	23.60%
Sliding distance (mm)	5.998 E-03	5.606 E-03
Frictional work (N·mm)	0.002218	0.0006492

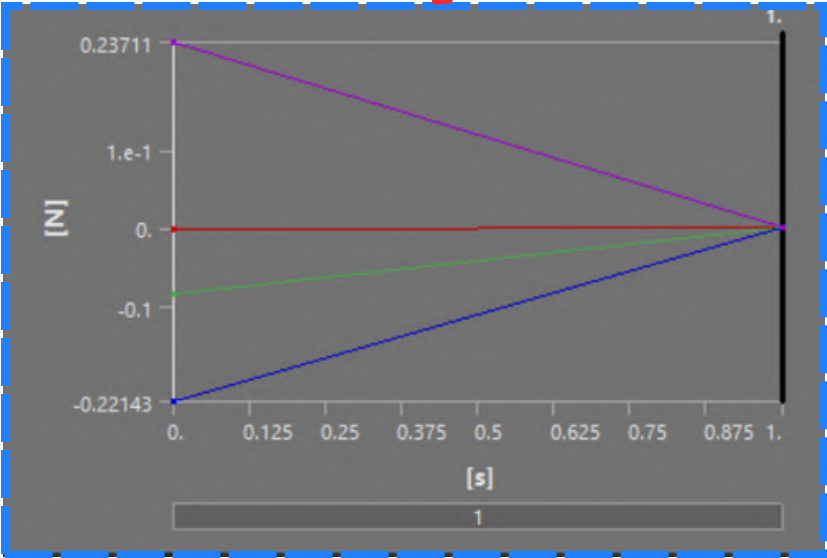
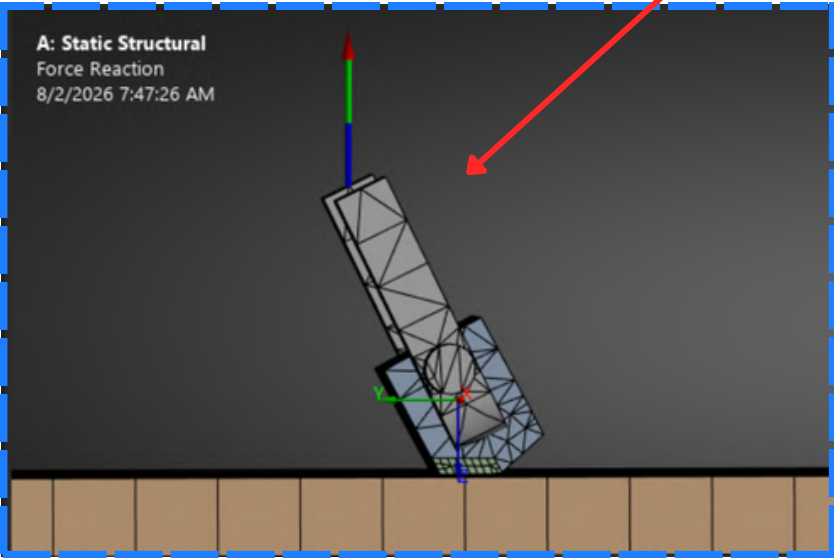
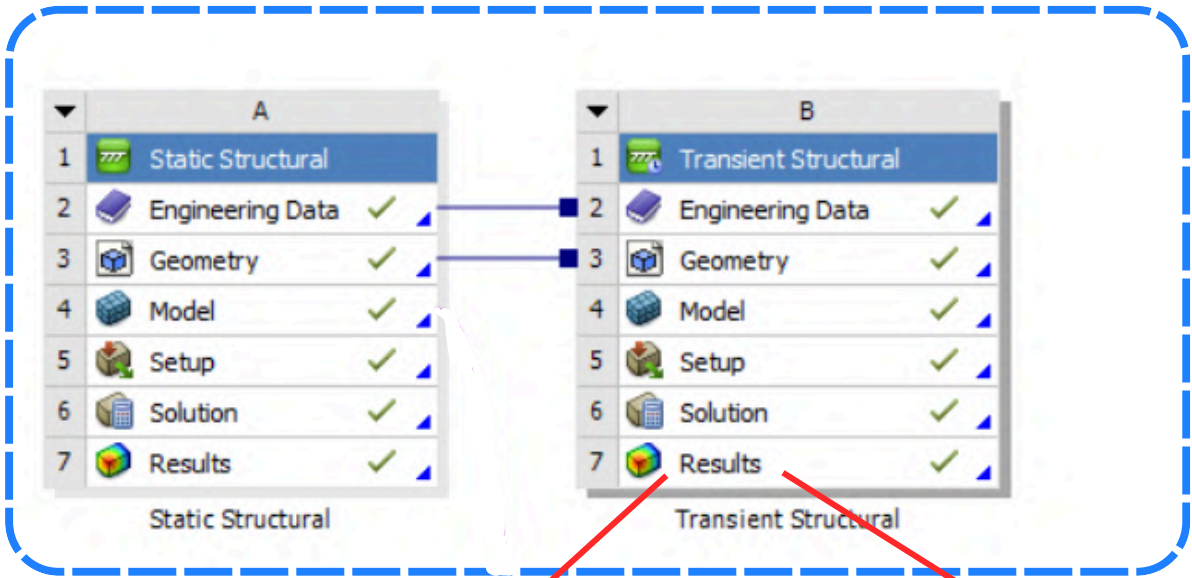
$$W_{\text{silicone-acrylic}} = 3.42 * W_{\text{PLA-acrylic}}$$

Stopper Angle Optimization and Stability Window

$\theta_c = (35^\circ)(\pm 5^\circ)$
Optimal Operating Stability Window

$\eta = 7.97$
Peak Directional Traction Ratio

$8.72\times$ Margin
Anchoring Safety Factor (SF_anchor)

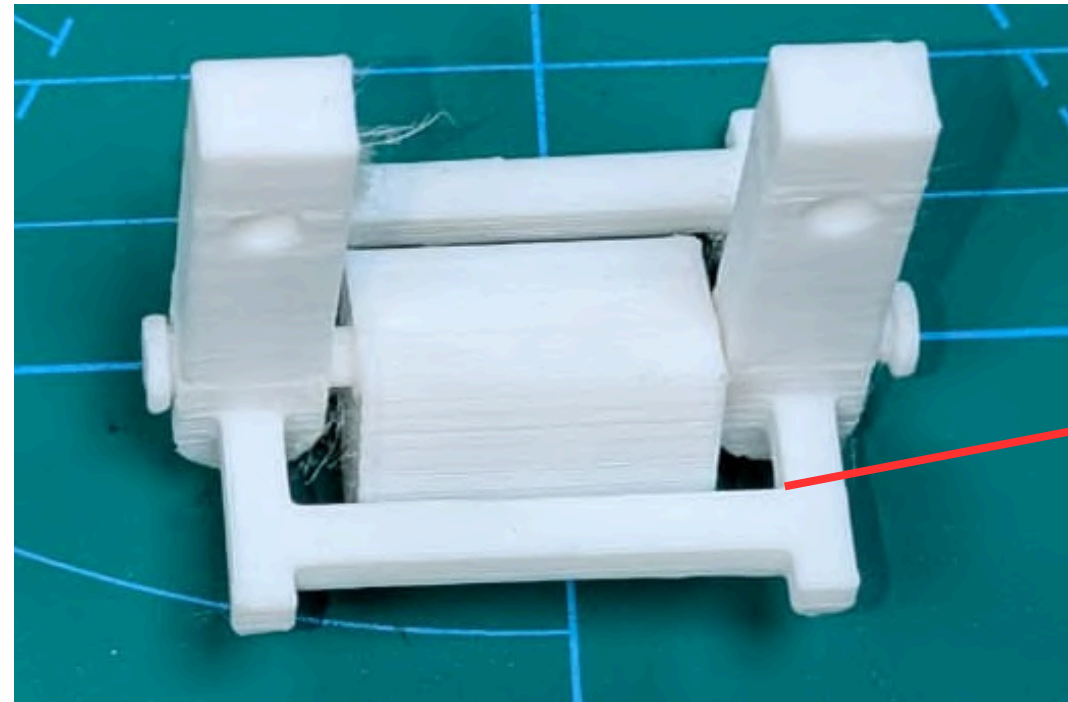


Stability Envelope: Within $[30^\circ, 40^\circ]$, anisotropic feet anchor reliably without **tipping** ($\theta_c \leq 15^\circ$) or local **deformation slip** ($\theta_c \geq 45^\circ$).

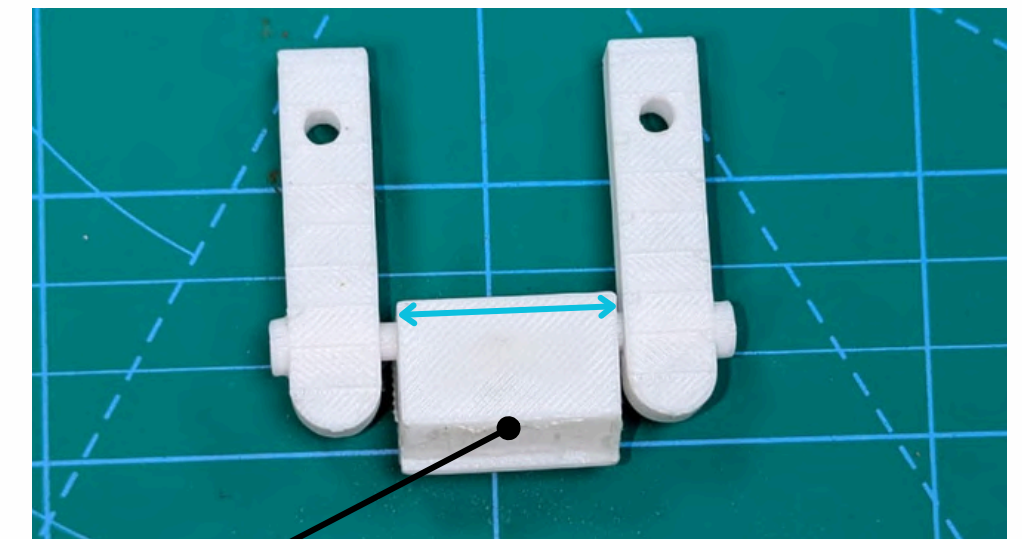
Parametric Evaluation ($d_s = 3.0\text{ mm}$)

Contact Angle θ_c ($^\circ$)	Glide Friction $F_{f,glide}$ (N)	Anchor Friction $F_{f,anchor}$ (N)	Traction Ratio η	Stability Status
15	Unstable	Unstable	Unstable	Tipping / Premature Slip
30	0.412	2.48	6.02	Stable ($\theta_c - 5^\circ$ boundary)
35	0.385	3.0681	7.97	Optimal Stable Configuration
40	0.401	2.71	6.76	Stable ($\theta_c + 5^\circ$ boundary)
45	Unstable	Unstable	Unstable	Slip
60	Unstable	Unstable	Unstable	Interfacial Failure

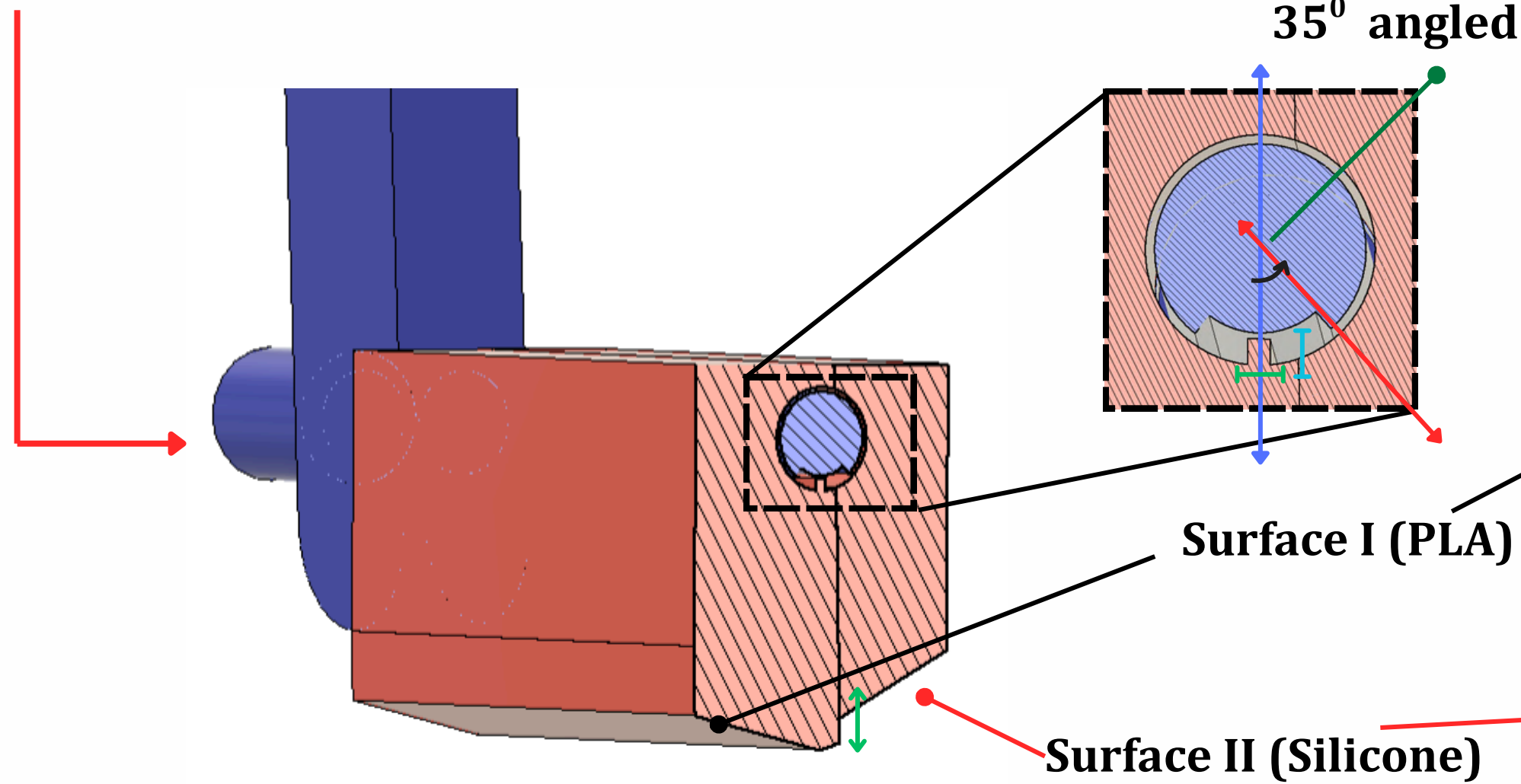
Iteration 3: PLA-Silicone in alternative contact area with a living hinge



**Stopper to limit the angle of rotation
of the living hinge (60°)**

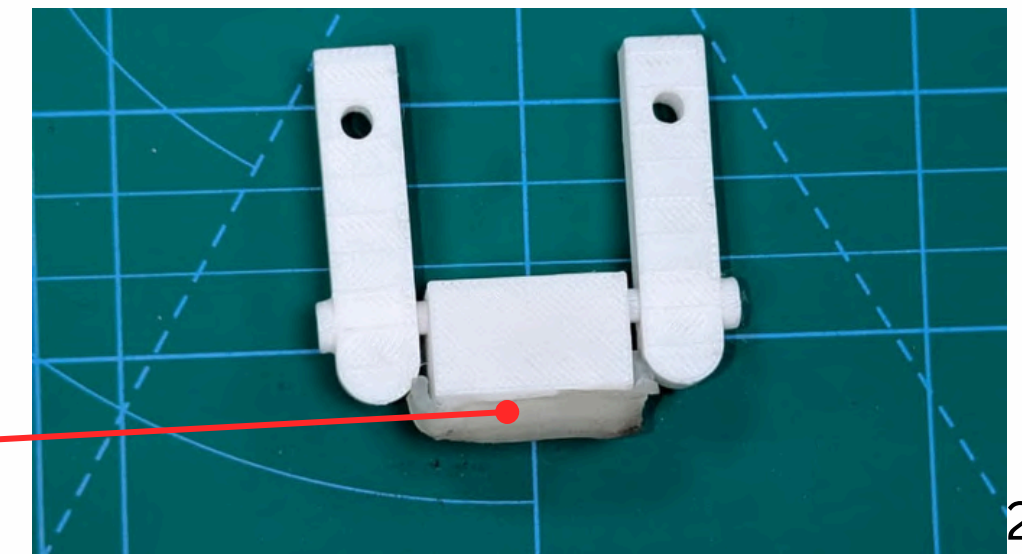


35° angled stopper



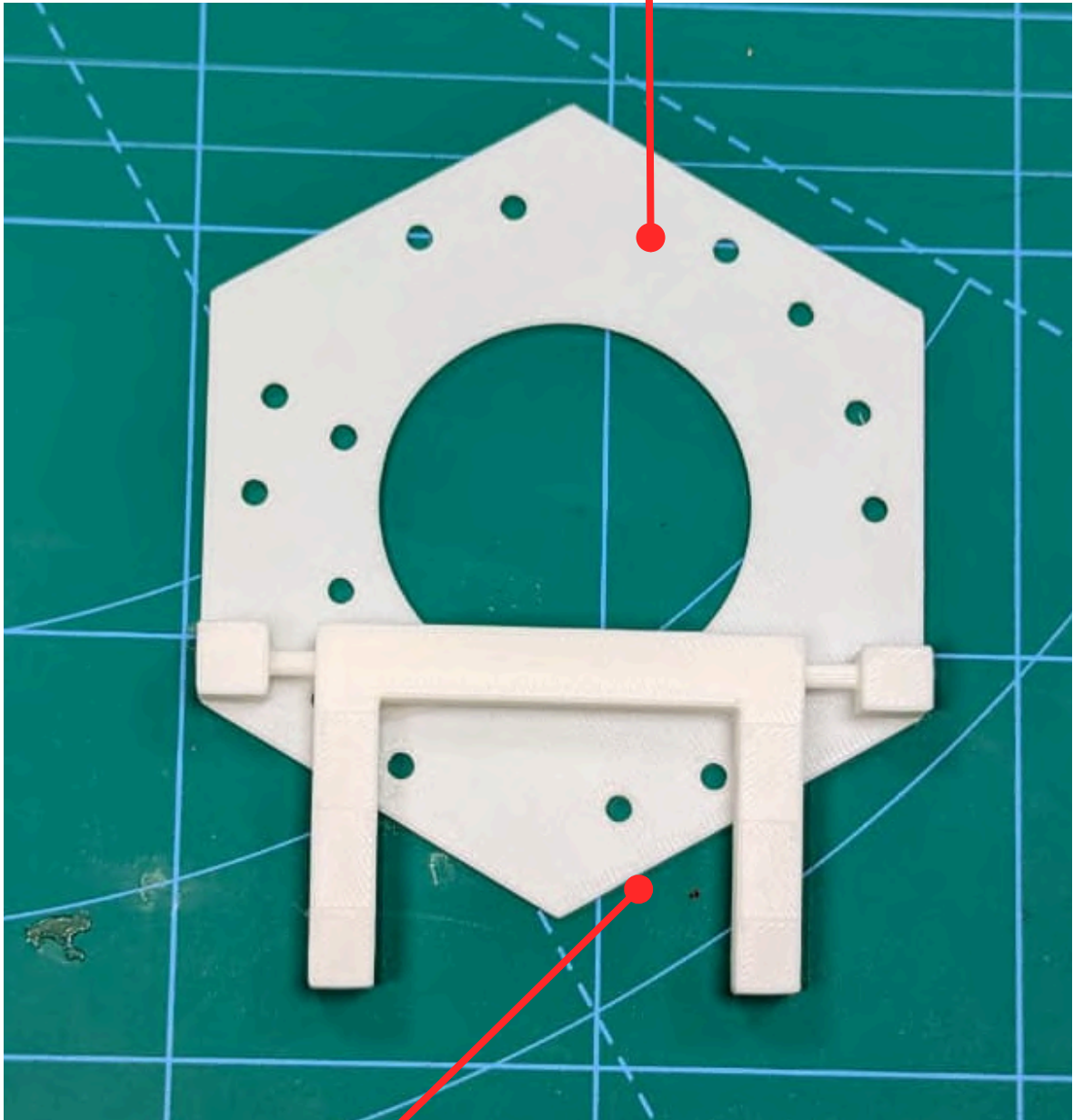
Surface I (PLA)

Surface II (Silicone)



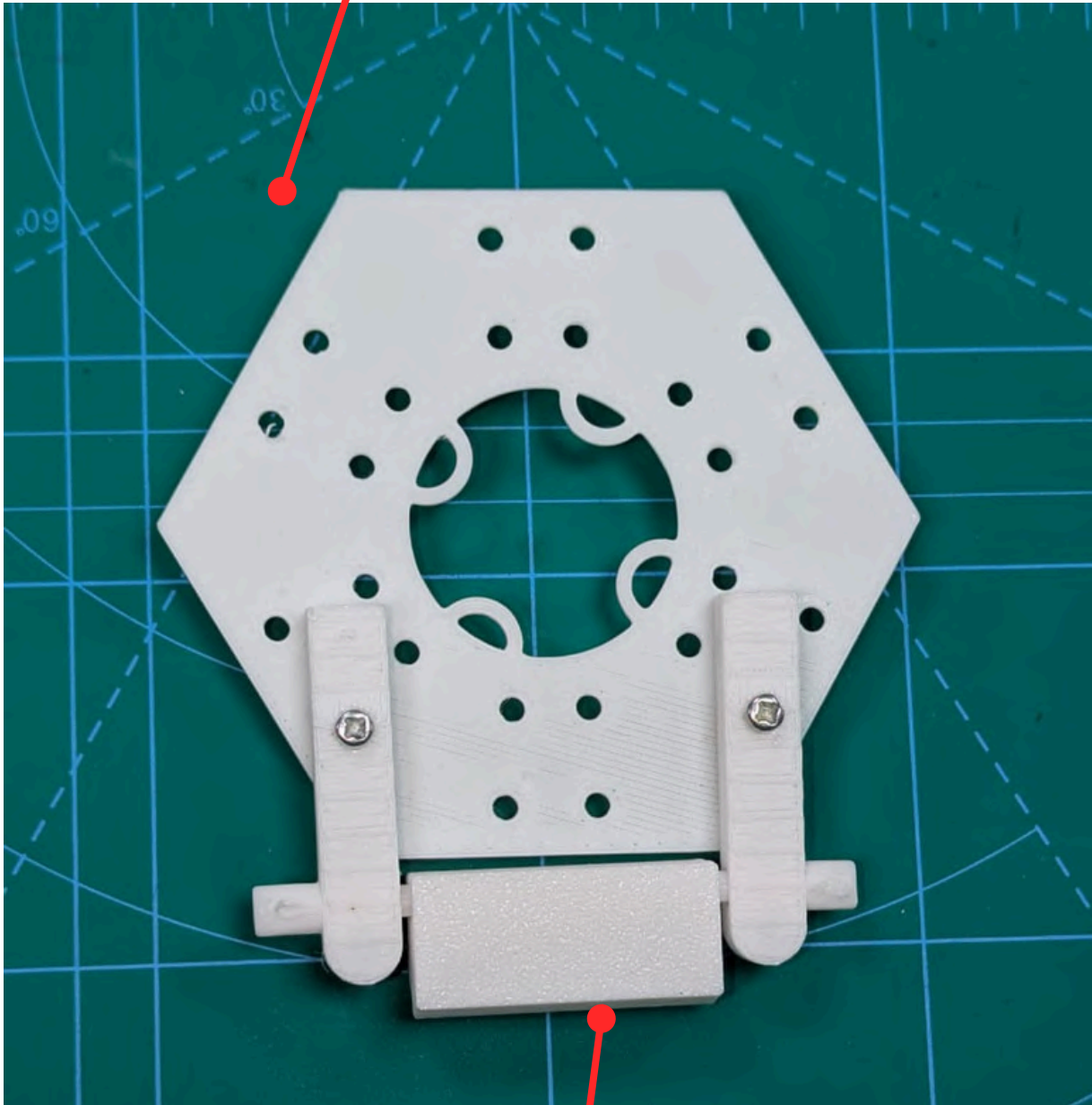
How is the chassis with the feet assembled?

Allows only one module to connect



Feet for balancing the module consisting of a single surface

Allows each pair of modules to connect while maintaining proper weight distribution

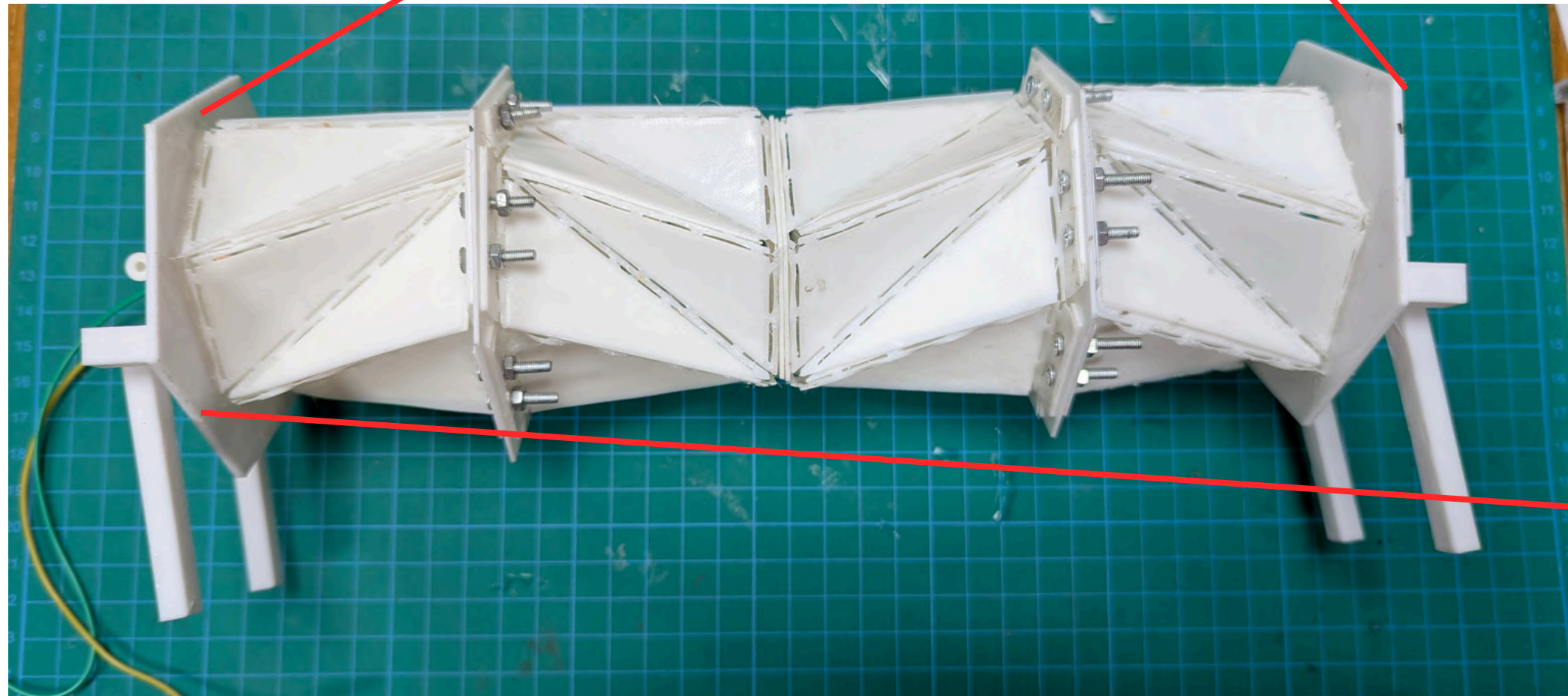


Feet consisting of two surfaces for anchoring and Sliding

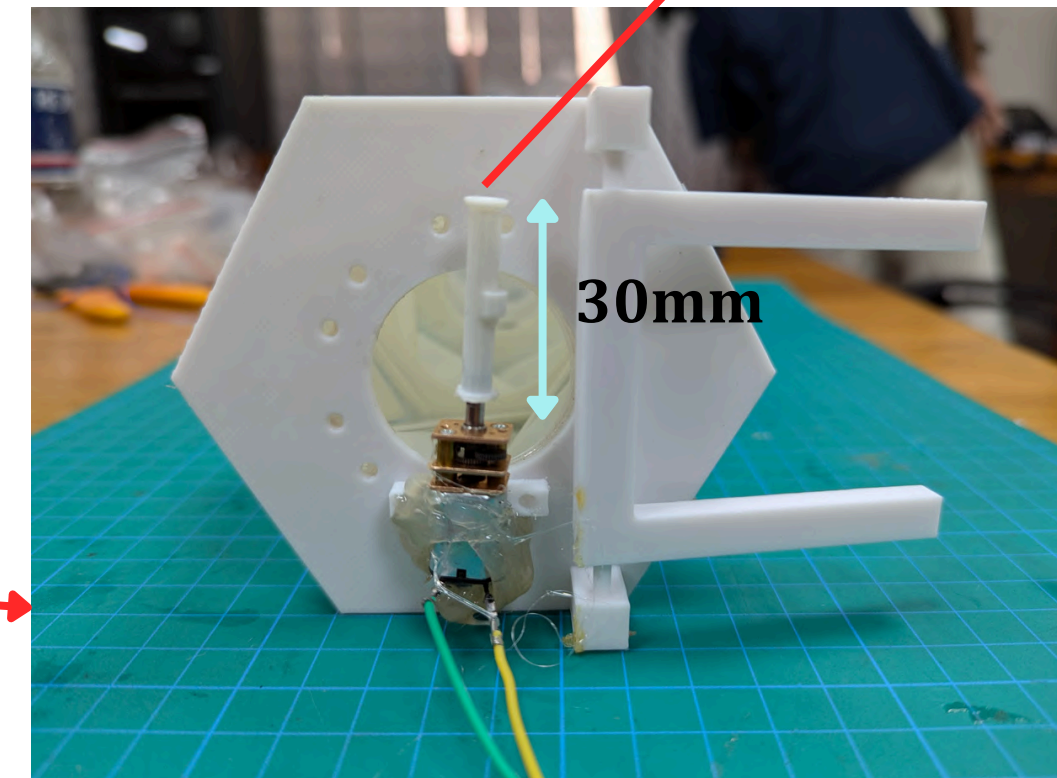
Optimization of the interated system

Prototype 1

2 plates at the end of modules

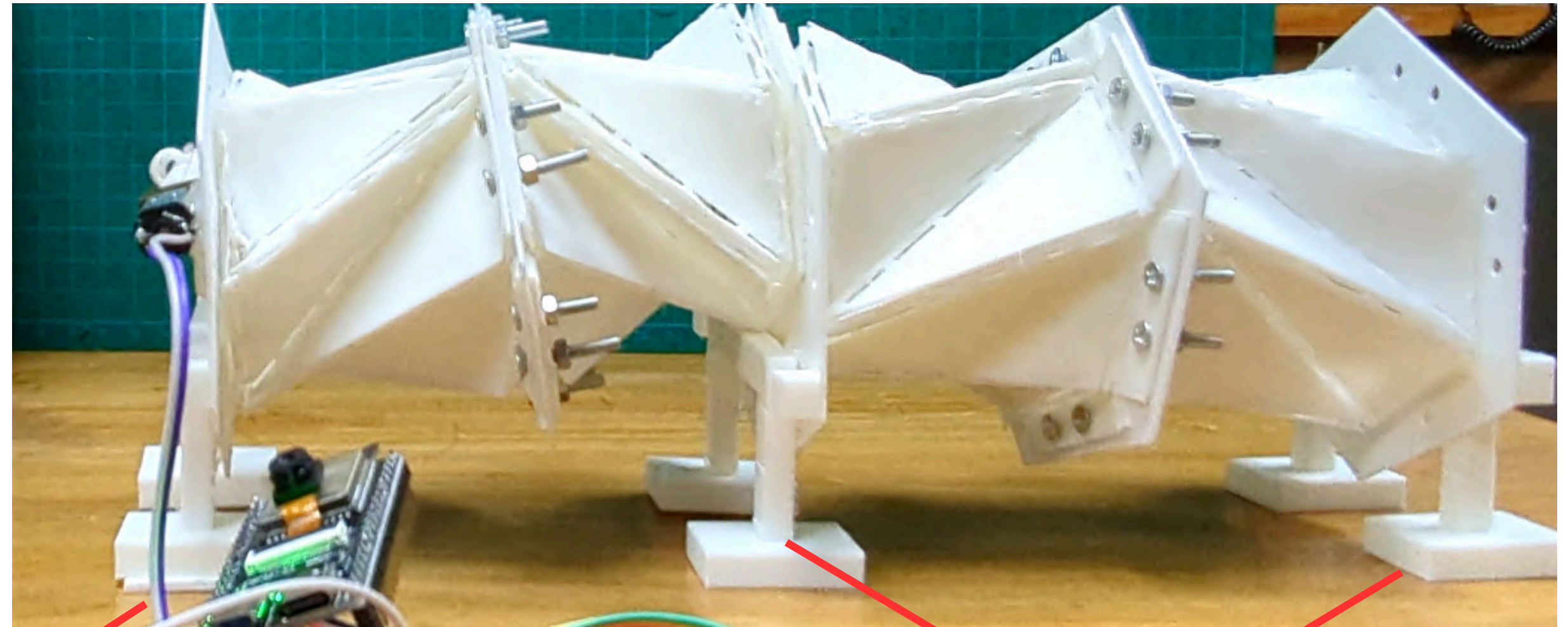
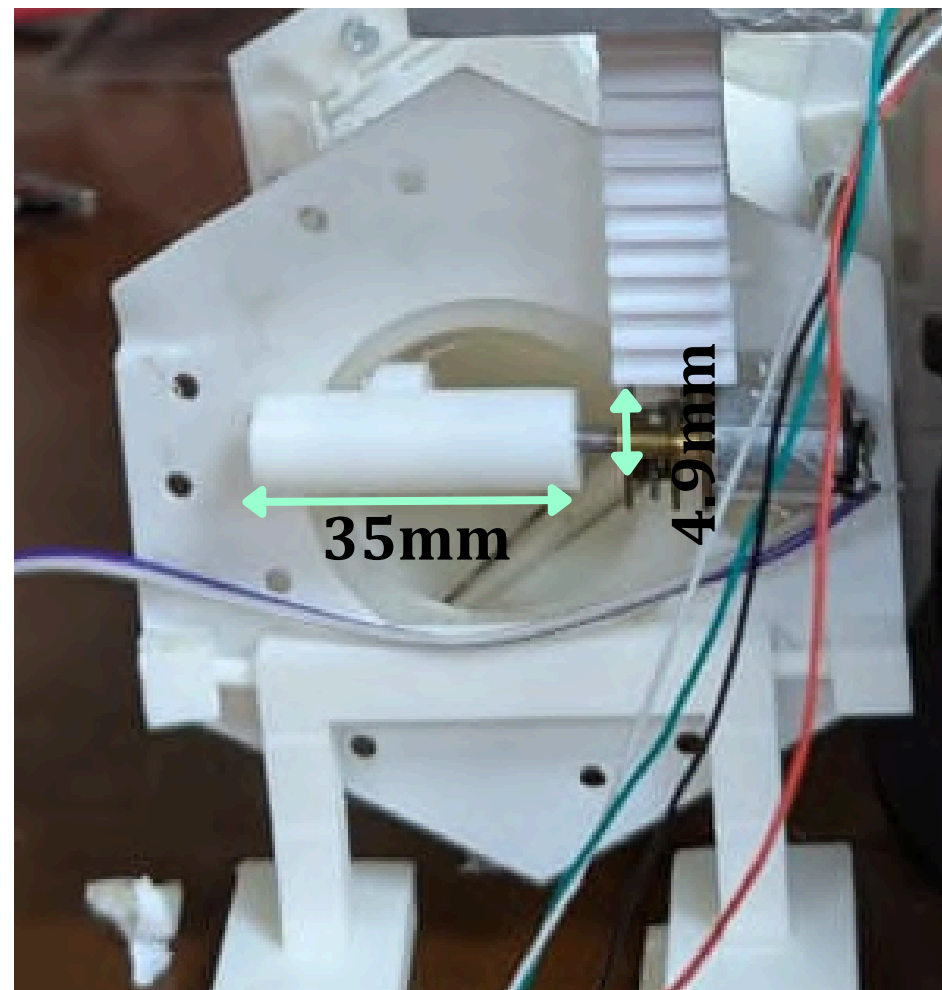


4.9mm diameter



Imbalance of the robot's **center of mass**

Prototype 2



6 feet

Each plate with 2 feet

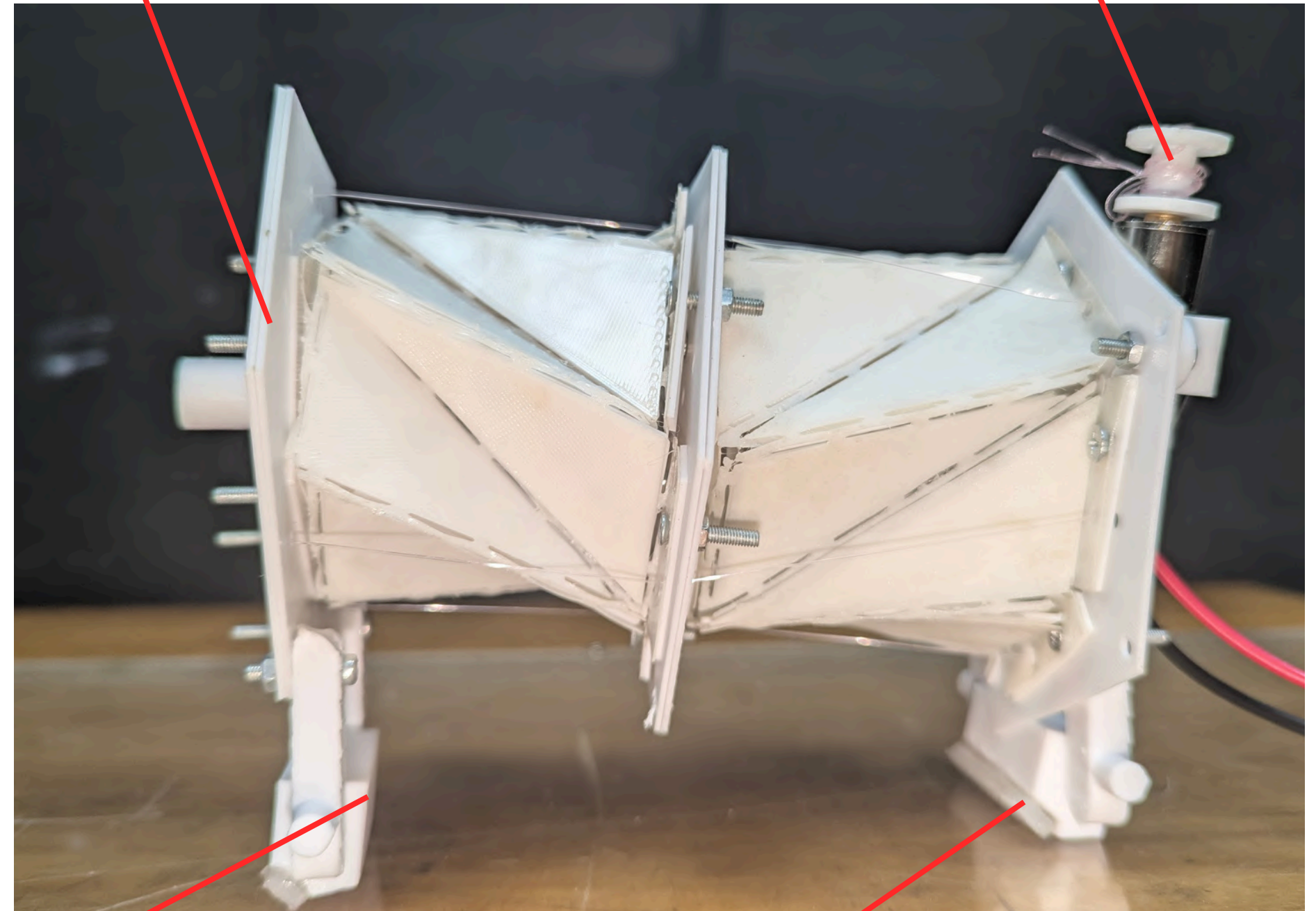
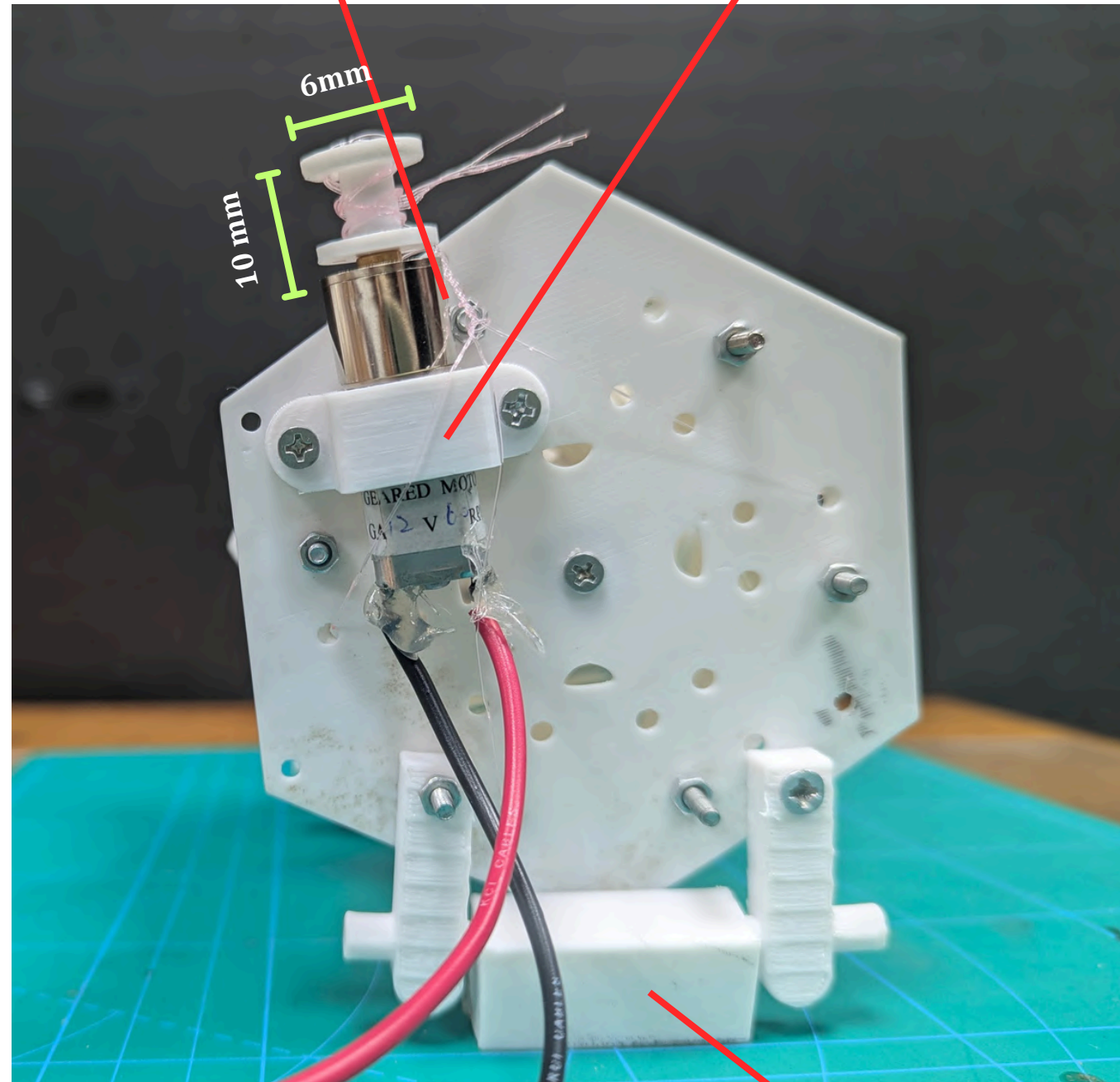
Lack of traction , couldn't move **forward**

Prototype 3

12 V, 60 RPM 16GA-050
DC gear motor

Motor holder attached to the front
plate

Spool rotates CW and CCW to actuate
locomotion



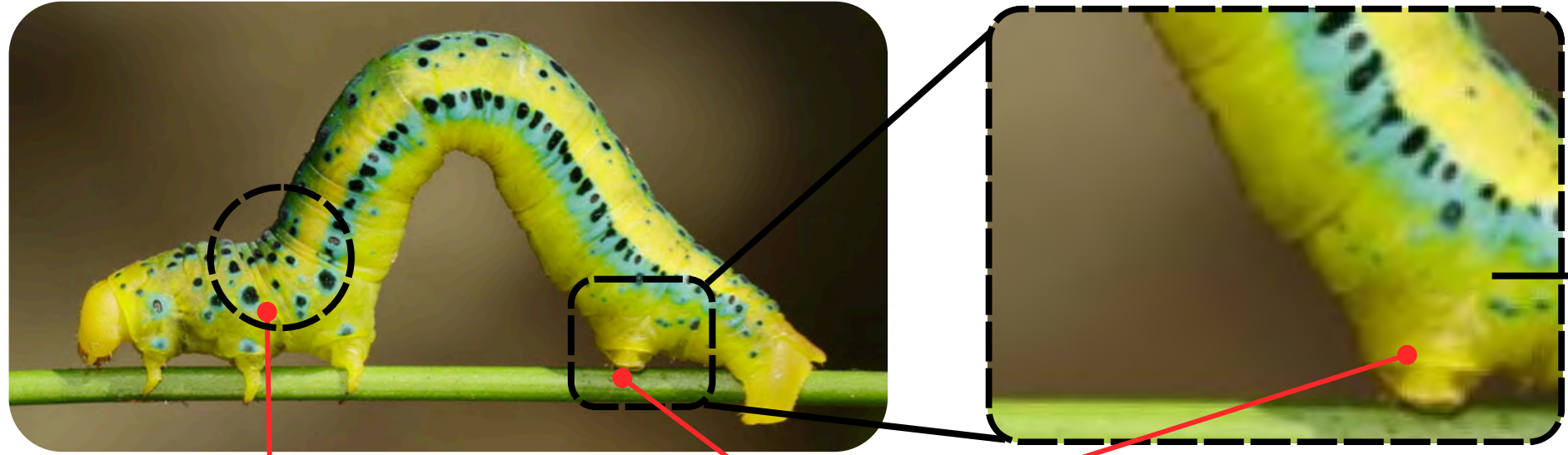
surface I (PLA)

surface II (Silicone)

Current functional prototype

How is the characteristics of an inchworm implemented in the integrated system ?

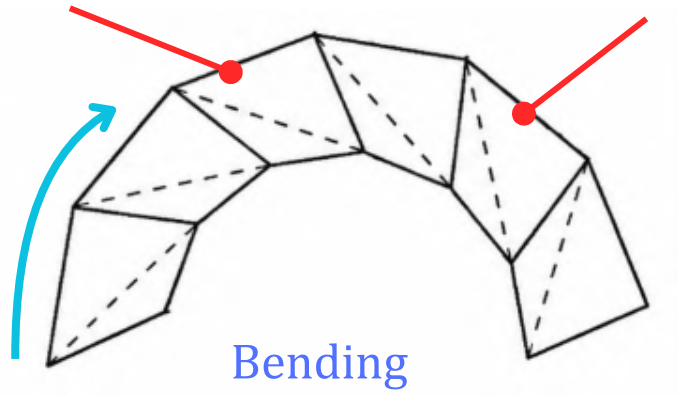
Biological Inspiration (Inchworm)



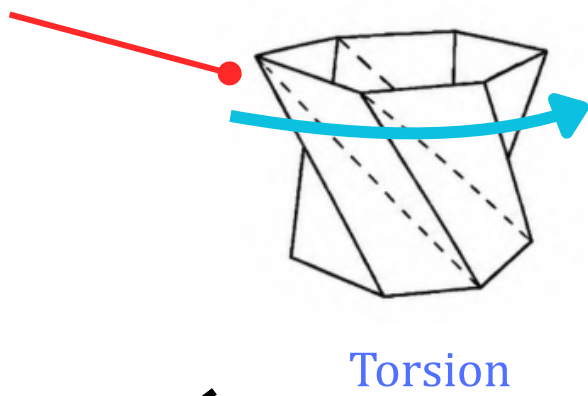
Setae (Rear Prolegs)

Anchors to the surface and provides high friction

Compliant Joints (Kresling Origami)

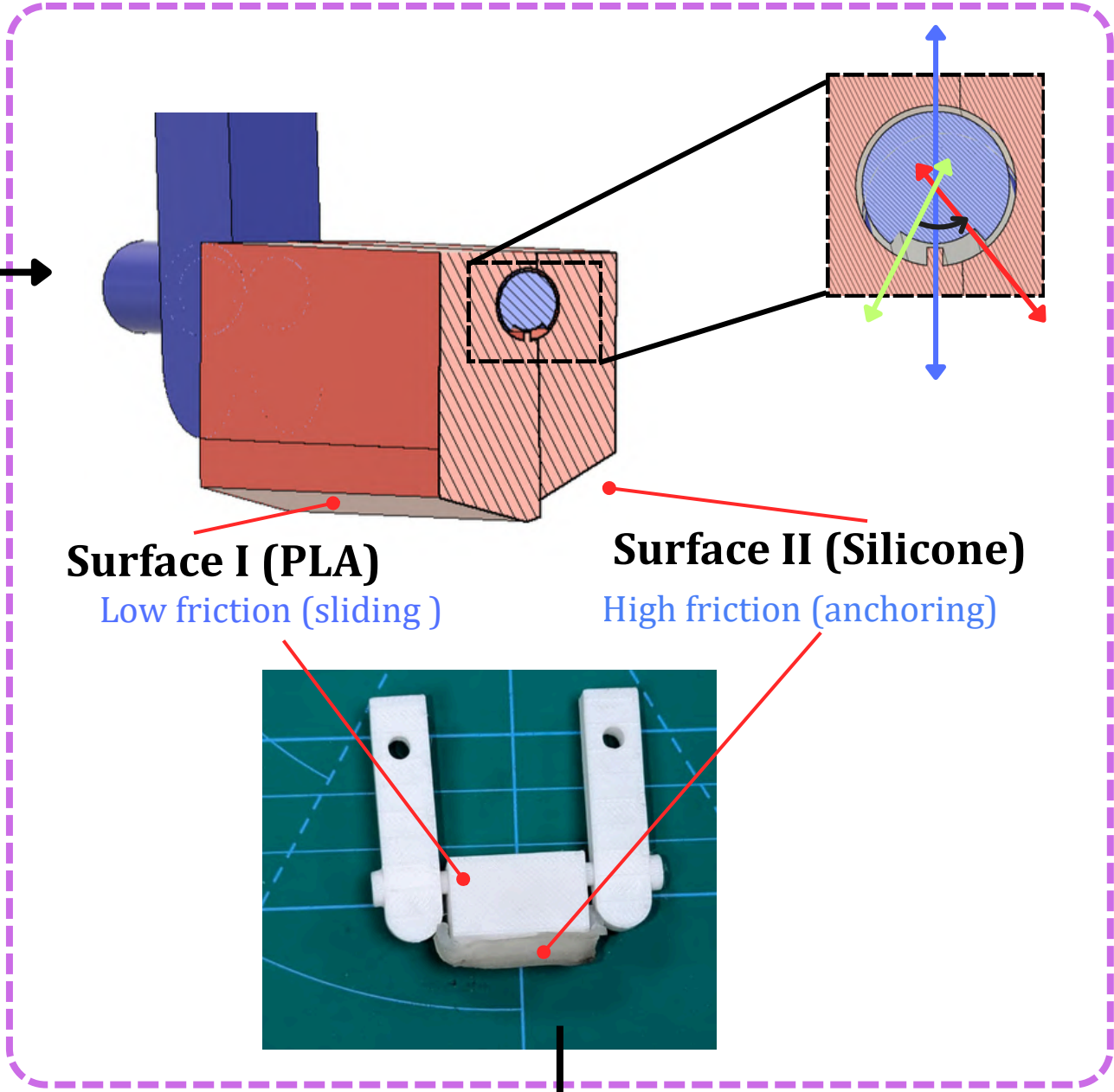


Bending



Torsion

Switchable Contact Surfaces

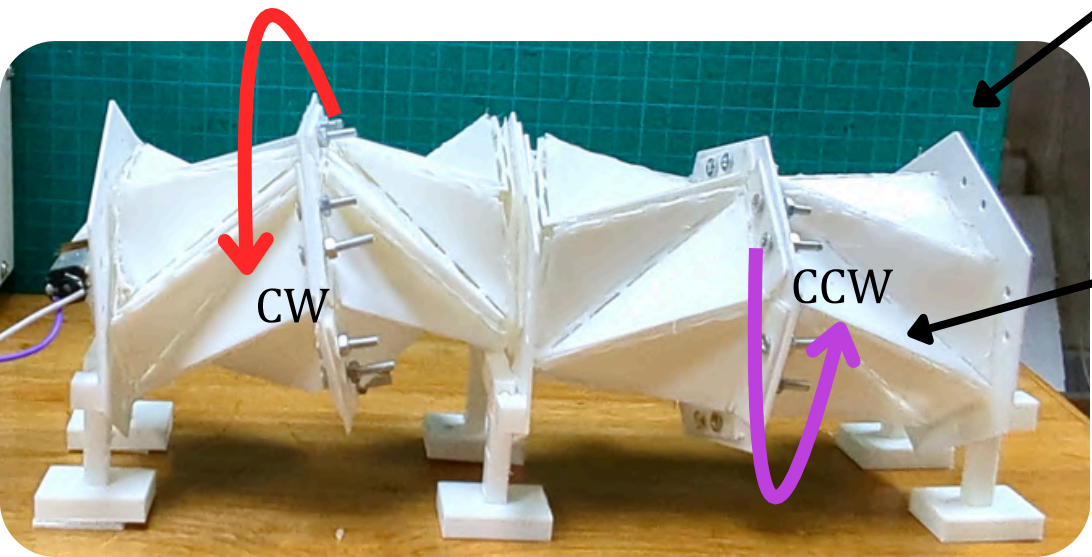


Surface I (PLA)

Low friction (sliding)

Surface II (Silicone)

High friction (anchoring)



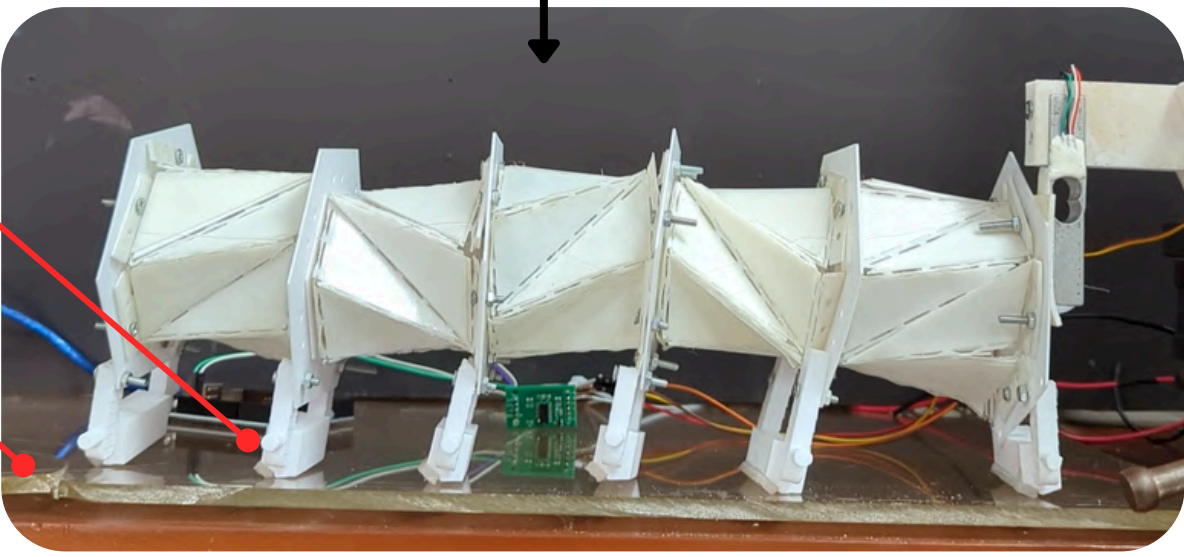
CW

CCW

Twist cancelling mirrored pair

Silicone Pads

Acrylic sheets



Does this system actually work?

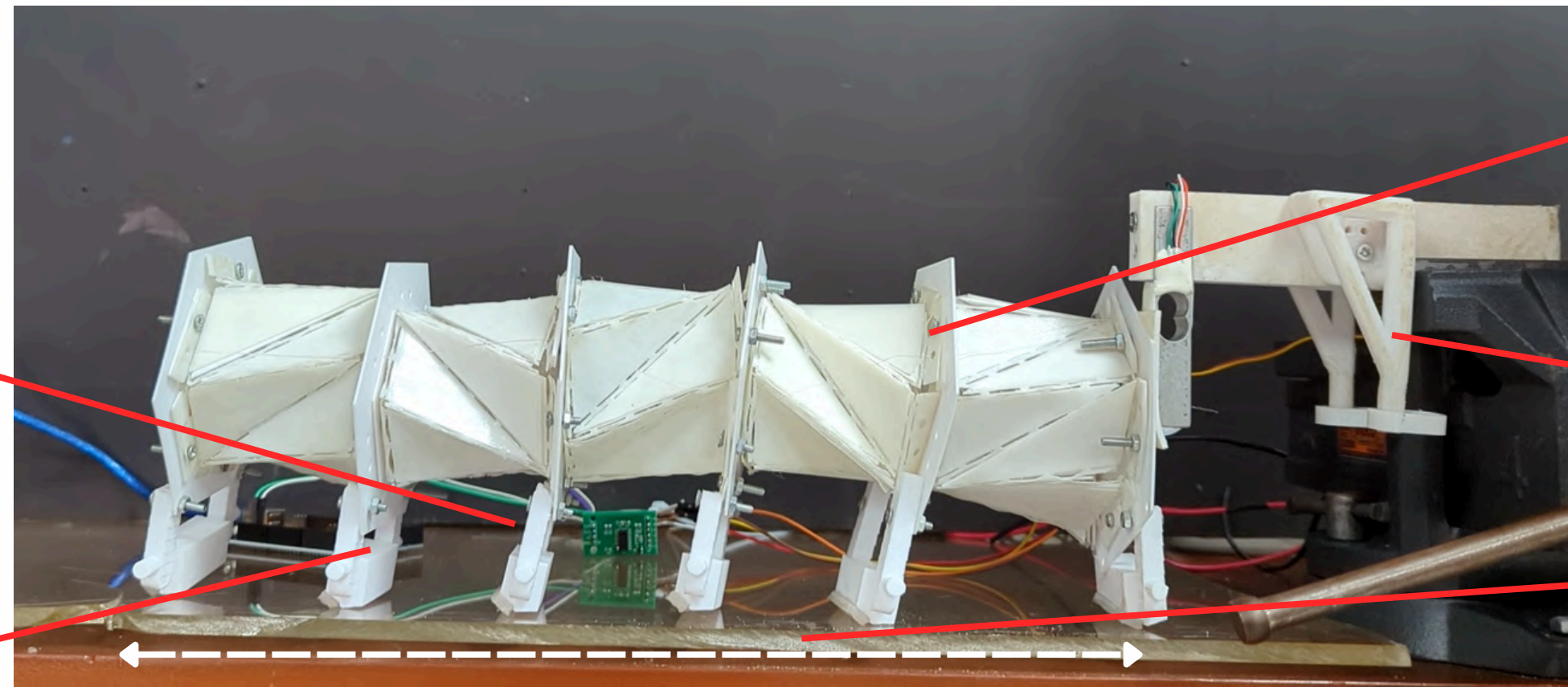
Silicone Pads
(2mm)

Acrylic Sheets
(3mm)

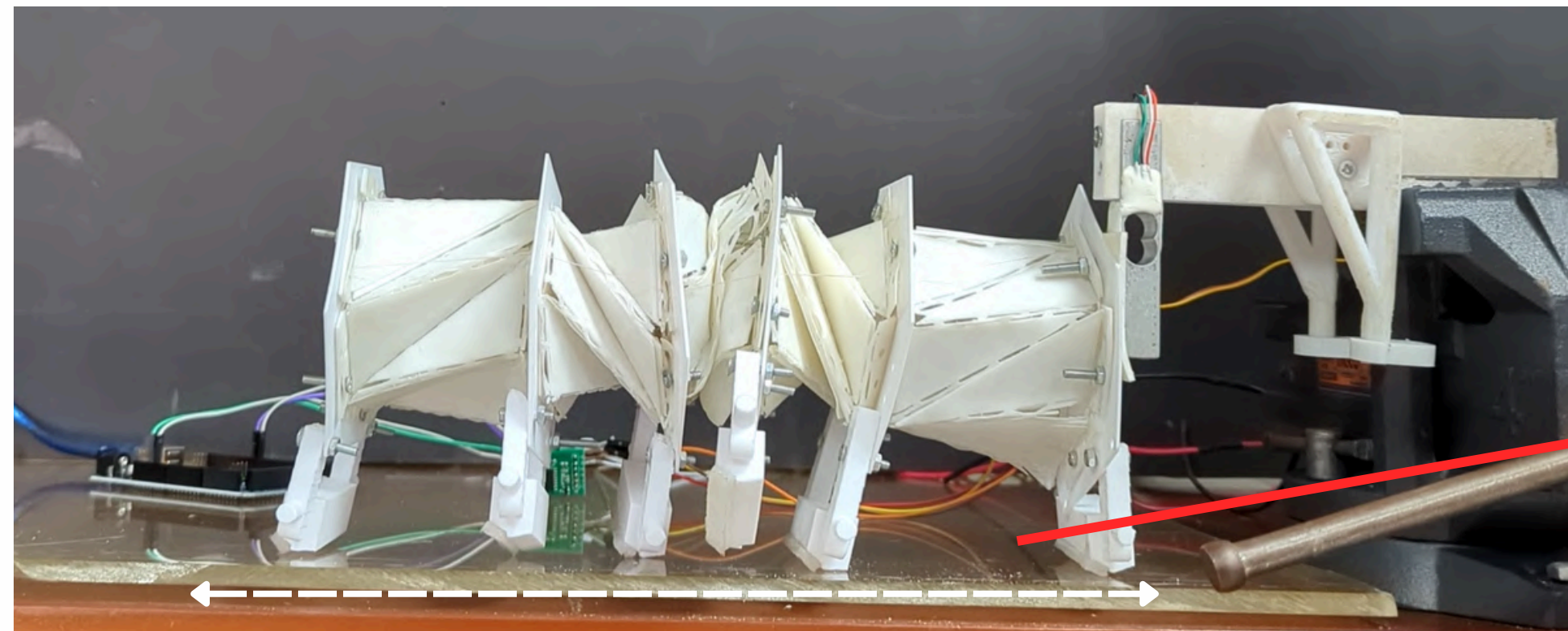
Passive module (TPU) that
allows lateral movement

Load Cell (5kg)

*Fully Expanded Robot
(33cm)*



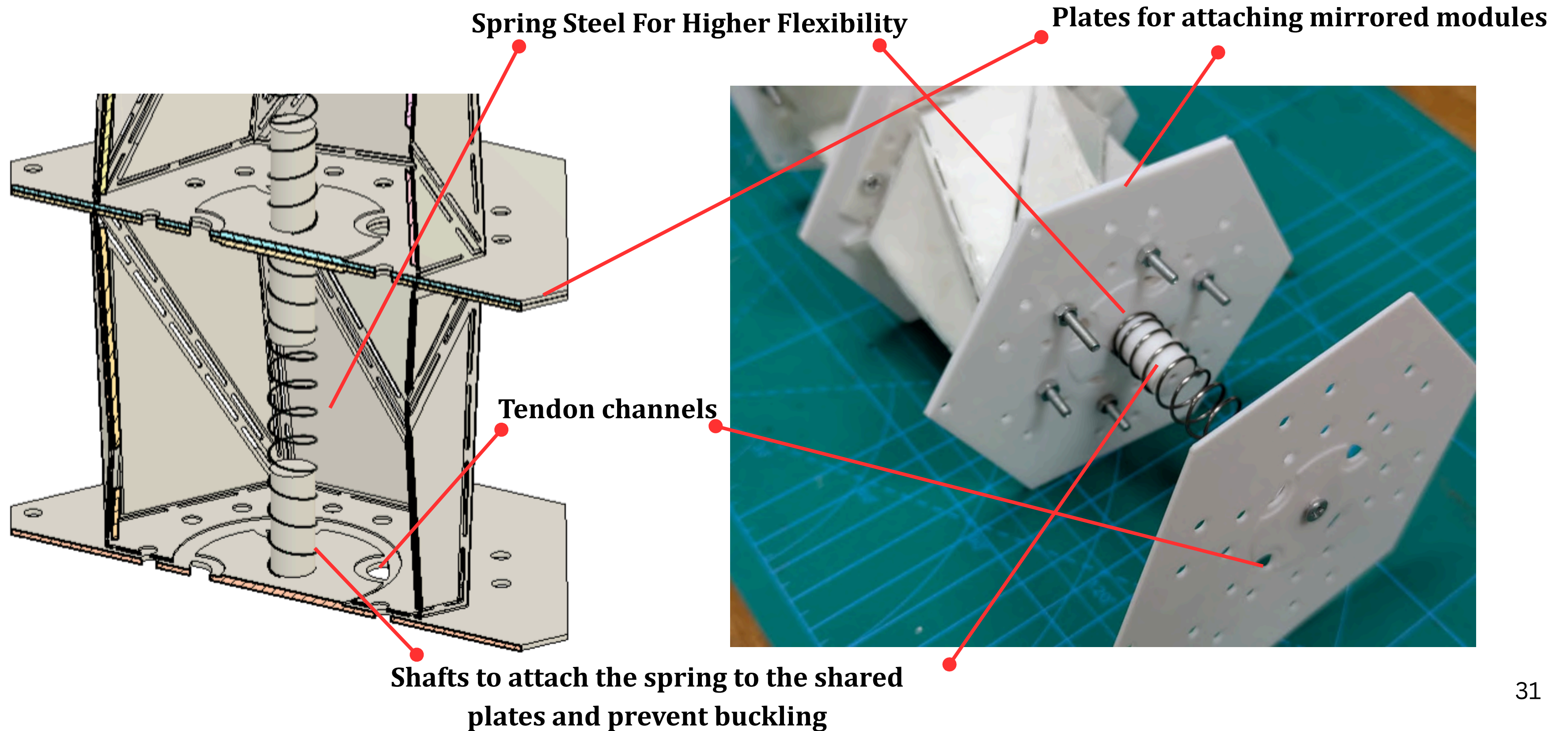
*Fully Contracted Robot
(18.5cm)*



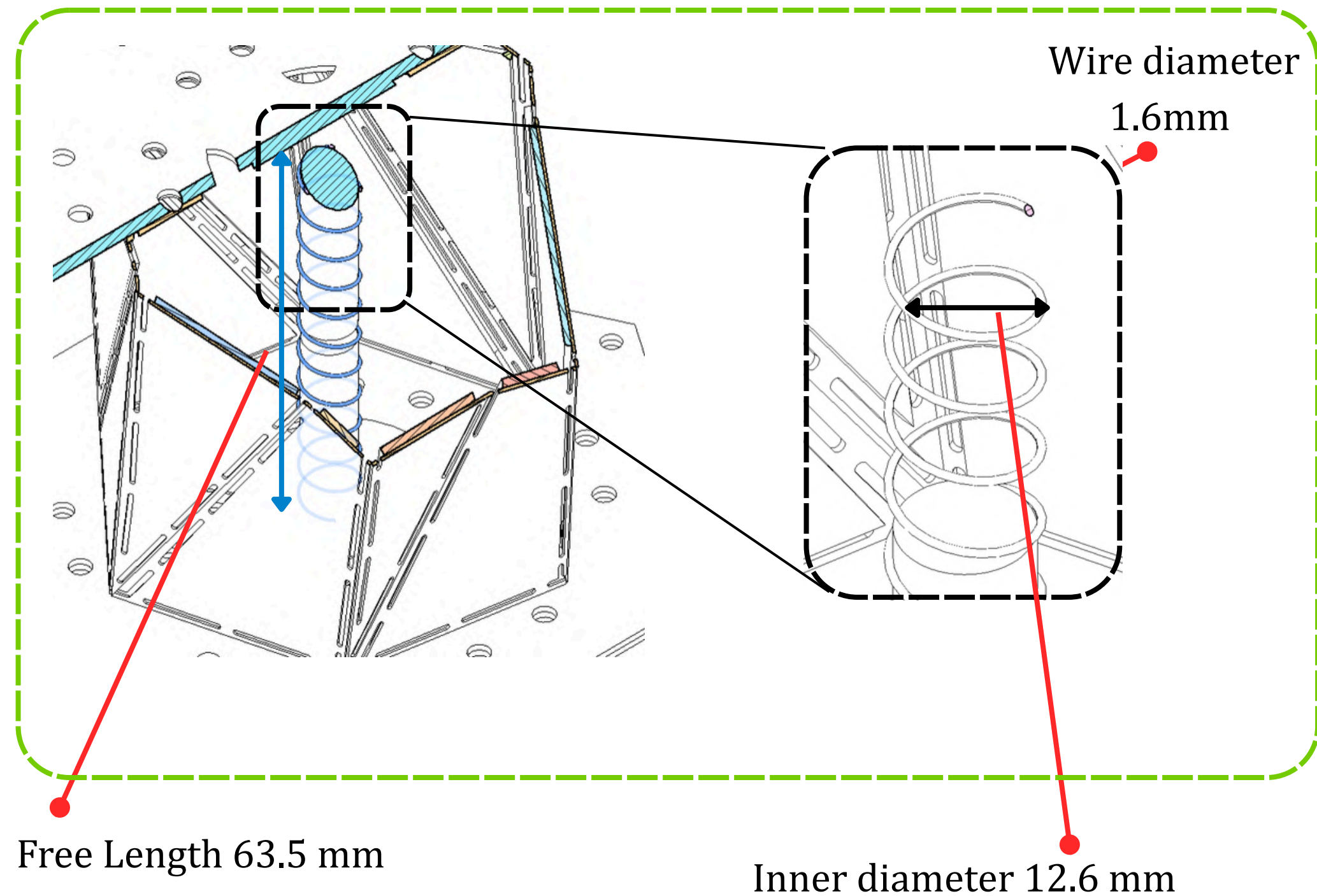
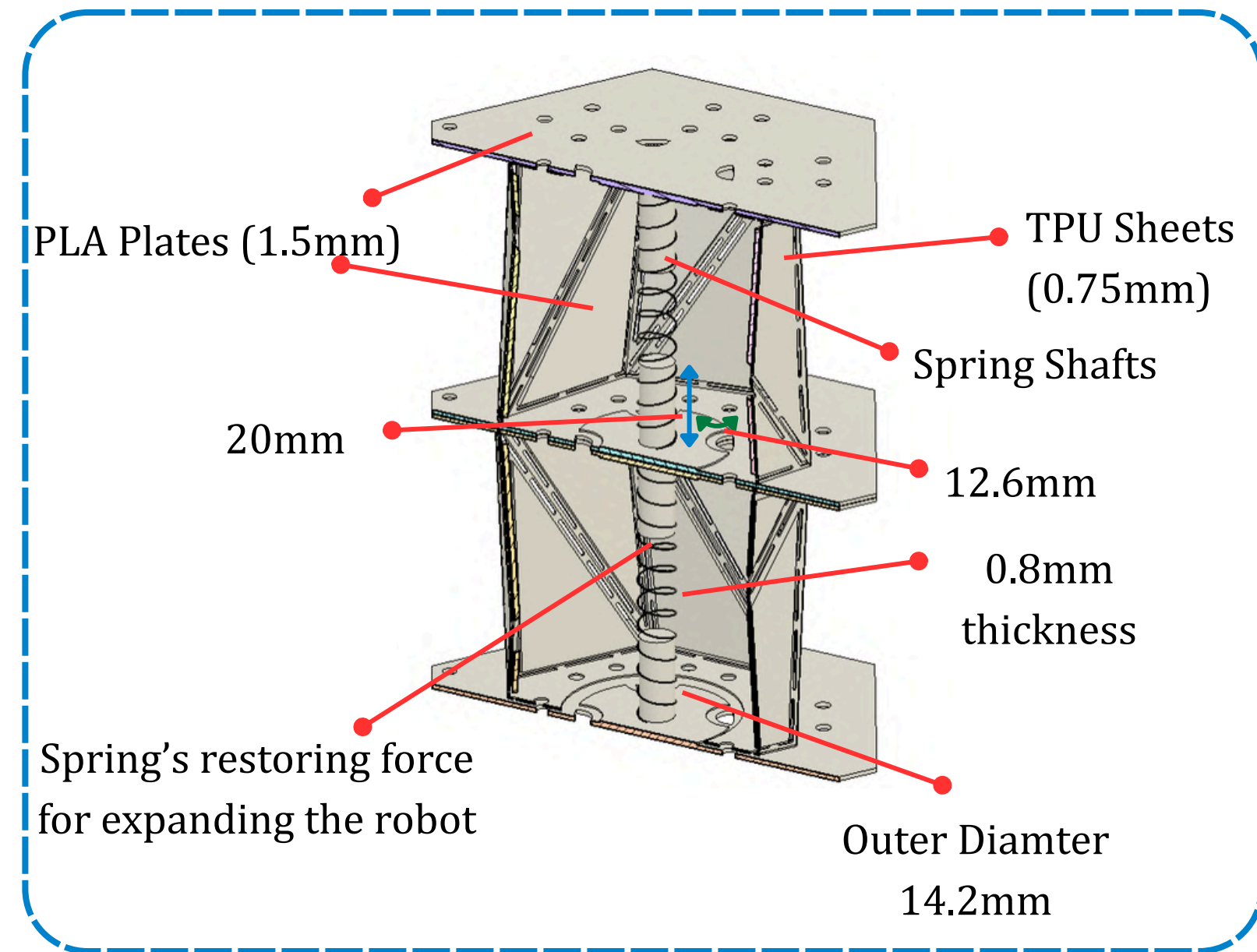
Unable to move forward, due to **lack of restoration force**

How can the restoration force of the chassis material be increased with the design constraints ?

Compression spring between successive plates



Design constraints of the compression springs

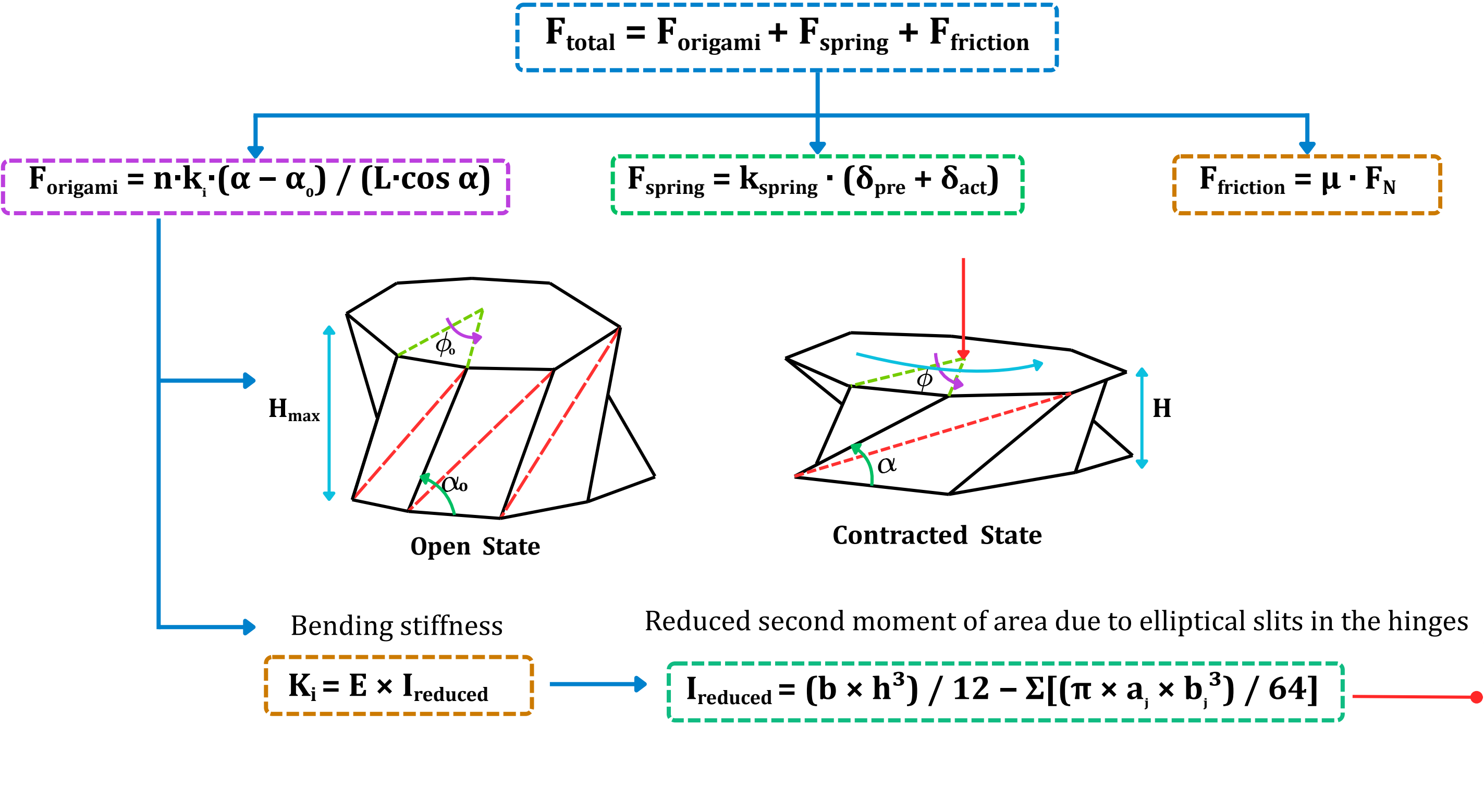


Spring index $\rightarrow 4 \leq C \leq 12 \rightarrow 8.875$

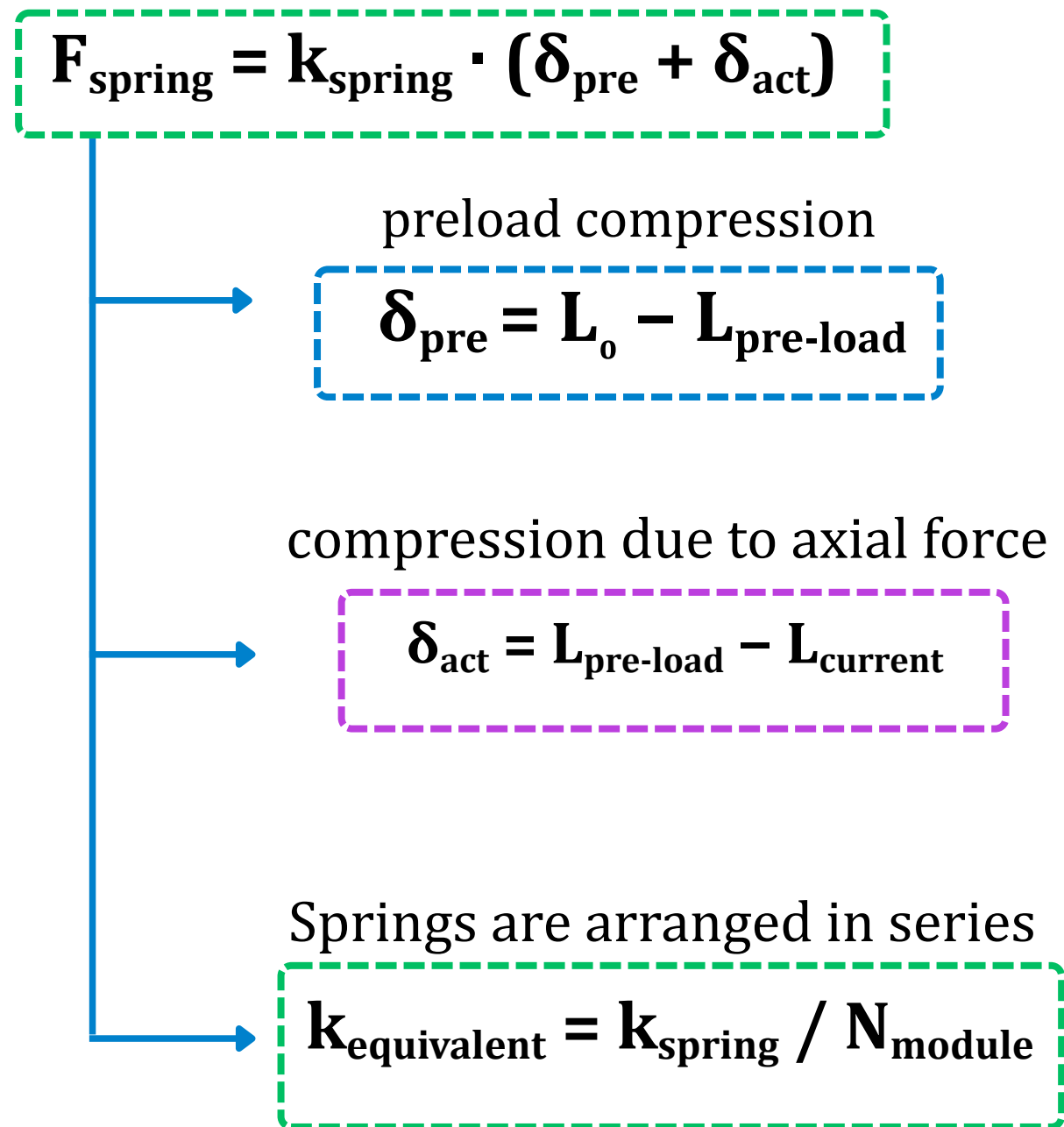
Pre compressed springs in each module **supplies the necessary restoration force**

Mechanical and Kinematic Analysis

For contraction, the actuator has to overcome the restoration force of the Kresling chassis, compression spring, and frictional resistance simultaneously



Mechanical and Kinematic Analysis : Motor Capability & Operating Margin



Torque produced by the system

$$T_{\text{actuator}} = (F_{\text{total}} \times r_p) \rightarrow 1.954\text{E-}03 \text{ Nm}$$

0.352 N

Stall to operating ratios

using a 12 V, 60 RPM 16GA-050 DC gear motor with I_{stall} 1.1A and T_{stall} 0.6276 Nm

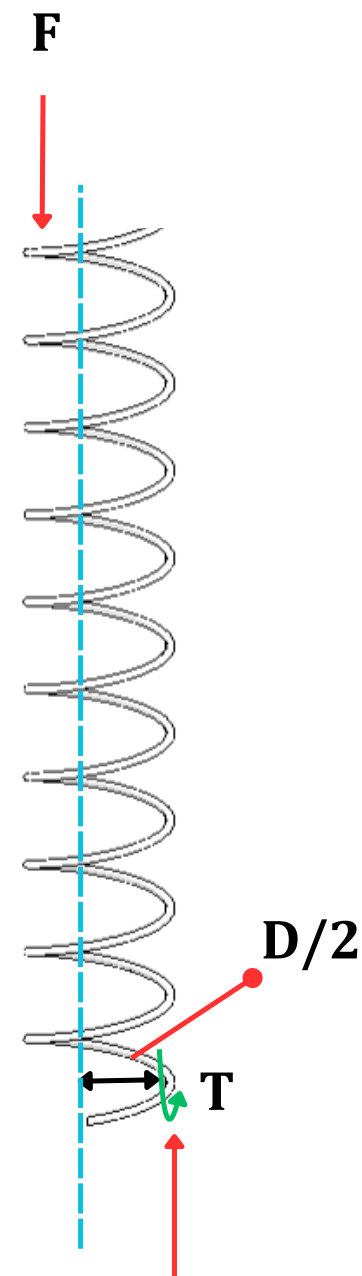
$$SF_{\text{current}} = I_{\text{stall}} / I_{\text{max}}$$

Stall current $\sim 5.39\times$ the maximum current experimentally observed during locomotion

$$SF_{\text{Torque}} = T_{\text{stall}} / T_{\text{actuator}}$$

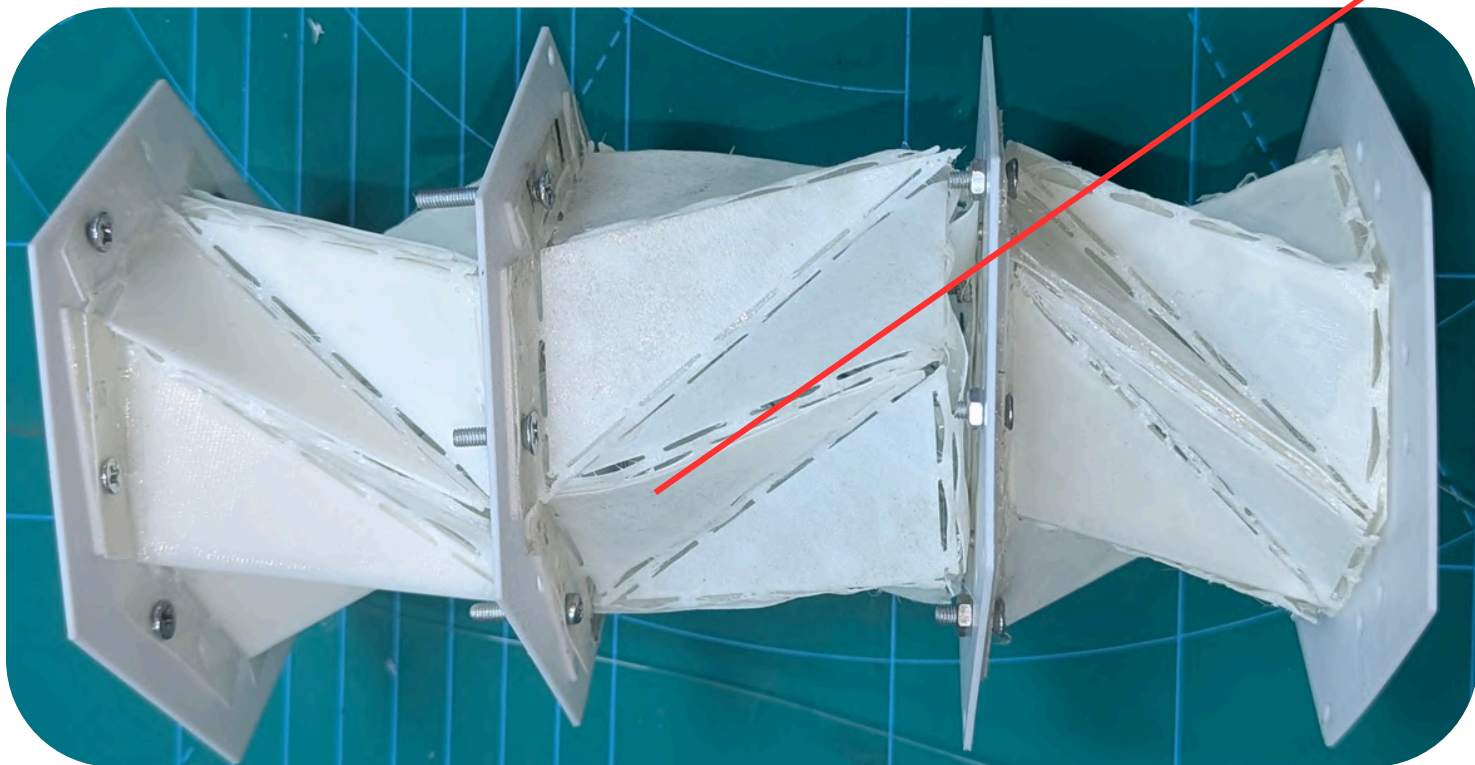
321 $\times$ stall-to-required-torque ratio

Axial Force



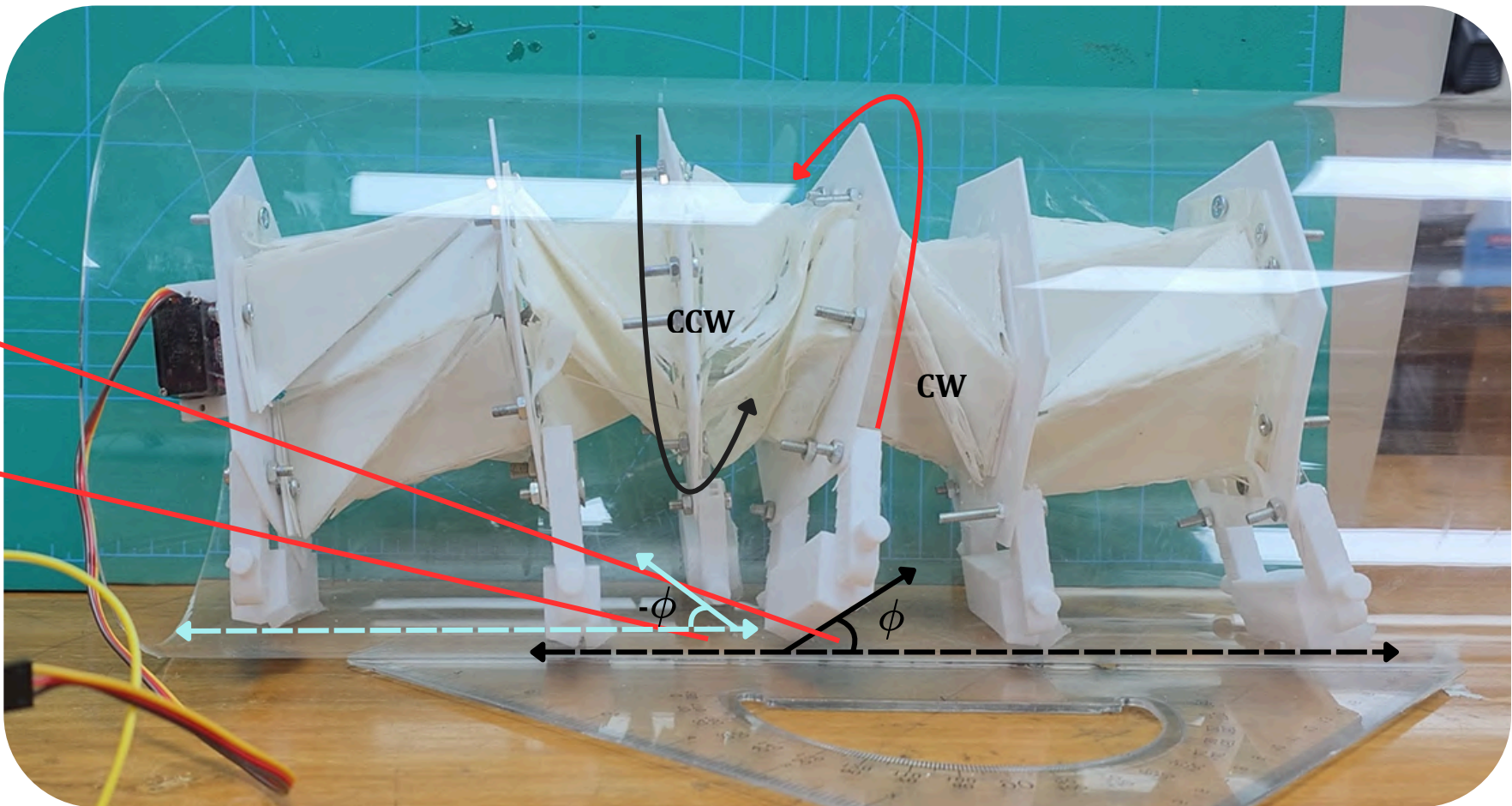
Integration of compliant passive module for yaw and pitch

Passive module between two active modules

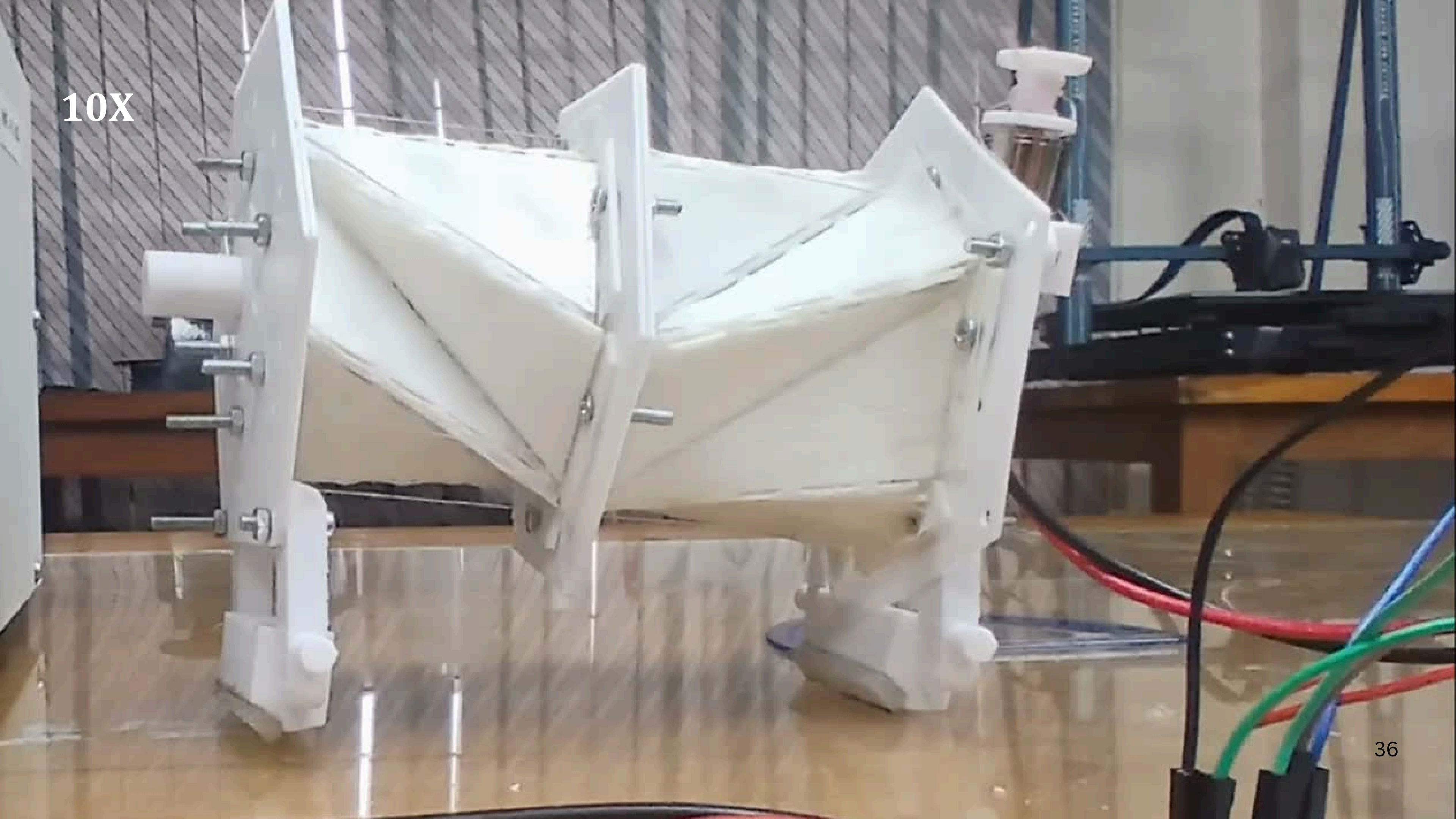


Allows Yaw and pitch

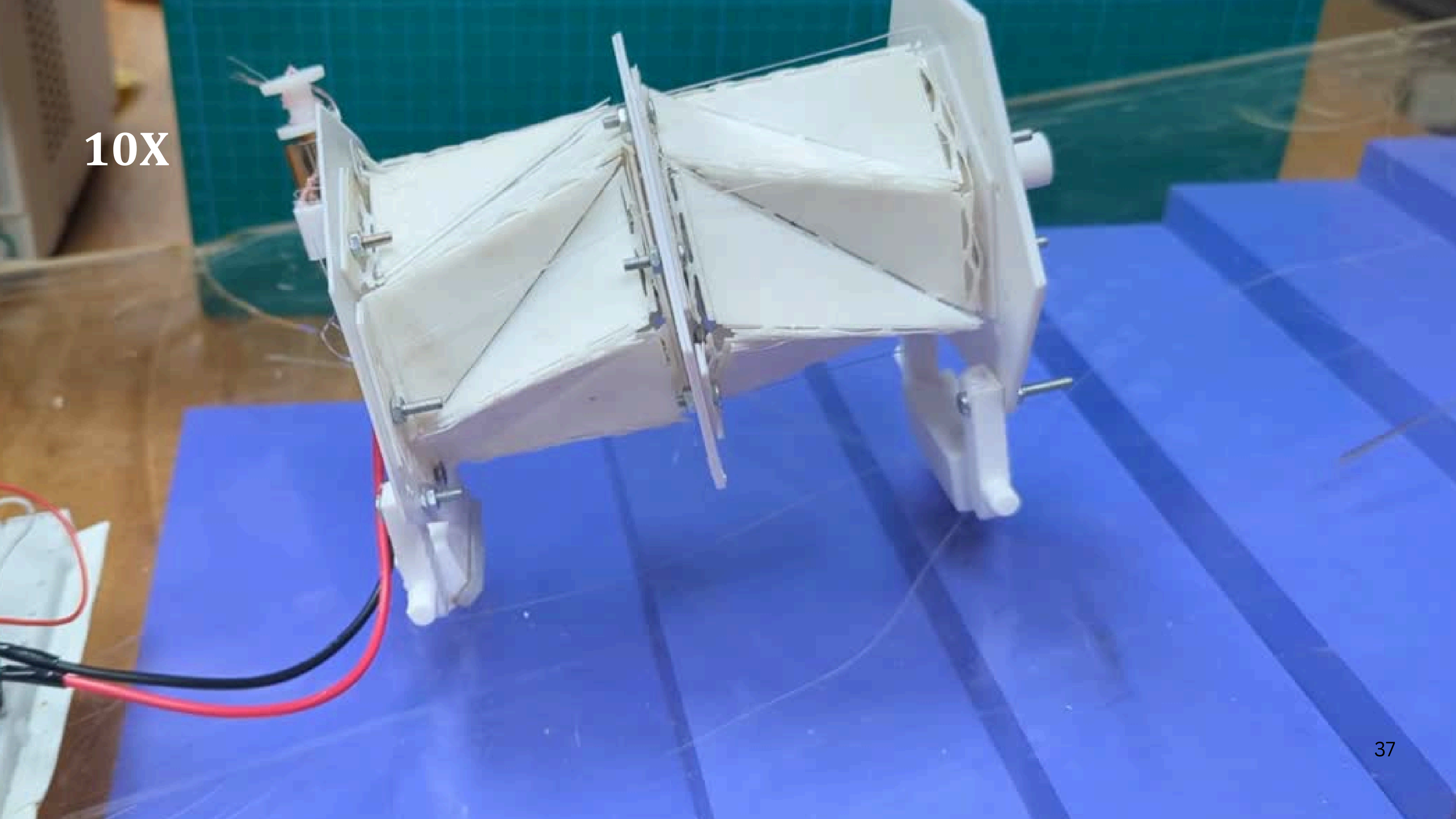
Peristalsis twisting led to the **misalignment of the foot**



10X



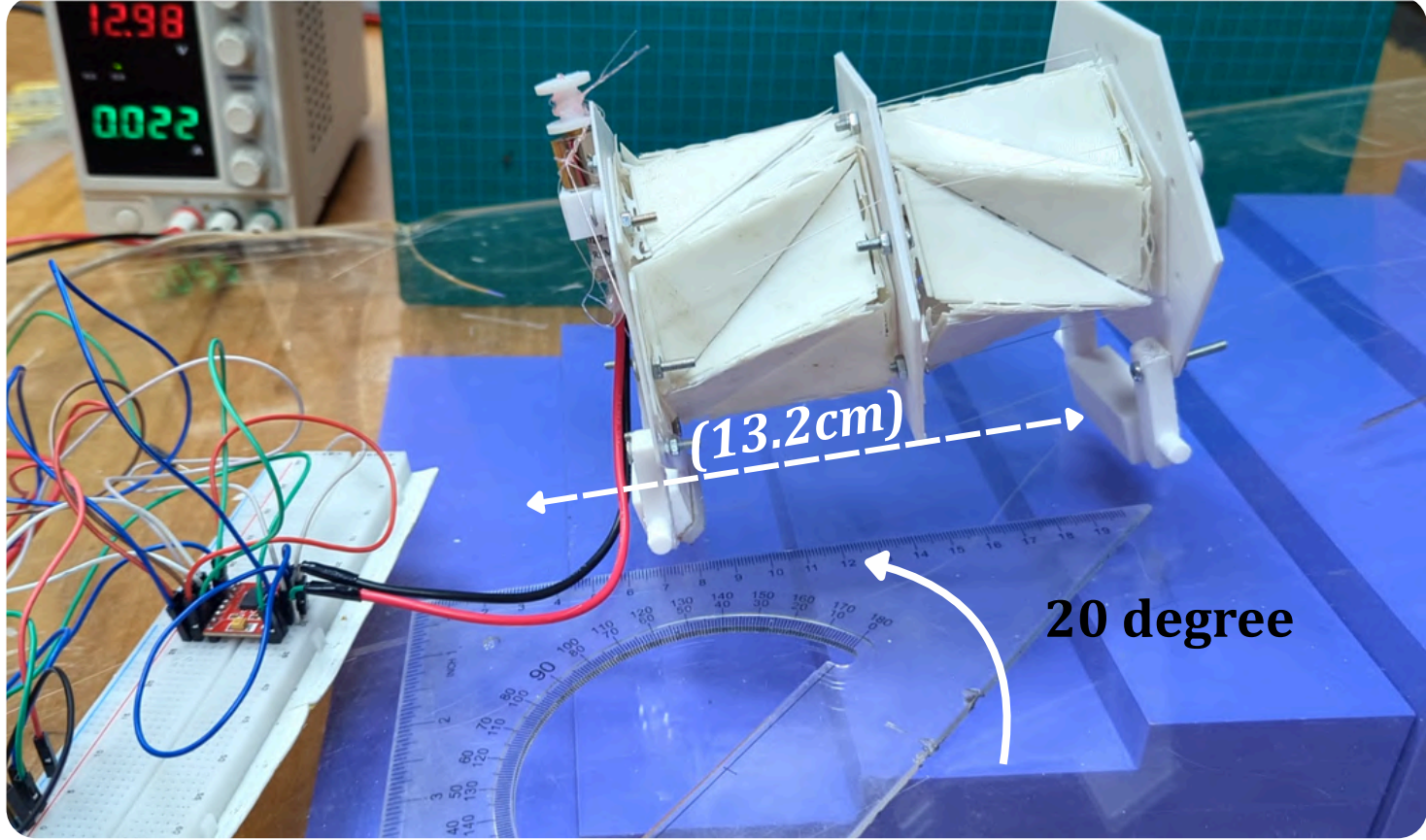
10X



Experimental Setup

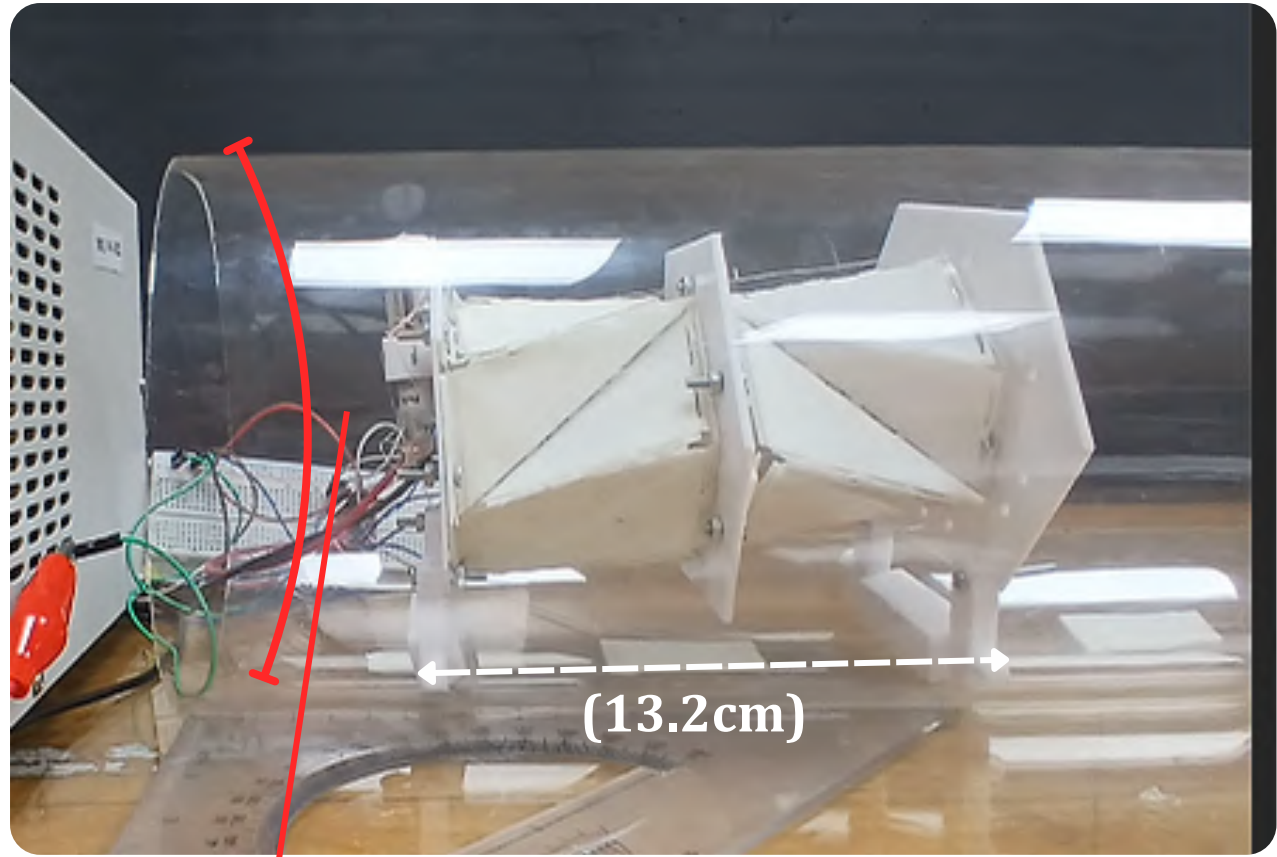


Horizontal Acrylic Sheet



Inclined Acrylic Sheet(20 degree)

Evaluates locomotion feasibility across flat, inclined, and constrained internal pipe geometries.



Horizontal Acrylic Pipe(5.6 inch)